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MATRIX ANALYSIS AND LINEAR ALGEBRA

Selected sections, adapted for

MA1522 Linear Algebra for Computing

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A reading companion to the course, not a replacement for its slides. The sections of Tyrtyshnikov's *Matrix Analysis and Linear Algebra* that bear on the MA1522 syllabus are arranged under the course's chapters and sections and rewritten in the course's notation and terminology. Material that goes beyond the syllabus is kept and marked BEYOND MA1522. Each subsection notes in the source the section of the book it comes from.

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Notation

The notation follows the MA1522 lecture slides.

Notation	Meaning
$\mathbf{u}, \mathbf{v}, \mathbf{w}$; $\mathbf{0}$	vectors in $\mathbb{R}^n$ (columns); the zero vector
$\mathbf{A} = \big(a_{ij}\big)_{m\times n}$	an $m \times n$ matrix with (i,j)-entry a_{ij}
$\mathbf{I}_n$, $\mathbf{0}_{m\times n}$	identity matrix of order n; zero matrix
$\mathbf{A}^T$, $\mathbf{A}^{-1}$	transpose; inverse
$(\mathbf{A} \mid \mathbf{b})$	augmented matrix of the system $\mathbf{Ax} = \mathbf{b}$
$R_i \leftrightarrow R_j$, $R_i + cR_j$, aR_i	elementary row operations ($a \neq 0$)
$\det(\mathbf{A})$, $\lvert\mathbf{A}\rvert$	determinant
$\mathbf{M}_{ij}$, A_{ij}	(i,j) matrix minor (a submatrix); (i,j)-cofactor $A_{ij} = (-1)^{i+j}\det\big(\mathbf{M}_{ij}\big)$
$\operatorname{adj}(\mathbf{A})$	adjoint of $\mathbf{A}$ (transpose of the matrix of cofactors)
$\mathbf{u}\cdot\mathbf{v}$, $\lVert\mathbf{u}\rVert$, $d(\mathbf{u},\mathbf{v})$	dot product, norm, distance $\lVert\mathbf{u}-\mathbf{v}\rVert$ in $\mathbb{R}^n$
$\operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\}$	linear span
$\dim(V)$	dimension of a subspace V
$[\mathbf{v}]_S$	coordinate vector of $\mathbf{v}$ relative to the basis S
$\mathbf{P}$ with $[\mathbf{w}]_S = \mathbf{P}[\mathbf{w}]_T$	transition matrix from T to S
$\operatorname{Col}(\mathbf{A})$, $\operatorname{Row}(\mathbf{A})$, $\operatorname{Null}(\mathbf{A})$	column space, row space, nullspace
$\operatorname{rank}(\mathbf{A})$, $\operatorname{nullity}(\mathbf{A})$	rank; nullity $= \dim(\operatorname{Null}(\mathbf{A}))$
$V^\perp$	orthogonal complement of a subspace V
$\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$	orthogonal projection $\mathbf{w}_p$ of $\mathbf{w}$ onto V, and $\mathbf{w}_n \perp V$
$\mathbf{A} = \mathbf{QR}$, $\mathbf{A} = \mathbf{LU}$	QR and LU factorizations
$\operatorname{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})$	characteristic polynomial
$E_\lambda = \operatorname{Null}(\lambda\mathbf{I} - \mathbf{A})$	eigenspace of the eigenvalue λ
$\mathbf{A} = \mathbf{PDP}^{-1}$, $\mathbf{A} = \mathbf{PDP}^T$	diagonalization; orthogonal diagonalization
$\mathbf{A} = \mathbf{U\Sigma V}^T$	singular value decomposition
$T : \mathbb{R}^n \to \mathbb{R}^m$	linear transformation, $T(\mathbf{u}) = \mathbf{Au}$ with standard matrix $\mathbf{A}$
$[T]_S$, $\mathrm{R}(T)$, $\ker(T)$	matrix of T relative to a basis S (its columns are $T\big(\mathbf{u}_j\big)$); range; kernel

Material marked BEYOND MA1522 works over the complex numbers or in general vector spaces. There $\mathbf{A}^*$ denotes the conjugate transpose ($\mathbf{A}^* = \mathbf{A}^T$ for real matrices), and $\langle \mathbf{u}, \mathbf{v} \rangle$ a general inner product, which for real column vectors is $\mathbf{u} \cdot \mathbf{v} = \mathbf{v}^T \mathbf{u}$.

Chapter 1. Linear Systems

1.1. Introduction to Linear Systems

Linear Systems

A system of equations of the form

$$\begin{cases} a_{11}x_1 + ... + a_{1k}x_k = b_1, \\ ... \\ a_{n1}x_1 + ... + a_{nk}x_k = b_n \end{cases} \tag{1}$$

in the variables (unknowns) $x_1, ..., x_k$ is called a *system of linear equations*, or a *linear system*. Using matrix notation (see Section 2.3), it can be written in the form

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} a_{11} & ... & a_{1k} \\ \vdots & \vdots & \vdots \\ a_{n1} & ... & a_{nk} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

We denote the set of matrices of size $n \times k$ with entries $a_{ij} \in \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers, by $\mathbb{R}^{n\times k}$. In accordance with this notation, $\mathbb{R}^{n\times 1}$ and $\mathbb{R}^{k\times 1}$ are the sets of column matrices having n and k entries, respectively. For brevity we shall write $\mathbb{R}^n = \mathbb{R}^{n\times 1}$ and $\mathbb{R}^k = \mathbb{R}^{k\times 1}$ and call column matrices *vectors*.

The matrix $\mathbf{A} \in \mathbb{R}^{n\times k}$ is called the *coefficient matrix*, the vector $\mathbf{b} \in \mathbb{R}^n$ is called the *constant vector* (the right-hand side), and a vector $\mathbf{x} \in \mathbb{R}^k$ is called a *solution* of the system (1).

1.2. Solving a Linear System and Row-Echelon Form

Row-Echelon Form

We shall say that a matrix $\mathbf{S} = \left(s_{ij}\right)_{m\times n}$ is in *row-echelon form* (REF) with k nonzero rows if there exist indices $1 \le j_1 < ... < j_k \le n$ for which:

- if $1 \le i \le k$, then $s_{ij} \ne 0$ for $j = j_i$ and $s_{ij} = 0$ for all $1 \le j \le j_i - 1$;
- if $k + 1 \le i \le m$, then $s_{ij} = 0$ for all $1 \le j \le n$.

Thus the first k rows of $\mathbf{S}$ are its nonzero rows and the zero rows are at the bottom; the entry s_{ij_i} is the *leading entry* of the i-th row, and the columns with indices $j_1, ..., j_k$ are the *pivot columns* of $\mathbf{S}$. A matrix $\mathbf{S}$ is said to be in *column-echelon form* with k nonzero columns if $\mathbf{S}^T$ is in row-echelon form with k nonzero rows.

Proposition. If a matrix $\mathbf{S}$ is in row-echelon form with k nonzero rows, or in column-echelon form with k nonzero columns, then $\operatorname{rank}(\mathbf{S}) = k$.

Proof. Recall that $\operatorname{rank}(\mathbf{S}) = \dim(\operatorname{Row}(\mathbf{S})) = \dim(\operatorname{Col}(\mathbf{S}))$ (see Section 4.2). Consider a matrix $\mathbf{S}$ in row-echelon form and let us prove that the dimension of its row space $\operatorname{Row}(\mathbf{S})$ (the span of the rows) is equal to the number k of nonzero rows. The zero rows of $\mathbf{S}$ do not contribute to $\operatorname{Row}(\mathbf{S})$, so we do not lose generality by assuming that $k = m$. Let us prove that the rows of $\mathbf{S}$ are linearly independent. Set their linear combination with coefficients $\alpha_1, ..., \alpha_k$ equal to zero:

$$(\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_k)\mathbf{S} = (0 \quad 0 \quad \cdots \quad 0).$$

From this,

$$\alpha_1 s_{1j_1} = 0 \quad \Rightarrow \quad \alpha_1 = 0.$$

Next,

$$0 \cdot s_{1j_2} + \alpha_2 s_{2j_2} = 0 \quad \Rightarrow \quad \alpha_2 = 0.$$

Continuing in a similar manner, we find $\alpha_1 = ... = \alpha_k = 0$. In the case of a matrix $\mathbf{S}$ in column-echelon form, $\dim(\operatorname{Col}(\mathbf{S}))$ obviously coincides with $\dim(\operatorname{Row}(\mathbf{S}^T))$, where $\mathbf{S}^T$ is in row-echelon form. □

1.3. Elementary Row Operations

Elementary Row Operations in Matrix Form

Each step of the elimination of unknowns (Gaussian elimination, described in Section 1.4) transforms a linear system $\mathbf{Ax} = \mathbf{b}$ into an equivalent system of the form $(\mathbf{PA})\mathbf{x} = \mathbf{Pb}$, where $\mathbf{P}$ is some invertible matrix. If k steps were required, then as a result a sequence of equivalent systems arises:

$$\mathbf{Ax} = \mathbf{b}, \quad (\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_1\mathbf{b}, \quad (\mathbf{P}_2\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_2\mathbf{P}_1\mathbf{b}, \quad ...,$$
$$..., \quad (\mathbf{P}_k...\mathbf{P}_2\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_k...\mathbf{P}_2\mathbf{P}_1\mathbf{b}.$$

The coefficient matrix of the last system has such a simple form that solving the corresponding system is carried out in an obvious way.

The goal of each step is to obtain additional zeros, or, as is often said, the *elimination of entries* in the matrix of the transformed system. To explain how this is done, consider a matrix $\mathbf{A}$ of size 4×6; entries known to be zero will be denoted by a "zero," arbitrary entries by a "cross," and entries known to be nonzero by a "boxed cross." If $a_{11} \neq 0$, then from each row with index $i > 1$ we can subtract the first row with a coefficient yielding zero in position $(i, 1)$ (the elementary row operations $R_i + c_i R_1$ with $c_i = -a_{i1}/a_{11}$, $i = 2, 3, 4$):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \end{pmatrix}.$$

Thereafter the first row and column remain unchanged. If in the second column there is a nonzero entry, we make the corresponding row the second row (a row interchange $R_2 \leftrightarrow R_i$) and use it to eliminate all the remaining entries of the second column (the operations $R_i + c_i R_2$, $i = 3, 4$):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \end{pmatrix}.$$

The first two columns and the first two rows will no longer change. It may happen that we obtain "unscheduled" zero columns. For instance, if the remaining "crosses" in the third and fourth columns turn out to be zeros, then elimination should be carried out using a nonzero entry in the fifth column (if there is one):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & 0 & 0 & \boxtimes & \times \\ 0 & 0 & 0 & 0 & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & 0 & 0 & \boxtimes & \times \\ 0 & 0 & 0 & 0 & 0 & \times \end{pmatrix}.$$

Each step obviously reduces to multiplying the matrix on the left by some matrix $\mathbf{P}_l$. Moreover, each of the matrices $\mathbf{P}_l$, $1 \le l \le k$, can be represented as the product of two matrices:

$$\mathbf{P}_l = \mathbf{Z}_l \mathbf{\Pi}_l,$$

where $\mathbf{Z}_l$ differs from $\mathbf{I}$ (the identity matrix) only in the positions below the main diagonal of some single (say, the j-th) column:

$$\mathbf{Z}_l = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & (\mathbf{Z}_l)_{j+1,j} & \ddots & \\ & & \vdots & & \\ & & (\mathbf{Z}_l)_{nj} & & 1 \end{pmatrix},$$

and $\mathbf{\Pi}_l$ is obtained from $\mathbf{I}$ by a permutation of columns (or rows). The matrix $\mathbf{\Pi}_l$ is called a *permutation matrix*, and $\mathbf{Z}_l$ a *row modification matrix*. Both are products of elementary matrices (matrices obtained from $\mathbf{I}$ by a single elementary row operation): $\mathbf{\Pi}_l$ of elementary matrices corresponding to row interchanges $R_i \leftrightarrow R_j$, and $\mathbf{Z}_l$ of those corresponding to the operations $R_i +$

$(\mathbf{Z}_l)_{ij} R_j$, $i = j+1, ..., n$. Their role in the elimination process is explained by the following facts:

- the matrix $\mathbf{\Pi}_l \mathbf{A}$ differs from $\mathbf{A}$ by a permutation of rows;
- if $\mathbf{Z}_l$ differs from the identity matrix in the j-th column, then the matrix $\mathbf{Z}_l \mathbf{A}$ has the same first j rows as the matrix $\mathbf{A}$, while the i-th row for $i > j$ is the sum of the i-th row and the j-th row of the matrix $\mathbf{A}$ taken with some coefficient (namely $(\mathbf{Z}_l)_{ij}$: the operation $R_i + (\mathbf{Z}_l)_{ij} R_j$).

Proposition 1. Any permutation matrix $\mathbf{\Pi}$ is invertible and, moreover,

$$\mathbf{\Pi}^{-1} = \mathbf{\Pi}^T.$$

Proposition 2. Any row modification matrix $\mathbf{Z} = \mathbf{Z}_l$ is invertible and, moreover, the inverse matrix is obtained from $\mathbf{Z}$ by changing the signs of the subdiagonal entries:

$$\mathbf{Z}^{-1} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -(\mathbf{Z}_l)_{j+1,j} & \ddots & \\ & & \vdots & & \\ & & -(\mathbf{Z}_l)_{nj} & & 1 \end{pmatrix}.$$

The proof reduces to a direct verification of the equalities $\mathbf{\Pi}\mathbf{\Pi}^{-1} = \mathbf{I}$, $\mathbf{Z}\mathbf{Z}^{-1} = \mathbf{I}$. (Proposition 2 expresses the fact that the operations $R_i + cR_j$ are reversed by the operations $R_i - cR_j$.)

1.4. Row Reduction, Gaussian and Gauss–Jordan Elimination

Gaussian Elimination

If a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given and one is required to find its general solution or establish its inconsistency, this is most conveniently done by *successive elimination of unknowns* (*Gaussian elimination*): if in some equation the coefficient of x_1 is nonzero, then one can eliminate x_1 from all the other equations by subtracting this equation, previously multiplied by suitably chosen coefficients; if among the equations that no longer contain x_1 there is an equation with a nonzero coefficient of x_2, then x_2 can be eliminated in a similar manner from all the other equations except the given one and the first equation containing x_1, and so on.

At each step of elimination, one obtains a new system that is obviously equivalent to the original one: in terms of the augmented matrix $(\mathbf{A} \mid \mathbf{b})$, each step consists of elementary row operations $R_i + cR_j$ (and row interchanges $R_i \leftrightarrow R_j$ if the equations are reordered), so the new augmented matrix is row

equivalent to $(\mathbf{A} \mid \mathbf{b})$. If an equation arises in which all the coefficients of the unknowns are equal to zero while the right-hand side is a nonzero number, then the system has no solutions (in a row-echelon form of $(\mathbf{A} \mid \mathbf{b})$, the last column is then a pivot column). Otherwise, the system is consistent, and the method described allows one to write out its general solution with ease (by back substitution, with the variables corresponding to the non-pivot columns as parameters).

Reduction to Row-Echelon Form

Theorem 1. For any $m \times n$ matrix $\mathbf{A}$ of rank r, there exists an invertible matrix $\mathbf{P}$ representable as a product of a finite number of permutation and row modification matrices such that the matrix $\mathbf{S} = \mathbf{PA}$ is in row-echelon form with r nonzero rows.

Proof. We follow the steps of Gaussian elimination. Denote by j_1 the index of the first column of the matrix $\mathbf{A}$ in which there is at least one nonzero entry. (If no such column exists, then $\mathbf{A} = \mathbf{0}$ and the theorem is already proved.) By multiplying on the left by some permutation matrix $\mathbf{\Pi}_1$ (a row interchange $R_1 \leftrightarrow R_i$, if necessary), a nonzero entry can be moved to position $(1, j_1)$. Next, by multiplying on the left by some row modification matrix $\mathbf{Z}_1$ (the operations $R_i + c_i R_1$, $i = 2, ..., m$), one can obtain a matrix with zeros in the subdiagonal positions of the j_1-th column while preserving the zeros in the preceding columns. Obviously, the transformed matrix has the block form (by $\mathbf{0}_{p\times q}$ we denote the zero block of size $p \times q$):

$$\mathbf{Z}_1\mathbf{\Pi}_1\mathbf{A} = \begin{pmatrix} \mathbf{0}_{1\times(j_1-1)} & \mathbf{u}^T \\ \mathbf{0}_{(m-1)\times(j_1-1)} & \mathbf{B} \end{pmatrix}, \quad \mathbf{u} \in \mathbb{R}^{(n-j_1+1)\times 1}.$$

Let us make the inductive hypothesis that there exists a matrix $\mathbf{Q}$ which is a product of permutation and row modification matrices of order $m-1$ and is such that $\mathbf{QB}$ is in row-echelon form. Consider the matrix

$$\mathbf{P} = \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix} \mathbf{Z}_1\mathbf{\Pi}_1.$$

It is easy to see that $\mathbf{P}$ is a product of permutation and row modification matrices and, moreover, $\mathbf{S} = \mathbf{PA}$ is in row-echelon form. Let the number of nonzero rows be equal to k. This means that $\dim(\mathrm{Row}(\mathbf{S})) = k \Rightarrow k = \mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{A}) = r$. □

Theorem 2. For any $m \times n$ matrix $\mathbf{A}$ of rank r, there exists an invertible matrix $\mathbf{Q}$ representable as a product of a finite number of permutation and row modification matrices such that the matrix $\mathbf{AQ}^T$ is in column-echelon form with r nonzero columns.

Proof. It suffices to apply Theorem 1 to the matrix $\mathbf{A}^T$ and note that if the matrix $\mathbf{QA}^T$ is in row-echelon form, then the matrix $\left(\mathbf{QA}^T\right)^T = \mathbf{AQ}^T$ will be in column-echelon form. □

Reduction to Diagonal Form BEYOND MA1522

Theorem. For any $m \times n$ matrix $\mathbf{A}$ of rank r, there exist invertible matrices $\mathbf{P}$ and $\mathbf{Q}$ representable as a product of a finite number of permutation and row modification matrices such that the matrix $\mathbf{B} = \mathbf{PAQ}^T$ has nonzero entries $b_{11}, ..., b_{rr}$, while all its other entries are equal to zero.

Proof. First we reduce $\mathbf{A}$ to row-echelon form $\mathbf{S} = \mathbf{PA}$, and then note that the column-echelon form $\mathbf{SQ}^T$, obtained according to the constructions of Theorem 2, will have the required diagonal form. □

Remark. Gauss–Jordan elimination uses elementary row operations only and reduces $\mathbf{A}$ to its reduced row-echelon form (RREF), which in general is not diagonal. The diagonal form $\mathbf{B}$ also requires operations on the columns of $\mathbf{A}$; these are performed by the factor $\mathbf{Q}^T$ on the right.

1.5. More on Linear Systems

Consistency of a Linear System

Theorem 1. A linear system $\mathbf{Ax} = \mathbf{b}$, $\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_k \end{pmatrix}$, is consistent if and only if

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\}.$$

Proof. In any case we have

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\}. \qquad (*)$$

If the system is consistent, then $\mathbf{b} \in \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. Consequently,

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \qquad (**)$$

The inclusions $(*)$ and $(**)$ prove the equality of the two spans. If $(**)$ holds, then it is obvious that $\mathbf{b} \in \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$, and this is precisely what the consistency of the system $\mathbf{Ax} = \mathbf{b}$ means. □

Theorem 2. If $n = k$, then when the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ are linearly independent, the linear system $\mathbf{Ax} = \mathbf{b}$ is consistent and has a unique solution.

Proof. It is obvious that

$$\mathbf{a}_1, ..., \mathbf{a}_n \in \operatorname{span}\{\mathbf{e}_1, ..., \mathbf{e}_n\},$$

where $\mathbf{e}_1, ..., \mathbf{e}_n$ are the columns of the identity matrix $\mathbf{I}_n$ (in the vector $\mathbf{e}_i$, the i-th entry is 1, and all other entries are equal to 0). By the Basis

Extension Lemma (Section 3.7), there exists a basis of $r \geq n$ vectors containing the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$. By the theorem on the dimension of a linear span (Section 3.6), $r \leq n$. For the same reason, the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ form a basis in $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\}$. Therefore $\mathbf{b} \in \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, which proves the consistency of the system. The uniqueness of the solution follows from Lemma 3 (Section 3.5). □

Problem 1. A linear system of the form

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_0 & a_1 \\ a_2 & a_1 & a_0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

has a solution with $x_1 \neq 0$. Prove that the columns of the coefficient matrix are linearly independent.

Chapter 2. Matrix Algebra

2.1. Definition and Special Types of Matrices

Linear Mappings and Matrices

In mathematics and other sciences one constantly studies the dependence of some quantities upon others. Usually such a dependence is described by various types of functions (mappings, operators). The simplest case is that of linear mappings. We shall give rigorous definitions later. For now, let us suppose that the variables $y_1, ..., y_m$ are expressed in terms of $x_1, ..., x_n$ as follows:

$$\begin{cases} y_1 = a_{11}x_1 + ... + a_{1n}x_n, \\ ... \\ y_m = a_{m1}x_1 + ... + a_{mn}x_n, \end{cases} \qquad (*)$$

where the coefficients are assumed to be given constant quantities. Let us collect all the constant coefficients into a rectangular table and denote it by the letter $\mathbf{A}$; let us also form column tables from the quantities $x_1, ..., x_n$ and $y_1, ..., y_m$:

$$\mathbf{A} = \begin{pmatrix} a_{11} & ... & a_{1n} \\ ... & ... & ... \\ a_{m1} & ... & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ ... \\ x_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ ... \\ y_m \end{pmatrix}.$$

Such tables are called *matrices*. We have three matrices in all: of sizes $m \times n$, $n \times 1$, and $m \times 1$. The relations $(*)$ describing the dependence of $\mathbf{y}$ on $\mathbf{x}$ may be written symbolically in the following way:

$$\mathbf{y} = \mathbf{A}\mathbf{x}. \qquad (**)$$

One gets the impression that the matrix $\mathbf{A}$ is *multiplied* by the column matrix $\mathbf{x}$, resulting in the column matrix $\mathbf{y}$. And so it will be if we say that the relations $(*)$ constitute the *definition* of the operation $(**)$ of multiplying $\mathbf{A}$ by $\mathbf{x}$.

If $m = n$, the matrix is called *square*. A square matrix of size $n \times n$ is also called a *matrix of order n*.

2.2. Matrix Algebra

Matrix Multiplication

Suppose that $y_1, ..., y_m$ are expressed in terms of $x_1, ..., x_n$, and at the same time $x_1, ..., x_n$ are expressed in terms of $z_1, ..., z_k$ as follows:

$$\begin{cases} y_1 = a_{11}x_1 + ... + a_{1n}x_n, \\ ... \\ y_m = a_{m1}x_1 + ... + a_{mn}x_n, \end{cases} \qquad \begin{cases} x_1 = b_{11}z_1 + ... + b_{1k}z_k, \\ ... \\ x_n = b_{n1}z_1 + ... + b_{nk}z_k. \end{cases}$$

Clearly, $y_1, ..., y_m$ are expressed in terms of $z_1, ..., z_k$ in a similar manner. Let us denote the matrix of constant coefficients of this dependence by $\mathbf{C}$. Then

$$\mathbf{y} = \mathbf{Ax}, \quad \mathbf{x} = \mathbf{Bz} \quad \text{and} \quad \mathbf{y} = \mathbf{Cz}.$$

To obtain the entries of the matrix $\mathbf{C}$, one must substitute the expressions for $x_1, ..., x_n$ in terms of $z_1, ..., z_k$ into the formulas expressing $y_1, ..., y_m$ in terms of $x_1, ..., x_n$, and collect the coefficients of the quantities $z_1, ..., z_k$. The result is:

$$\mathbf{C} = (c_{ij}), \quad \text{where} \quad c_{ij} = \sum_{l=1}^{n} a_{il}b_{lj}. \qquad (*)$$

Definition. The matrix $\mathbf{C}$ of the form $(*)$ is called the *product* of the matrices $\mathbf{A}$ and $\mathbf{B}$ and is denoted by $\mathbf{C} = \mathbf{AB}$.

Corollary. $\mathbf{y} = \mathbf{A}(\mathbf{Bz}) = (\mathbf{AB})\mathbf{z}$.

It is often said that matrices are multiplied according to the "row-by-column" rule. The number of columns in the first factor must, of course, coincide with the number of rows in the second. When we write $\mathbf{C} = \mathbf{AB}$, we automatically mean that the matrices $\mathbf{A}$ and $\mathbf{B}$ are not altogether arbitrary.

Associativity of Matrix Multiplication

Theorem. $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.

Proof. Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ have sizes $m \times n$, $n \times k$, $k \times l$. Then

$$\{(\mathbf{AB})\mathbf{C}\}_{ij} = \sum_{p=1}^{k} \{\mathbf{AB}\}_{ip}c_{pj} = \sum_{p=1}^{k} \left(\sum_{q=1}^{n} a_{iq}b_{qp} \right) c_{pj} =$$

$$= \sum_{q=1}^{n} a_{iq} \left(\sum_{p=1}^{k} b_{qp}c_{pj} \right) = \{\mathbf{A}(\mathbf{BC})\}_{ij}. \qquad \square$$

Noncommutativity of Matrix Multiplication

In the general case $\mathbf{AB} \neq \mathbf{BA}$ even for square matrices. For example,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Matrix Addition and Multiplication by a Scalar

A matrix $\mathbf{C} = \big(c_{ij}\big)$ is called the *sum* of the matrices $\mathbf{A} = \big(a_{ij}\big)$ and $\mathbf{B} = \big(b_{ij}\big)$ if

$$c_{ij} = a_{ij} + b_{ij} \quad \text{for all} \quad i, j.$$

The matrices $\mathbf{A}, \mathbf{B}$ and $\mathbf{C} = \mathbf{A} + \mathbf{B}$ have the same dimensions. The operation of matrix addition possesses two pleasant properties at once:

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C} \quad \text{(associativity)},$$
$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A} \quad \text{(commutativity)}.$$

It is also useful to introduce the operation of multiplying a matrix by a scalar (number). If α is a scalar, then the matrix $\mathbf{C} = \alpha\mathbf{A}$ is defined as the matrix of the same dimensions with entries $c_{ij} = \alpha a_{ij}$.

2.3. Linear System and Matrix Equation

Multiplication of Block Matrices

Suppose that the matrices $\mathbf{A}$ and $\mathbf{B}$ are partitioned into blocks $\mathbf{A}_{ij}$ and $\mathbf{B}_{ij}$:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \dots & \mathbf{A}_{1q} \\ \dots & \dots & \dots \\ \mathbf{A}_{p1} & \dots & \mathbf{A}_{pq} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \dots & \mathbf{B}_{1r} \\ \dots & \dots & \dots \\ \mathbf{B}_{q1} & \dots & \mathbf{B}_{qr} \end{pmatrix},$$

where $\mathbf{A}_{ij}$ is $m_i \times n_j$ and $\mathbf{B}_{ij}$ is $n_i \times k_j$. Then the product $\mathbf{C} = \mathbf{AB}$ exists and can be computed using the operations of multiplication and addition of block submatrices:

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_{11} & \dots & \mathbf{C}_{1r} \\ \dots & \dots & \dots \\ \mathbf{C}_{p1} & \dots & \mathbf{C}_{pr} \end{pmatrix}, \quad \text{where} \quad \mathbf{C}_{ij} = \sum_{l=1}^{q} \mathbf{A}_{il}\mathbf{B}_{lj} \quad \text{is} \quad m_i \times k_j.$$

Prove this!

One may say that block matrices are multiplied by the "block-row by block-column" rule. We shall see very soon how useful block multiplication can be.

Linear Combinations

For understanding the essence of the matter, the following interpretation of the linear system (1) of Section 1.1, $\mathbf{Ax} = \mathbf{b}$, is exceptionally useful. By the definition of multiplying a matrix by a scalar, if $\alpha \in \mathbb{R}$, then

$$\alpha \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \equiv \begin{pmatrix} \alpha b_1 \\ \vdots \\ \alpha b_n \end{pmatrix}.$$

Let $\mathbf{a}_1, ..., \mathbf{a}_k$ be the columns of the matrix $\mathbf{A}$:

$$\mathbf{A} = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_k), \quad \mathbf{a}_1, ..., \mathbf{a}_k \in \mathbb{R}^n.$$

Then the relations (1) are equivalent to the vector equality

$$x_1\mathbf{a}_1 + ... + x_k\mathbf{a}_k = \mathbf{b}. \tag{2}$$

The expression $x_1\mathbf{a}_1 + ... + x_k\mathbf{a}_k$ is called a *linear combination* of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$, and the numbers $x_1, ..., x_k$ are called the *coefficients of the linear combination*. The set of all possible linear combinations of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$,

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = \{\alpha_1\mathbf{a}_1 + ... + \alpha_k\mathbf{a}_k : \alpha_1, ..., \alpha_k \in \mathbb{R}\},$$

is called the *linear span* of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$.

Thus, the equality (2) means that

$$\mathbf{b} \in \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \tag{3}$$

In other words, the system (1) has a solution (is consistent) if and only if the right-hand side $\mathbf{b}$ belongs to the linear span (is a linear combination) of the columns of the coefficient matrix.

2.4. Inverse of Matrices

Inverse Matrix

A matrix $\mathbf{A}$ of order n is called *invertible* if there exists a matrix $\mathbf{X}$ of order n such that

$$\mathbf{AX} = \mathbf{XA} = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix of order n; $\mathbf{X}$ is called the *inverse matrix* for $\mathbf{A}$.

There can exist only one inverse matrix: if $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$ and $\mathbf{AY} = \mathbf{YA} = \mathbf{I}$, then $\mathbf{X} = \mathbf{X}(\mathbf{AY}) = (\mathbf{XA})\mathbf{Y} = \mathbf{Y}$.

Notation for the inverse matrix: $\mathbf{X} = \mathbf{A}^{-1}$.

Problem 1. Let $\mathbf{A}, \mathbf{B}$ be arbitrary matrices of order n; $\mathbf{I}$ and $\mathbf{0}$ be the identity and zero matrices of order n. Prove that

$$\begin{pmatrix} \mathbf{I} & \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{B} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A} & \mathbf{AB} \\ \mathbf{0} & \mathbf{I} & -\mathbf{B} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix}.$$

(It follows from this that any algorithm for computing the inverse of a matrix of order n with operation count $s(n)$ generates an algorithm for multiplying matrices of order n with operation count $s(3n)$.)

Problem 2. Given a square matrix $\mathbf{A}$ such that $\mathbf{A}^3 = \mathbf{0}$. Prove that the matrix $\mathbf{I} - \mathbf{A}$ is invertible.

Problem 3. Find all invertible matrices $\mathbf{A}$ of order n for which all entries of $\mathbf{A}$ and $\mathbf{A}^{-1}$ are nonnegative. Prove that the set of all such matrices forms a group with respect to the operation of matrix multiplication.

Problem 4. Given matrices $\mathbf{P}_1, ..., \mathbf{P}_n$ of order n, each differing from the identity matrix by a permutation of columns (such matrices are called *permutation matrices*). Let $\mathbf{P}_1 + ... + \mathbf{P}_n = \mathbf{E}$, where $\mathbf{E}$ is the matrix all of whose entries are equal to 1. Furthermore, let $\mathbf{P}_i\mathbf{P}_j = \mathbf{P}_j\mathbf{P}_i$ for all i, j. Prove that the set of matrices $\mathbf{P}_1, ..., \mathbf{P}_n$ forms a group with respect to the operation of matrix multiplication.

Inversion and Transposition

Proposition. $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Proof. $(\mathbf{AB})_{ij} = \sum_k (\mathbf{A})_{ik}(\mathbf{B})_{kj} = \sum_k (\mathbf{B})_{kj}(\mathbf{A})_{ik} = \sum_k (\mathbf{B}^T)_{jk}(\mathbf{A}^T)_{ki} = (\mathbf{B}^T\mathbf{A}^T)_{ji}$. □

From the equality $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$ we obtain: $\mathbf{X}^T\mathbf{A}^T = \mathbf{A}^T\mathbf{X}^T = \mathbf{I}$. Thus, *a matrix is invertible if and only if its transpose is invertible*. Moreover,

$$\mathbf{X}^T = (\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T.$$

Notation: $\mathbf{A}^{-T} \equiv (\mathbf{A}^{-1})^T$.

As a corollary, we obtain the "row" analogue of the matrix invertibility criterion (Section 2.6): *the invertibility of a square matrix is equivalent to the linear independence of its rows.*

Group of Invertible Matrices

The set of invertible $(n \times n)$-matrices forms a group with respect to the operation of multiplication. For the proof we have everything except the fact that the product of invertible matrices is invertible. But this is verified directly: if $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{IB} = \mathbf{I},$$
$$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = \mathbf{AIA}^{-1} = \mathbf{I}.$$

Hence

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

Problem 5. Let H be the group of invertible upper triangular matrices of order n, and G the group of all invertible matrices of order n. Is it true that $\mathbf{ABA}^{-1} \in H$ for any matrices $\mathbf{B} \in H$ and $\mathbf{A} \in G$ (in which case the subgroup H is called a *normal subgroup* of the group G)?

Computing the Inverse Matrix BEYOND MA1522

The inverse matrix can be computed using the constructions of the same Gaussian elimination method (one way to organize this is the course's row reduction of $(\mathbf{A} \mid \mathbf{I})$ to $(\mathbf{I} \mid \mathbf{A}^{-1})$). Once the LU factorization $\mathbf{A} = \mathbf{LU}$ (Section 2.7) has been obtained, since $\mathbf{A}^{-1} = \mathbf{U}^{-1}\mathbf{L}^{-1}$, it suffices to learn how to compute the inverses of upper and lower triangular matrices. The total number of arithmetic operations will be $\mathcal{O}(n^3)$.

However, in 1965 Strassen published a paper entitled "Gaussian Elimination Is Not Optimal," in which it was shown for the first time that faster algorithms also exist. Suppose we have an algorithm for multiplying two $(n \times n)$-matrices with a number of operations $\leq cn^\gamma$ (for example, Strassen's algorithm, for which $\gamma = \log_2 7 < 3$; see Tyrtyshnikov, §§1.10–1.11). Then in

the case of a strictly regular matrix $\mathbf{A}$, one can construct an algorithm for computing $\mathbf{A}^{-1}$ with $\mathcal{O}(n^\gamma)$ operations.

For simplicity, suppose that $n = 2^p$. Partition $\mathbf{A}$ into blocks of order $n/2$ and consider the following equality:

$$\begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{W} \end{pmatrix},$$

$$\mathbf{W} = \mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}.$$

From the invertibility of $\mathbf{A}$ and $\mathbf{A}_{11}$ follows the invertibility of the block $\mathbf{W}$. Moreover, the blocks $\mathbf{A}_{11}$ (which is obvious) and $\mathbf{W}$ (prove this!) inherit the strict regularity of the matrix $\mathbf{A}$. It is not hard to verify that

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{W} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{W}^{-1} \\ \mathbf{0} & \mathbf{W}^{-1} \end{pmatrix}.$$

Thus,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{W}^{-1} \\ \mathbf{0} & \mathbf{W}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix}. \qquad (*)$$

The expressions for the blocks of the matrix $\mathbf{A}^{-1}$ obtained from $(*)$ are sometimes called the *Frobenius formulas*.

For our purpose, formula $(*)$ is interesting in that it shows how inverting a matrix of order n reduces to two analogous problems for the matrices $\mathbf{A}_{11}$ and $\mathbf{W}$ of order $n/2$. To carry out this reduction, one must perform several multiplications of matrices of order $n/2$. The total cost across all steps of the reduction is proportional to

$$\Big(\frac{n}{2}\Big)^\gamma + 2\Big(\frac{n}{2^2}\Big)^\gamma + 2^2\Big(\frac{n}{2^3}\Big)^\gamma + \ldots \le \frac{n^\gamma}{2^\gamma} \cdot \frac{1}{1 - 1/2^{\gamma-1}} = \frac{n^\gamma}{2^\gamma - 2}.$$

Following Strassen's discovery, a paper by other authors appeared under the title "Strassen's Method Is Not Optimal." Yet the leadership of the new algorithm did not last very long. The competition to construct ever faster algorithms for inverting matrices and solving systems continues to this day, and the question of an optimal algorithm from the standpoint of the number of operations remains open.

Even less clear is the question of an algorithm with the minimum number of parallel steps (even in the model of infinite parallelism). Quite a long time ago, an algorithm was devised in which the number of parallel steps in the case of a general matrix is $\mathcal{O}(\log_2^2 n)$. Nobody knows whether a faster parallel algorithm can be constructed. Curiously, the algorithm presented has nothing in common with either Gaussian elimination or Strassen's method. Furthermore, even for a triangular matrix, no algorithm with a smaller number of parallel steps (in order of dependence on n) is known.

Problem 6. Prove that the determinant of a strictly regular matrix of order n can be computed in $\mathcal{O}\left(n^{\log_2 7}\right)$ arithmetic operations.

2.5. Elementary Matrices

Elementary row operations in matrix form (permutation and row modification matrices) is treated in Section 1.3.

Equivalent Matrices BEYOND MA1522

Matrices $\mathbf{A}$ and $\mathbf{B}$ are called *equivalent* (not to be confused with *row equivalent*) if there exist invertible matrices $\mathbf{P}$ and $\mathbf{Q}$ such that $\mathbf{B} = \mathbf{PAQ}$.

Theorem. Matrices $\mathbf{A}$ and $\mathbf{B}$ are equivalent if and only if they have the same dimensions and the same ranks.

Proof. By virtue of the theorem on reduction to diagonal form (Section 1.4), each of the matrices $\mathbf{A}$ and $\mathbf{B}$ is equivalent to a rectangular diagonal matrix—let us denote them by $\mathbf{D}_A$ and $\mathbf{D}_B$. Moreover, it is obvious that $\mathbf{D}_A$ and $\mathbf{D}_B$ are equivalent if and only if they have the same number of nonzero diagonal entries. The latter means that $\operatorname{rank}(\mathbf{D}_A) = \operatorname{rank}(\mathbf{D}_B)$. It remains to take into account that $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{D}_A)$ and $\operatorname{rank}(\mathbf{B}) = \operatorname{rank}(\mathbf{D}_B)$. □

2.6. Equivalent Statements for Invertibility

Invertibility Criterion for a Matrix

Theorem. A square matrix is invertible if and only if its columns form a linearly independent system.

Proof. Let a matrix $\mathbf{A}$ of order n have linearly independent columns. According to the results of Section 1.5 on the consistency of linear systems, each of the systems

$$\mathbf{Ax}_1 = \mathbf{e}_1, \quad \mathbf{Ax}_2 = \mathbf{e}_2, \quad ..., \quad \mathbf{Ax}_n = \mathbf{e}_n,$$

where $\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n$ are the columns of the identity matrix, has a unique solution. Let $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \end{pmatrix}$. Then $\mathbf{AX} = \mathbf{I}$.

The columns of the matrix $\mathbf{X}$ are linearly independent. Indeed, suppose some linear combination of these columns is equal to zero:

$$\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + ... + \alpha_n\mathbf{x}_n = \mathbf{0}.$$

This means that

$$\mathbf{X}\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathbf{0}.$$

Consequently,

$$\mathbf{A}\mathbf{X}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\\ \alpha_n\end{pmatrix} = \begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\\ \alpha_n\end{pmatrix} = \mathbf{0}.$$

Hence $\alpha_1 = \alpha_2 = ... = \alpha_n = 0$. Thus, for $\mathbf{X}$ there exists a matrix $\mathbf{Y}$ such that $\mathbf{XY} = \mathbf{I}$. Let us prove that $\mathbf{Y} = \mathbf{A}$. Indeed, $\mathbf{A} = \mathbf{A}(\mathbf{XY}) = (\mathbf{AX})\mathbf{Y} = \mathbf{Y}$.[1]

Now suppose that $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$ is an invertible matrix, and consider a linear combination of its columns equal to zero:

$$\alpha_1\mathbf{a}_1 + \alpha_2\mathbf{a}_2 + ... + \alpha_n\mathbf{a}_n = \mathbf{0}.$$

Let us write this equality in the following form:

$$\mathbf{A}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\\ \alpha_n\end{pmatrix} = \mathbf{0}.$$

Multiplying both sides on the left by $\mathbf{A}^{-1}$, we find: $\alpha_1 = ... = \alpha_n = 0$. Consequently, the columns $\mathbf{a}_1, ..., \mathbf{a}_n$ are linearly independent. □

Invertibility and the Determinant

Theorem. A square matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Proof. Suppose that for the matrix $\mathbf{A}$ there exists an inverse matrix $\mathbf{A}^{-1}$. Then $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and, by virtue of the theorem on the determinant of a product of matrices (Section 2.10),

$$\det(\mathbf{A}) \cdot \det(\mathbf{A}^{-1}) = \det(\mathbf{I}) = 1 \quad \Rightarrow \quad \det(\mathbf{A}) \neq 0.$$

If $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible by the theorem on the inversion of a matrix with nonzero determinant (the adjoint formula, Section 2.10). □

Corollary. The columns of a matrix $\mathbf{A}$ are linearly independent if and only if $\det(\mathbf{A}) \neq 0$.

Problem 1. Let $\mathbf{I}_n$ and $\mathbf{I}_m$ be the identity matrices of orders n and m, respectively. Prove that for any matrices $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $n \times m$, the invertibility of $\mathbf{I}_m - \mathbf{AB}$ implies the invertibility of $\mathbf{I}_n - \mathbf{BA}$. Prove also that the invertibility of each of these matrices is equivalent to the invertibility of the matrix of order $m + n$ with the block partitioning of the form

[1] In essence, this reproduces part of the proof of a more general proposition connected with the redundancy of the definition of a group (see Tyrtyshnikov, Supplement to Lecture 2, §$\mathcal{D}$2.5).

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix}.$$

Problem 2. Given numbers a and b such that $1 - ab \neq 0$. Prove that the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & a & a^2 & \dots & a^n \\ b & 1 & a & \dots & a^{n-1} \\ b^2 & b & 1 & \dots & a^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b^n & b^{n-1} & b^{n-2} & \dots & 1 \end{pmatrix}$$

is invertible and its inverse matrix is tridiagonal.

Problem 3. Suppose that in the statement of the preceding problem a and b are square blocks $\mathbf{a}$ and $\mathbf{b}$ of order n, and 1 is replaced by the identity matrix $\mathbf{I}$ of the same order. Prove that if the block $\mathbf{I} - \mathbf{ab}$ is an invertible matrix, then the block matrix $\mathbf{A}$ with blocks of order n is invertible and, moreover, its inverse matrix is block-tridiagonal.

2.7. LU Factorization

Gaussian Elimination and LU Factorization

The method of successive elimination of unknowns described in Section 1.4 is usually called *Gaussian elimination* (or *Gauss's method*). Suppose it is applied to a linear system $\mathbf{Ax} = \mathbf{b}$ with an invertible matrix $\mathbf{A}$. In this case, the row-echelon form to which the matrix $\mathbf{A}$ is reduced turns out to be an upper triangular matrix.

Gaussian elimination can be interpreted as a method of eliminating entries of a matrix with the aim of reducing it to a simpler form. If one can do without permuting equations (row interchanges $R_i \leftrightarrow R_j$), then Gaussian elimination for a matrix of order n consists in successively eliminating the entries in columns from the 1-st to the $(n-1)$-st and leads to an equivalent system

$$(\mathbf{Z}_{n-1}...\mathbf{Z}_2\mathbf{Z}_1\mathbf{A})\mathbf{x} = \mathbf{Z}_{n-1}...\mathbf{Z}_2\mathbf{Z}_1\mathbf{b}, \qquad (*)$$

where $\mathbf{Z}_1, ..., \mathbf{Z}_{n-1}$ are row modification matrices (Section 1.3), with $\mathbf{Z}_i$ differing from $\mathbf{I}$ precisely in the i-th column. Each of the matrices $\mathbf{Z}_1, ..., \mathbf{Z}_{n-1}$ is lower triangular, and therefore their product

$$\hat{\mathbf{L}} = \mathbf{Z}_{n-1}...\mathbf{Z}_1$$

is also a lower triangular matrix. The coefficient matrix of the system $(*)$

$$\mathbf{U} = \mathbf{Z}_{n-1}...\mathbf{Z}_1\mathbf{A} = \hat{\mathbf{L}}\mathbf{A}$$

is upper triangular. The matrix $\mathbf{L} = \hat{\mathbf{L}}^{-1}$ is lower triangular. Consequently, Gaussian elimination generates a decomposition of the matrix $\mathbf{A}$ into a product of a lower and an upper triangular matrix:

$$\mathbf{A} = \mathbf{LU}.$$

Moreover, $\mathbf{L}$ has ones on the main diagonal (it is a *unit lower triangular* matrix), and $\mathbf{U}$ is an invertible matrix (by virtue of the invertibility of $\mathbf{A}$). Such a decomposition, with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ a row-echelon form of $\mathbf{A}$, is called an *LU factorization* of $\mathbf{A}$.

Let us count the number of arithmetic operations in reducing $\mathbf{A}$ to the upper triangular matrix $\mathbf{U}$. At the i-th step, it is required to obtain $n-i$ zeros below the diagonal in the i-th column. When obtaining a zero at the intersection of the i-th column and the l-th row for $l > i$, from the l-th row one subtracts the i-th row, previously multiplied by a coefficient whose choice ensures obtaining this zero. Since in the rows under consideration there can be only $n-i$ nonzero entries (beyond the i-th position), the number of multiplications (and subtractions) in obtaining zeros in the i-th column is equal to $(n-i)^2$. In total, this requires

$$(n-1)^2 + (n-2)^2 + ... + 1^2 = \frac{1}{3}n^3 + \mathcal{O}(n^2)$$

multiplications and the same number of subtractions; by $\mathcal{O}(n^2)$ we denote a polynomial in n of degree 2.

To find the solution of the system $\mathbf{Ax} = \mathbf{b}$, one must perform two more steps:

- compute the vector $\mathbf{Z}_{n-1}...\mathbf{Z}_1\mathbf{b}$;
- find the solution of the system with the upper triangular matrix $\mathbf{U}$.

Each of these steps requires only $\mathcal{O}(n^2)$ arithmetic operations—an order of magnitude less than the reduction to upper triangular form.

> **Problem 1.** An invertible matrix and its inverse are partitioned into blocks of identical dimensions:
>
> $$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}.$$
>
> Prove that the block $\mathbf{A}_{11}$ is invertible if and only if the block $\mathbf{B}_{22}$ is invertible.

LU Factorization and Strictly Regular Matrices BEYOND MA1522

Suppose that an invertible matrix $\mathbf{A}$ has an LU factorization: $\mathbf{A} = \mathbf{LU}$. Denote by $\mathbf{A}_k, \mathbf{L}_k, \mathbf{U}_k$ the submatrices of order k located in the upper left corner of the matrices $\mathbf{A}, \mathbf{L}, \mathbf{U}$, respectively, and consider the equality of block matrices

$$\mathbf{A} \equiv \begin{pmatrix} \mathbf{A}_k & \mathbf{P} \\ \mathbf{Q} & \tilde{\mathbf{A}}_k \end{pmatrix} = \begin{pmatrix} \mathbf{L}_k & \mathbf{0} \\ \mathbf{V} & \tilde{\mathbf{L}}_k \end{pmatrix} \begin{pmatrix} \mathbf{U}_k & \mathbf{W} \\ \mathbf{0} & \tilde{\mathbf{U}}_k \end{pmatrix}.$$

From this it follows that

$$\mathbf{A}_k = \mathbf{L}_k\mathbf{U}_k, \quad k = 1, ..., n.$$

Obviously, the matrices $\mathbf{L}_k$ and $\mathbf{U}_k$ are invertible (as triangular matrices with a nonzero diagonal). Therefore, the submatrix $\mathbf{A}_k$ must be invertible. A matrix $\mathbf{A}$ in which all the leading principal submatrices $\mathbf{A}_k$ are invertible is called

strictly regular. Thus, for the existence of an LU factorization of an invertible matrix $\mathbf{A}$, it is necessary that it be strictly regular.

It can be proved that this condition is also sufficient. Indeed, suppose that an LU factorization has already been constructed for the submatrix $\mathbf{A}_k = \mathbf{L}_k\mathbf{U}_k$. Then

$$\begin{pmatrix} \mathbf{L}_k^{-1} & \mathbf{0} \\ -\mathbf{Q}\mathbf{A}_k^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_k & \mathbf{P} \\ \mathbf{Q} & \tilde{\mathbf{A}}_k \end{pmatrix} = \begin{pmatrix} \mathbf{U}_k & \mathbf{L}_k^{-1}\mathbf{P} \\ \mathbf{0} & \mathbf{W} \end{pmatrix}, \quad \mathbf{W} = \tilde{\mathbf{A}}_k - \mathbf{Q}\mathbf{A}_k^{-1}\mathbf{P}. \ (\#)$$

The block $\mathbf{W}$ is called the *Schur complement* of the block $\mathbf{A}_k$ in the matrix $\mathbf{A}$. From equality (#) and the strict regularity of $\mathbf{A}$, one can deduce that $\mathbf{W}$ is also a strictly regular matrix. Suppose that for $\mathbf{W}$ an LU factorization $\mathbf{W} = \tilde{\mathbf{L}}_k\tilde{\mathbf{U}}_k$ has already been constructed. Then we set

$$\mathbf{L} = \begin{pmatrix} \mathbf{L}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{Q}\mathbf{A}_k^{-1}\mathbf{L}_k & \tilde{\mathbf{L}}_k \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{U}_k & \mathbf{L}_k^{-1}\mathbf{P} \\ \mathbf{0} & \tilde{\mathbf{U}}_k \end{pmatrix}.$$

The matrix $\mathbf{L}$ obtained in this way is lower triangular. The equality $\mathbf{LU} = \mathbf{A}$ is verified by direct computation.

Problem 2. Let $\mathbf{A}$ be an invertible matrix of order n and $\mathbf{A}(I, J)$ be its invertible submatrix on the rows and columns determined by the index sets $I = (i_1, ..., i_k)$ and $J = (j_1, ..., j_k)$, respectively (the notation of Section 2.8). Let $k < n$, and let I' and J' be the complementary index sets. Prove that

$$\det(\mathbf{A}^{-1}(J', I')) = (-1)^{i_1+...+i_k+j_1+...+j_k} \frac{\det(\mathbf{A}(I, J))}{\det(\mathbf{A})}.$$

2.8. Determinant by Cofactor Expansion

Linear Dependence Indicator

Consider a system of vectors $\mathbf{a}_1, ..., \mathbf{a}_n \in \mathbb{R}^n$ and let us try to construct a linear dependence indicator—a function $f(\mathbf{a}_1, ..., \mathbf{a}_n)$ that equals zero when the given system is linearly dependent. At the same time, the function f should have as simple a form as possible: let f be *linear* in each argument when the values of the remaining arguments are fixed.

Let us give a precise formulation of the requirements on the function f:

- A) for every $1 \le i \le n$, the function is linear in the i-th argument (the function must have a "simple form"):

$$f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \alpha\mathbf{a} + \beta\mathbf{b}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n) =$$
$$= \alpha f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \mathbf{a}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n) +$$
$$+\beta f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \mathbf{b}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n)$$

  for any vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ and scalars $\alpha, \beta \in \mathbb{R}$;
- B) if the system of vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ is linearly dependent, then $f(\mathbf{a}_1, ..., \mathbf{a}_n) = 0$;

- C) the function takes a prescribed nonzero value on a given linearly independent system (normalization condition):

$$f(\mathbf{e}_1, ..., \mathbf{e}_n) = 1,$$

where $\mathbf{e}_1, ..., \mathbf{e}_n$ are the columns of the identity matrix of order n.

A function f with the stated properties will be called a linear dependence indicator. To construct it, we shall need the concept of a permutation.

Permutations

An invertible mapping $\sigma : \boldsymbol{N} \to \boldsymbol{N}$, where $\boldsymbol{N} = \{1, 2, ..., n\}$, is called a *permutation* (also a *substitution*) of degree n. To denote a permutation σ, one often uses the table

$$\sigma = \begin{pmatrix} 1 & 2 & ... & n \\ \sigma(1) & \sigma(2) & ... & \sigma(n) \end{pmatrix},$$

in which the numbers $\sigma(1), \sigma(2), ..., \sigma(n)$ are the numbers $1, 2, ..., n$ written in some order (this is equivalent to the invertibility of the mapping σ).

Let us define the product of permutations a and b as the mapping obtained by the sequential execution (composition) of the mappings b and a:

$$(ab)(i) = a(b(i)), \quad i \in \boldsymbol{N}.$$

This is an algebraic operation on the set of all permutations of degree n, with respect to which it is a group. Indeed, associativity is obvious (the composition of mappings always possesses this property). The role of the identity element is played by the identity mapping

$$e = \begin{pmatrix} 1 & 2 & ... & n \\ 1 & 2 & ... & n \end{pmatrix},$$

and the inverse element for σ is the inverse mapping σ^{-1}.

The group of permutations of degree n is called the *symmetric group* of degree n and is denoted by S_n. This is one of the most important examples of finite groups (groups with a finite number of elements; the number of elements is called the *order* of the group). It is not hard to verify that the order of the group S_n is equal to $n! = 1 \cdot 2 \cdot ... \cdot n$.

The name of the symmetric group is inspired by the definition of *symmetric functions*: a function $F(x_1, ..., x_n)$ is so called if it is invariant under all permutations of its arguments:

$$F(x_1, ..., x_n) = F\left(x_{\sigma(1)}, ..., x_{\sigma(n)}\right) \quad \forall \sigma \in S_n.$$

An example of a symmetric function (determined by a numerical parameter k):

$$F_k(x_1, ..., x_n) = \sum_{i=1}^{n} x_i^k.$$

Problem 1. Given two systems of numbers $x_1 \le x_2 \le ... \le x_n$ and $y_1 \le y_2 \le ... \le y_n$, prove that for any permutation $\sigma \in S_n$ the inequality

$$\sum_{i=1}^{n}|x_i - y_i| \le \sum_{i=1}^{n}\left|x_i - y_{\sigma(i)}\right|$$

holds.

Cycles and Transpositions BEYOND MA1522

A permutation $a \in S_n$ is called a *cycle* of length k if there exist k pairwise distinct indices $i_1, ..., i_k \in \boldsymbol{N}$ such that:

1) $a(i_1) = i_2, a(i_2) = i_3, ..., a(i_{k-1}) = i_k, a(i_k) = i_1$,

2) $a(i) = i \ \forall i \in \boldsymbol{N} \setminus \{i_1, i_2, ..., i_k\}$.

To denote the cycle a, it is convenient to use the notation

$$a = (i_1, ..., i_k).$$

A cycle of length 2 is also called a *transposition*.

Cycles $a = (i_1, ..., i_k)$ and $b = (j_1, ..., j_m)$ are called *independent* (or *disjoint*) if

$$\{i_1, ..., i_k\} \cap \{j_1, ..., j_m\} = \varnothing.$$

Properties. 1) *Any independent cycles a and b commute: $ab = ba$.*

2) *Every permutation $\sigma \in S_n$ can be represented as a product of independent cycles uniquely up to the order of the factors.*

3) *Every cycle of length k can be represented as a product of $k-1$ transpositions.*

4) *Every permutation can be represented as a product of transpositions.*

Statement (1) is verified directly: in the case of independent cycles $a = (i_1, ..., i_k)$ and $b = (j_1, ..., j_m)$, we find

$$(ab)(i) = (ba)(i) = a(i) \text{ for } i \in \{i_1, ..., i_k\},$$
$$(ab)(i) = (ba)(i) = b(i) \text{ for } i \in \{j_1, ..., j_m\},$$
$$(ab)(i) = (ba)(i) = i \quad \text{for } i \notin \{i_1, ..., i_k\} \cup \{j_1, ..., j_m\}.$$

To prove (2), let us take an arbitrary index j and consider the sequence of indices $j, \sigma(j), \sigma^2(j), ...$ There are only n distinct values—therefore, for some $k < l$ we must have $\sigma^k(j) = \sigma^l(j)$, whence we obtain $\sigma^{l-k}(j) = j$. Let k be the smallest positive index such that $\sigma^k(j) = j$. Then we obtain the cycle

$$a = \big(j, \sigma(j), \sigma^2(j), ..., \sigma^{k-1}(j)\big),$$

for which

$$\sigma(i) = a(i) \quad \text{for } i \in \big\{j, \sigma(j), \sigma^2(j), ..., \sigma^{k-1}(j)\big\}.$$

Clearly, the permutation $\sigma_1 = \sigma a^{-1}$ leaves fixed the indices

$$i \in \big\{j, \sigma(j), \sigma^2(j), ..., \sigma^{k-1}(j)\big\}.$$

Next, we take $j_1 \notin \big\{j, \sigma(j), \sigma^2(j), ..., \sigma^{k-1}(j)\big\}$ and in a similar manner construct a cycle b performing transformations of the form

$$j_1 \to \sigma_1(j_1) \to \sigma_1^2(j_1) \to ...$$

(Note that $\sigma_1^l(j_1) = \sigma^l(j_1)$ for all l.) Continuing such constructions, we inevitably arrive at the identity permutation

$$\sigma a^{-1} b^{-1} ... c^{-1} = e,$$

whence

$$\sigma = c...ba.$$

By construction, the cycles $a, b, ..., c$ are independent.

Statement (3) is proved by verifying, for example, the following equality:

$$(i_1, ..., i_k) = (i_1, i_2)(i_2, i_3)...(i_{k-1}, i_k).$$

Statement (4) obviously follows from (2) and (3).

Problem 2. Prove that the entire set of permutations of degree n can be ordered in such a way that each subsequent permutation is obtained from the preceding one by multiplying on the right by some transposition.

Parity of a Permutation BEYOND MA1522

A permutation can be decomposed into a product of transpositions in many ways. For example,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 5 & 3 & 1 & 2 & 4 & 6 \end{pmatrix} =$$

$$= (1,7)(7,6)(6,4)(2,5) = (1,7)(7,6)(6,4)(7,2)(7,5)(7,2).$$

However, the number of transpositions in any decomposition of one and the same permutation possesses the following important property.

Lemma (Number of Transpositions)**.** The parity of the number of transpositions does not depend on the way a permutation is represented as a product of transpositions.

Proof. For a given permutation $\sigma \in S_n$,

$$\sigma = \begin{pmatrix} 1 & 2 & ... & n \\ \sigma(1) & \sigma(2) & ... & \sigma(n) \end{pmatrix},$$

let us call a pair (i, j) an *inversion* if $i < j$ but $\sigma(i) > \sigma(j)$. Let $\delta(\sigma)$ be the total number of inversions for σ. Let us prove that for any transposition τ, the difference $\delta(\sigma\tau) - \delta(\sigma)$ is an odd number.

Let $\tau = (i, j)$, $i < j$. Then

$$\sigma\tau = \begin{pmatrix} 1 & ... & i-1 & i & i+1 & ... & j-1 & j & j+1 & ... & n \\ \sigma(1) & ... & \sigma(i-1) & \sigma(j) & \sigma(i+1) & ... & \sigma(j-1) & \sigma(i) & \sigma(j+1) & ... & \sigma(n) \end{pmatrix}.$$

Suppose that the permutation σ has k inversions among pairs of the form

$$(i, l), \quad \text{where} \quad l \in \{i+1, i+2, ..., j-1\}, \qquad (\dagger)$$

m inversions among pairs of the form

$$(l, j), \quad \text{where} \quad l \in \{i+1, i+2, ..., j-1\}, \tag{‡}$$

and s more inversions among all other pairs. Then $\sigma\tau$ will have $j-i-1-m$ inversions among pairs of the form (†) and $j-i-1-k$ inversions among pairs of the form (‡). Furthermore, among all other pairs the permutation $\sigma\tau$ will have $s+1$ inversions if the pair (i, j) was not an inversion, and $s-1$ inversions otherwise. Thus,

$$\delta(\sigma) = k + m + s,$$

$$\delta(\sigma\tau) = (j-i-1-k) + (j-i-1-m) + s \pm 1.$$

Hence

$$\delta(\sigma\tau) - \delta(\sigma) = 2(j-i-1-k-m) \pm 1. \qquad \square$$

Corollary. The parity of the number of transpositions in a decomposition of a permutation coincides with the parity of its number of inversions.

Definition. A permutation is called *even* if it is a product of an even number of transpositions, and *odd* otherwise.

Remark. Consider the function

$$\Delta(x_1, ..., x_n) = \prod_{i<j} (x_j - x_i).$$

Then for any permutation $\sigma \in S_n$ one of the two holds: either

$$\Delta(x_1, ..., x_n) = \Delta\big(x_{\sigma(1)}, ..., x_{\sigma(n)}\big)$$

or

$$\Delta(x_1, ..., x_n) = -\Delta\big(x_{\sigma(1)}, ..., x_{\sigma(n)}\big).$$

Even permutations, and only they, preserve the sign (the first case); odd permutations, and only they, reverse the sign (the second case).

The set of all even permutations in S_n forms a subgroup (prove this!), which is called the *alternating group* of degree n and is denoted by A_n.

Problem 3. Prove that every even permutation of degree $n \geq 3$ can be represented as a product of cycles of length 3.

Uniqueness of the Linear Dependence Indicator

Let us return to the construction of the linear dependence indicator—a function $f(\mathbf{a}_1, ..., \mathbf{a}_n)$ of the vectors

$$\mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{pmatrix}, \quad ..., \quad \mathbf{a}_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{pmatrix},$$

satisfying requirements (A), (B), (C). It is easy to see that

$$\mathbf{a}_1 = \sum_{i_1=1}^{n} a_{i_1 1}\mathbf{e}_{i_1}, \quad \mathbf{a}_2 = \sum_{i_2=1}^{n} a_{i_2 2}\mathbf{e}_{i_2}, \quad ..., \quad \mathbf{a}_n = \sum_{i_n=1}^{n} a_{i_n n}\mathbf{e}_{i_n},$$

where $\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n$ are the columns of the identity matrix of size $n \times n$.

If the sought function f exists, then property (A) of linearity in each argument leads to the expression

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{i_1=1}^{n} ... \sum_{i_n=1}^{n} a_{i_1 1}a_{i_2 2}...a_{i_n n} f\big(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, ..., \mathbf{e}_{i_n}\big).$$

According to requirement (B), $f = 0$ on any linearly dependent system of vectors. Obviously, the system of vectors $\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, ..., \mathbf{e}_{i_n}$ is linearly dependent if and only if there are equal vectors among them (if all these vectors are pairwise distinct, then they form a permutation of the columns of the identity matrix). Consequently, excluding the terms that are certainly zero from the summation, we find

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{\sigma \in S_n} a_{\sigma(1)1}a_{\sigma(2)2}...a_{\sigma(n)n} \times$$
$$\times f\big(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\big).$$

Furthermore, from requirements (A) and (B) it follows that f must change sign when any two arguments are interchanged. Let us prove this, for example, for the first and second arguments. Taking into account that $f = 0$ in the case of equal arguments, and using linearity in each argument, we have

$$\begin{aligned} 0 &= f(\mathbf{a}_1 + \mathbf{a}_2, \mathbf{a}_1 + \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) = \\ &= f(\mathbf{a}_1, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n) + f(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) + \\ &+ f(\mathbf{a}_2, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n) + f(\mathbf{a}_2, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n). \end{aligned}$$

The first and fourth terms have coinciding vectors and are therefore equal to zero. Hence

$$f(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) = -f(\mathbf{a}_2, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n).$$

Consequently, if the permutation σ is a transposition, then

$$f\big(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\big) = -f(\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n) = -1.$$

In the general case, a permutation σ can be decomposed into a product of transpositions. Let $\delta(\sigma)$ be the number of transpositions in any one of the decompositions. Then

$$f\left(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\right) = (-1)^{\delta(\sigma)}.$$

By the lemma on the number of transpositions, the parity of the number $\delta(\sigma)$ does not depend on the specific decomposition into a product of transpositions; therefore the quantity $(-1)^{\delta(\sigma)}$ depends only on σ. We call it the *sign of the permutation* and denote it by $\operatorname{sgn}(\sigma)$. Finally:

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} ... a_{\sigma(n)n}. \qquad (*)$$

We have proved an important result:

Proposition. If a linear dependence indicator function exists, then it is given by formula $(*)$.

The Determinant

Definition. A function of the form $(*)$ is called the *determinant* of the matrix $\mathbf{A}$ with columns $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n$ and is denoted by $\det(\mathbf{A})$ or $|\mathbf{A}|$.

Thus, if $\mathbf{A} = \left(a_{ij}\right)$ is a matrix of size $n \times n$, then

$$\det(\mathbf{A}) = |\mathbf{A}| = \begin{vmatrix} a_{11} & ... & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{n1} & ... & a_{nn} \end{vmatrix} = \qquad (\triangle)$$

$$= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} ... a_{\sigma(n)n}.$$

Special cases:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12},$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} -$$

$$-a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

In the general case, the sum $(\triangle)$ (the *Leibniz formula* for the determinant) contains $n!$ terms; in each of them n entries of the matrix are multiplied together, and no two entries in a single product belong to the same row or the same column.

Despite the fact that the determinant is introduced as a function of a matrix, historically the concept of a determinant took shape in the 18th century (first in the works of Leibniz and Cramer, and then the theory of determinants was developed in the works of Vandermonde, Laplace, Cauchy, and C. Jacobi)—much earlier than the concept of a matrix, which was introduced into algebra by Hamilton and Cayley in the middle of the 19th century.

Of course, from the very beginning the determinant was associated with a square $n \times n$ table of numbers (which is why one spoke of a determinant of *order* n). These were, in particular, the coefficient tables of a "square" linear system. Yet such tables came to be called matrices only later—when the operation of multiplication was introduced for them.

Submatrices and $k \times k$ Minors

For a given matrix $\mathbf{A} = \big(a_{ij}\big)$, one can select some of its rows and columns and form the table of entries located at the intersections of the selected rows and columns. Such a table is called a *submatrix* of the matrix $\mathbf{A}$.

Let $\mathbf{A}$ be a square matrix of order n. To specify a square submatrix of order k, one must indicate the indices of the rows $1 \le i_1 < ... < i_k \le n$ and columns $1 \le j_1 < ... < j_k \le n$ containing it. Let us denote by $\boldsymbol{N}_k$ the set of all systems of indices $(i_1, ..., i_k)$ ordered in ascending order: $1 \le i_1 < ... < i_k \le n$. Then specifying a submatrix is equivalent to choosing two specific systems of indices

$$I = (i_1, ..., i_k) \in \boldsymbol{N}_k, \quad J = (j_1, ..., j_k) \in \boldsymbol{N}_k.$$

The submatrix on the rows with indices from I and columns with indices from J is denoted by

$$\mathbf{A}(I, J) = \big(a_{i_p j_q}\big), \quad 1 \le p \le k, \quad 1 \le q \le k.$$

Let $I' = (i'_1, ..., i'_m)$ be another system of indices ordered in ascending order: $1 \le i'_1 < ... < i'_m \le n$. We call the system I' *complementary* to $I = (i_1, ..., i_k)$ if

$$\{i_1, ..., i_k\} \cap \{i'_1, ..., i'_m\} = \varnothing,$$

$$\{i_1, ..., i_k\} \cup \{i'_1, ..., i'_m\} = \{1, ..., n\}.$$

Obviously, in this case $k + m = n$.

Suppose we are given systems of row and column indices $I, J \in \boldsymbol{N}_k$, and let I' and J' be the complementary systems for I and J, respectively. The submatrix $\mathbf{A}(I', J')$ of order $m = n - k$ is called the *complementary* submatrix with respect to the submatrix $\mathbf{A}(I, J)$ of order k.

The determinant $\det(\mathbf{A}(I, J))$ of a submatrix of order k is called a $k \times k$ *minor* (or a *minor of order* k), and the determinant of the corresponding complementary submatrix is called the *complementary minor*. A $k \times k$ minor is a number; it must not be confused with the course's (i, j) *matrix minor* $\mathbf{M}_{ij}$, which is the submatrix obtained from $\mathbf{A}$ by deleting the i-th row and the j-th column. In the notation above, $\mathbf{M}_{ij} = \mathbf{A}(I', J')$ is the complementary submatrix of the 1×1 submatrix $\big(a_{ij}\big) = \mathbf{A}(I, J)$, $I = (i)$, $J = (j)$.

Laplace's Theorem

Theorem (Laplace's Theorem)**.** Let $\mathbf{A}$ be a square matrix of order n. Fix any system of k columns by choosing $J \in \boldsymbol{N}_k$. Then the computation of the determinant of the matrix $\mathbf{A}$ reduces to computing the $k \times k$ minors on the fixed k columns and their complementary minors:

$$\det(\mathbf{A}) = \sum_{I \in \boldsymbol{N}_k} \det(\mathbf{A}(I, J)) \det(\mathbf{A}(I', J'))(-1)^{\nu(I)+\nu(J)},$$

where $\nu(I) = i_1 + ... + i_k$ for $I = (i_1, ..., i_k)$, and likewise $\nu(J) = j_1 + ... + j_k$.

Proof. Relying on the lemma on partitioning the set of permutations of degree n into the subsets of permutations $\sigma(\pi, \tau)$, $\pi \in S_k$, $\tau \in S_m$ (see Tyrtyshnikov, §5.6), we find

$$\begin{aligned}
\det(\mathbf{A}) = \sum_{I \in \boldsymbol{N}_k} \Bigg(\sum_{\pi \in S_k} \sum_{\tau \in S_m} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \times \\
\times \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\sigma(\pi, \tau)) \Bigg) = \\
= \sum_{I \in \boldsymbol{N}_k} \Bigg(\sum_{\pi \in S_k} \sum_{\tau \in S_m} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \times \\
\times \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\pi) \operatorname{sgn}(\tau) \Bigg) (-1)^{\nu(I)+\nu(J)} = \\
= \sum_{I \in \boldsymbol{N}_k} \left(\sum_{\pi \in S_k} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \operatorname{sgn}(\pi) \right) \times \\
\times \left(\sum_{\tau \in S_m} \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\tau) \right) (-1)^{\nu(I)+\nu(J)}.
\end{aligned}$$

It remains to note that the first and second parenthesized sums yield $\det(\mathbf{A}(I, J))$ and $\det(\mathbf{A}(I', J'))$, respectively. □

The quantity $\det(\mathbf{A}(I', J'))(-1)^{\nu(I)+\nu(J)}$ is called the *cofactor* (or *algebraic complement*) of the $k \times k$ minor $\det(\mathbf{A}(I, J))$. Thus, Laplace's theorem asserts that for any choice of a system of columns, *the determinant of a matrix is equal to the sum of all possible $k \times k$ minors located in the given columns, multiplied by their cofactors.*

Since $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ (Section 2.10), the following version of Laplace's theorem also holds: *for any choice of a system of rows, the determinant of a matrix is equal to the sum of all possible $k \times k$ minors located in the given rows, multiplied by their cofactors.*

For $k = 1$, Laplace's theorem is the *cofactor expansion*. Take $J = (j)$ and $I = (i)$: the 1×1 minor is the entry a_{ij}, its complementary submatrix is the (i, j) matrix minor $\mathbf{M}_{ij}$, and its cofactor is the (i, j)-*cofactor* $A_{ij} = (-1)^{i+j} \det(\mathbf{M}_{ij})$. Hence

$$\det(\mathbf{A}) = a_{1j}A_{1j} + a_{2j}A_{2j} + ... + a_{nj}A_{nj}$$

(cofactor expansion along the j-th column) and, by the row version,

$$\det(\mathbf{A}) = a_{i1}A_{i1} + a_{i2}A_{i2} + ... + a_{in}A_{in}$$

(cofactor expansion along the i-th row). MA1522 takes the cofactor expansion as the definition of the determinant; Laplace's theorem shows that it agrees with formula ($\triangle$).

Problem 4. A matrix $\mathbf{B}$ with determinant $b = \det(\mathbf{B})$ is obtained from a matrix $\mathbf{A}$ with determinant $a = \det(\mathbf{A})$ by adding a number $c \neq 0$ to each entry. Find the sums of the cofactors of all entries (submatrices of order 1) for $\mathbf{A}$ and for $\mathbf{B}$.

Problem 5. In an $(n \times n)$-matrix there is a unique $r \times r$ minor, $r < n$, that is nonzero. Prove that all $k \times k$ minors with $k \geq r + 1$ are equal to zero.

Problem 6. Let $\mathbf{A}$ be a matrix of order n with entries $a_{ij} = \pm 1$. Prove that if $n = 4$, then $|\det(\mathbf{A})| \leq 16$, and construct a matrix for which $\det(\mathbf{A}) = 16$.[2]

2.9. Determinant by Reduction

The Determinant as a Function of the Columns (Rows) of a Matrix

Property 1. The determinant as a function of the columns of a matrix is a linear function with respect to each column: if $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$ and $\mathbf{a}_i = \alpha\mathbf{p} + \beta\mathbf{q}$ is a linear combination of the columns $\mathbf{p}$ and $\mathbf{q}$, then

$$\det(\mathbf{A}) = \alpha \det(\mathbf{A}_p) + \beta \det(\mathbf{A}_q),$$

where the matrices $\mathbf{A}_p$ and $\mathbf{A}_q$ are obtained from $\mathbf{A}$ by replacing the column $\mathbf{a}_i$ with $\mathbf{p}$ and $\mathbf{q}$, respectively.

Proof. In accordance with the definition,

[2] In the general case one can prove that if $|a_{ij}| \leq 1$, then $|\det(\mathbf{A})| \leq n^{\frac{n}{2}}$ (see Section 5.4). Matrices with entries ± 1 for which $|\det(\mathbf{A})| = n^{\frac{n}{2}}$ are called *Hadamard matrices*. It is not hard to establish that Hadamard matrices do not exist for all n. There is a conjecture that Hadamard matrices exist for all n divisible by 4.

$$\begin{aligned}\det(\mathbf{A}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \ldots a_{\sigma(i)i} \ldots a_{\sigma(n)n} = \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \ldots \big(\alpha p_{\sigma(i)i} + \beta q_{\sigma(i)i}\big) \ldots a_{\sigma(n)n} = \\ &= \alpha \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \ldots p_{\sigma(i)i} \ldots a_{\sigma(n)n} + \\ &+ \beta \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \ldots q_{\sigma(i)i} \ldots a_{\sigma(n)n} = \\ &= \alpha \det\big(\mathbf{A}_p\big) + \beta \det\big(\mathbf{A}_q\big). \end{aligned}$$

□

Property 2. The determinant changes sign when two columns are interchanged.

Proof. Suppose the matrix $\mathbf{B} = \big(b_{ij}\big)$ differs from $\mathbf{A}$ by the interchange of the columns $\mathbf{a}_k$ and $\mathbf{a}_l$. Then for any permutation $\sigma \in S_n$,

$$a_{\sigma(1)1} \ldots a_{\sigma(n)n} = b_{(\sigma\tau)(1)1} \ldots b_{(\sigma\tau)(n)n},$$

where $\tau = (k, l)$, and since a transposition changes the sign of a permutation,

$$\operatorname{sgn}(\sigma\tau) = -\operatorname{sgn}(\sigma).$$

It is easy to see that the mapping $\sigma \to \sigma\tau$ defines a one-to-one correspondence between permutations. Each term of the sum of the form ($\triangle$) of Section 2.8 is determined by one and only one permutation. The permutations σ and $\sigma\tau$ in the expansions of $\det(\mathbf{A})$ and $\det(\mathbf{B})$ determine terms with the product of the very same entries (in a different order), but with opposite signs. Hence $\det(\mathbf{A}) = -\det(\mathbf{B})$. □

Property 3. If the columns of a matrix are linearly dependent, then its determinant is equal to zero.

Proof. First of all, let us note that a determinant with two equal columns is equal to zero, since by statement (2) it is equal to itself with the opposite sign.

If the columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ are linearly dependent, then at least one of them is linearly expressible in terms of the others. Suppose

$$\mathbf{a}_i = \sum_{k\neq i} \alpha_k \mathbf{a}_k.$$

Denote by $\mathbf{B}$ the matrix obtained from $\mathbf{A}$ by replacing the column $\mathbf{a}_i$ with

$$\mathbf{a}_i - \sum_{k\neq i} \alpha_k \mathbf{a}_k = \mathbf{0}.$$

Relying on property (1) already established, we find

$$0 = \det(\mathbf{B}) = \det(\mathbf{A}) - \sum_{k\neq i} \alpha_k \det(\mathbf{A}_k),$$

where the matrix $\mathbf{A}_k$ is obtained from $\mathbf{A}$ by replacing the i-th column with $\mathbf{a}_k$. Clearly, in $\mathbf{A}_k$ the i-th and k-th columns are equal, so $\det(\mathbf{A}_k) = 0$. Thus, $\det(\mathbf{A}) = \det(\mathbf{B}) = 0$. □

Property 4. The determinant as a function of the rows of a matrix possesses properties analogous to (1), (2), (3).

Proof. It suffices to take into account that $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ (Section 2.10) and consider $\det(\mathbf{A})$ as a function of the columns of the matrix $\mathbf{A}^T$. □

Problem 1. Given column matrices $\mathbf{u}_1, ..., \mathbf{u}_k, \mathbf{v}_1, ..., \mathbf{v}_k \in \mathbb{R}^n$ and $\mathbf{A} = \mathbf{u}_1\mathbf{v}_1^T + ... + \mathbf{u}_k\mathbf{v}_k^T$, prove that $\det(\mathbf{A}) = 0$ if $k < n$.

Problem 2. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and let $\mathbf{I}$ be the identity matrix. Prove that $\det(\mathbf{I} + \mathbf{u}\mathbf{v}^T) = 1 + \mathbf{v}^T\mathbf{u}$.

Existence of the Linear Dependence Indicator

Theorem. The linear dependence indicator (a function endowed with properties (A), (B), (C) from Section 2.8) exists, is unique, and is the determinant.

Properties (A) and (B) of the linear dependence indicator coincide with properties (1) and (3) of the determinant established above. Property (C) means that the determinant of the identity matrix is equal to 1, and is a consequence of the following more general statement.

Proposition. The determinant of a diagonal matrix is equal to the product of its diagonal entries:

$$\det \begin{pmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix} = a_{11} \cdot ... \cdot a_{nn}.$$

Proof. For a diagonal matrix, in the sum ($\triangle$) of Section 2.8 for its determinant there is only one nonzero term, equal to the product of the entries on the main diagonal. □

2.10. Properties of Determinant

Determinant of the Transpose Matrix

Suppose we have a rectangular matrix of size $m \times n$:

$$\mathbf{A} = (a_{ij}), \quad 1 \le i \le m, \quad 1 \le j \le n.$$

If we interchange its rows and columns, we obtain a new matrix—of size $n \times m$. It is called the *transpose* with respect to $\mathbf{A}$ and is denoted by $\mathbf{A}^T$:

$$\mathbf{A}^T = \big(a_{ji}\big), \quad 1 \le j \le n, \quad 1 \le i \le m.$$

Proposition. For any square matrix, $\det(\mathbf{A}^T) = \det(\mathbf{A})$.

Proof. By the definition of the transpose matrix and formula ($\triangle$) of Section 2.8 for the determinant of a matrix of order n,

$$\begin{aligned}\det(\mathbf{A}^T) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)}\ldots a_{n\sigma(n)} = \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma^{-1}(1)1}\ldots a_{\sigma^{-1}(n)n} = \\ &= \sum_{\sigma^{-1}\in S_n} \operatorname{sgn}(\sigma^{-1}) a_{\sigma^{-1}(1)1}\ldots a_{\sigma^{-1}(n)n} = \det(\mathbf{A}).\end{aligned}$$

In the last equality we took into account that $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$. □

Problem 1. Prove that the columns of a real rectangular matrix $\mathbf{A}$ are linearly independent if and only if the columns of the matrix $\mathbf{A}^T\mathbf{A}$ are linearly independent.

Determinant of a Product of Matrices

Theorem. The determinant of a product of square matrices is equal to the product of their determinants.

Proof. Let $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$ be a matrix of order n with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ and $\mathbf{B} = \big(b_{ij}\big)$. Then any column of the matrix $\mathbf{AB}$ is a linear combination of the columns of the matrix $\mathbf{A}$ with coefficients from the corresponding column of the matrix $\mathbf{B}$:

$$\mathbf{AB} = \Big(\textstyle\sum_{i_1=1}^n b_{i_1 1}\mathbf{a}_{i_1} \ \cdots \ \sum_{i_n=1}^n b_{i_n n}\mathbf{a}_{i_n}\Big).$$

Using the linearity of the determinant in each column, we obtain

$$\begin{aligned}\det(\mathbf{AB}) &= \sum_{i_1=1}^n \ldots \sum_{i_n=1}^n b_{i_1 1}\ldots b_{i_n n} \det\big(\mathbf{a}_{i_1} \ \cdots \ \mathbf{a}_{i_n}\big) = \\ &= \sum_{\sigma\in S_n} b_{\sigma(1)1}\ldots b_{\sigma(n)n} \det\big(\mathbf{a}_{\sigma(1)} \ \cdots \ \mathbf{a}_{\sigma(n)}\big) = \\ &= \left(\sum_{\sigma\in S_n} b_{\sigma(1)1}\ldots b_{\sigma(n)n} \operatorname{sgn}(\sigma)\right) \det(\mathbf{A}) = \\ &= \det(\mathbf{B}) \cdot \det(\mathbf{A}).\end{aligned}$$

□

Problem 2. Let $\mathbf{P}$ be an invertible matrix of order n. Prove that for any columns $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{n\times 1}$ the equality $\det(\mathbf{I} + \mathbf{uv}^T) = \det\Big(\mathbf{I} + (\mathbf{Pu})(\mathbf{P}^{-T}\mathbf{v})^T\Big)$ holds.

Problem 3. Prove that the determinant of a tridiagonal matrix does not change if every superdiagonal entry is multiplied and every subdiagonal entry is divided by one and the same nonzero number.

The Adjoint Formula for the Inverse

In this subsection we consider square matrices with a nonzero determinant.

Let $\tilde{\mathbf{A}} = \big(A_{ij}\big)$ be the matrix of order n in which the entry A_{ij} is the cofactor of the entry a_{ij} in the matrix $\mathbf{A}$, that is, the (i,j)-cofactor $A_{ij} = (-1)^{i+j}\det\big(\mathbf{M}_{ij}\big)$ (Section 2.8). The matrix $\operatorname{adj}(\mathbf{A}) = \tilde{\mathbf{A}}^T$, whose (i,j)-entry is the (j,i)-cofactor A_{ji}, is called the *adjoint* (also the *adjugate* or *classical adjoint*) of the matrix $\mathbf{A}$.

Theorem. If a matrix $\mathbf{A}$ of order n has $\det(\mathbf{A}) \neq 0$, then it is invertible and, moreover,

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A}),$$

where $\operatorname{adj}(\mathbf{A})$ is the adjoint of $\mathbf{A}$.

Proof. Note that

$$\sum_{j=1}^{n} a_{ij}A_{kj} = \begin{cases} \det(\mathbf{A}), & k = i, \\ 0, & k \neq i. \end{cases} \qquad (*)$$

For $k = i$, equality $(*)$ is obtained by applying Laplace's theorem, that is, by the cofactor expansion of the determinant of the matrix $\mathbf{A}$ along the k-th row. For $k \neq i$, the left-hand side of equality $(*)$ represents the cofactor expansion along the k-th row of the determinant of the matrix obtained from $\mathbf{A}$ by replacing the k-th row with the i-th row. Such a determinant is equal to zero, as the determinant of a matrix with two identical rows. For example, if $n = 3$, $i = 1$, $k = 2$, then

$$a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} =$$

$$= a_{11}\left(-\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}\right) + a_{12}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\left(-\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}\right) =$$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0.$$

Furthermore,

$$\sum_{j=1}^{n} a_{ij}A_{kj} = \sum_{j=1}^{n} a_{ij}\big(\operatorname{adj}(\mathbf{A})\big)_{jk}, \quad 1 \leq i, k \leq n.$$

Therefore, by virtue of the relations $(*)$,

$$\mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \begin{pmatrix} \det(\mathbf{A}) & & \\ & \ddots & \\ & & \det(\mathbf{A}) \end{pmatrix} = \det(\mathbf{A})\mathbf{I}.$$

Using Laplace's theorem for expanding the determinant along the k-th column (cofactor expansion along a column), we find

$$\sum_{i=1}^{n} a_{ij}A_{ik} = \begin{cases} \det(\mathbf{A}), & k = j, \\ 0, & k \neq j \end{cases} \quad \Rightarrow$$

$$\Rightarrow \quad \mathrm{adj}(\mathbf{A})\mathbf{A} = \begin{pmatrix} \det(\mathbf{A}) & & 0 \\ & \ddots & \\ 0 & & \det(\mathbf{A}) \end{pmatrix} = \det(\mathbf{A})\mathbf{I}. \qquad \square$$

Problem 4. Prove that any invertible matrix can be made singular by changing just one of its entries.

Cramer's Rule

Theorem. Let $\mathbf{A}$ be a matrix of order n with $\det(\mathbf{A}) \neq 0$. Then the linear system $\mathbf{Ax} = \mathbf{b}$ has a unique solution $\mathbf{x}$ with components

$$x_i = \frac{\det(\mathbf{A}_i(\mathbf{b}))}{\det(\mathbf{A})}, \quad 1 \leq i \leq n,$$

where $\mathbf{A}_i(\mathbf{b})$ is the matrix obtained from $\mathbf{A}$ by replacing the i-th column with $\mathbf{b}$.

Proof. According to the theorem on the inversion of a matrix with nonzero determinant (the adjoint formula),

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{\det(\mathbf{A})}\,\mathrm{adj}(\mathbf{A})\mathbf{b} \quad \Rightarrow \quad x_i = \frac{1}{\det(\mathbf{A})}\sum_{j=1}^{n} A_{ji}b_j, \quad 1 \leq i \leq n.$$

It remains to note that the sum $\sum_{j=1}^{n} A_{ji}b_j$ is the cofactor expansion along the i-th column of the determinant of the matrix $\mathbf{A}_i(\mathbf{b})$. $\square$

Determinant of a Block-Triangular Matrix BEYOND MA1522

Consider a block-triangular matrix of order n:

$$\mathbf{A} = \begin{pmatrix} \mathbf{P} & \mathbf{R} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}, \quad \mathbf{P} \in \mathbb{R}^{k\times k}, \quad \mathbf{Q} \in \mathbb{R}^{m\times m}, \quad k + m = n.$$

Applying Laplace's theorem (Section 2.8) to the system of the first k columns (or rows) immediately gives the useful formula

$$\det(\mathbf{A}) = \det(\mathbf{P})\det(\mathbf{Q}).$$

Chapter 3. Euclidean Vector Spaces

3.1. Euclidean Vector Spaces

Linear Operations on Vectors

Sum of vectors: let $\overrightarrow{AB} \in \mathbf{a}$ and $\overrightarrow{BC} \in \mathbf{b}$; then $\mathbf{c} = \mathbf{a} + \mathbf{b}$ is defined as the vector generated by the directed segment $\overrightarrow{AC}$.

It is important that the vector $\mathbf{c}$ obtained in this way does not depend on the choice of the point A. Indeed, let $\overrightarrow{PQ} \in \mathbf{a}$ and $\overrightarrow{QR} \in \mathbf{b}$. Then from the congruence of the triangles $\triangle ABC$ and $\triangle PQR$ there follows the equality of lengths and codirectionality of the directed segments $\overrightarrow{AC}$ and $\overrightarrow{PR}$, and hence their equality.

The set of free vectors forms an abelian group with respect to the operation of vector addition. The role of the identity element is played by the zero vector $\mathbf{0} = \overrightarrow{AA}$. For $\mathbf{a} = \overrightarrow{AB}$, the inverse element is $\mathbf{b} = \overrightarrow{BA}$. In this context, the vector $\mathbf{b}$ is called the *opposite vector* for $\mathbf{a}$ and is denoted by $\mathbf{b} = -\mathbf{a}$.

Multiplication of a vector by a scalar: let $\overrightarrow{AB} \in \mathbf{a}$; then $\alpha\mathbf{a}$ is defined as the vector generated by the directed segment $\overrightarrow{AC}$ that has length $|AC| = |\alpha|\ |AB|$ and, if $\alpha \neq 0$, is codirectional with $\overrightarrow{AB}$ for $\alpha > 0$ and oppositely directed for $\alpha < 0$. It is not difficult to verify that the vector $\alpha\mathbf{a}$ does not depend on the choice of the point A.

One can check that $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a}$ for any real numbers α, β. It is also useful to note that $1\mathbf{a} = \mathbf{a}$, and the vector $(-1)\mathbf{a}$ is the opposite of $\mathbf{a}$.

Furthermore, the operations of vector addition and multiplication of a vector by a scalar are linked by the following distributive laws:

$$(\alpha + \beta)\mathbf{a} = (\alpha\mathbf{a}) + (\beta\mathbf{a}), \quad \alpha(\mathbf{a} + \mathbf{b}) = (\alpha\mathbf{a}) + (\alpha\mathbf{b}).$$

Abstract Vector Spaces

When studying the linear dependence of vectors, linear spans, bases, and dimensions (Sections 2.3 and 3.5–3.7), we assume that vectors are column vectors with real entries. However, when studying the rank of a matrix (Section 4.2), we shall also speak of the linear dependence and independence of the rows of a matrix. Of course, from a formal standpoint, rows can be transposed so that we once again deal with column vectors. Yet all the concepts listed above and many of the facts obtained can be applied without any changes to the case where vectors are understood to be matrices of any fixed size. This fact alone prompts us to consider introducing a more general (and more abstract) concept of a vector.

Moreover, when studying bases and dimensions, we deal exclusively with linear spans of vectors, and this is not always convenient: for example, the set of all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ is, of course, the linear span of the vectors of a basis for the solution space, but it would be useful to have the right to study the properties of this set without mentioning the system of vectors that generates it.

Let us say that vectors are elements of some nonempty set V on which two operations are defined: addition of vectors (if $\mathbf{a}, \mathbf{b} \in V$, then $\mathbf{a} + \mathbf{b} \in V$) and multiplication of vectors by real numbers (if $\mathbf{a} \in V$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{a} \in V$). Let us require that these operations possess the following properties:

- $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}) \quad \forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in V$ (associativity of vector addition);
- there exists a special vector $\mathbf{0}$, called the *zero vector*, such that
$$\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a} \quad \forall \mathbf{a} \in V;$$
- for any vector $\mathbf{a} \in V$ there exists a vector $\mathbf{b} \in V$ such that
$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = \mathbf{0};$$
- $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} \quad \forall \mathbf{a}, \mathbf{b} \in V$ (commutativity of vector addition);
- $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a} \quad \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{a} \in V$;
- $(\alpha + \beta)\mathbf{a} = \alpha\mathbf{a} + \beta\mathbf{a} \quad \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{a} \in V$ (distributivity);
- $\alpha(\mathbf{a} + \mathbf{b}) = \alpha\mathbf{a} + \alpha\mathbf{b} \quad \forall \alpha \in \mathbb{R}, \forall \mathbf{a}, \mathbf{b} \in V$ (distributivity);
- $1\mathbf{a} = \mathbf{a} \quad \forall \mathbf{a} \in V$.[3]

A set V possessing all these properties is called a (real) *vector space*, or a vector space over $\mathbb{R}$. A frequently encountered synonymous term is *linear space*.

Note that the set V with respect to the operation of vector addition is an abelian group. The role of the identity element is played by the zero vector. A vector $\mathbf{b}$ such that $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = \mathbf{0}$ is called the *opposite* (or *negative*) of the vector $\mathbf{a}$ and is denoted by $\mathbf{b} = -\mathbf{a}$.

Certain familiar properties of the operations we have introduced, which were previously applied freely to column vectors, *require proofs* in the more abstract setting considered here.

Proposition 1. $0\mathbf{a} = \mathbf{0} \quad \forall \mathbf{a} \in V$.

Proof. By distributivity,
$$0\mathbf{a} = (0 + 0)\mathbf{a} = (0\mathbf{a}) + (0\mathbf{a}).$$
Further, let $\mathbf{b} = -(0\mathbf{a})$ (the vector opposite to the vector $0\mathbf{a}$). Then $\mathbf{0} = \mathbf{b} + (0\mathbf{a}) = (\mathbf{b} + (0\mathbf{a})) + (0\mathbf{a}) \Rightarrow \mathbf{0} = 0\mathbf{a}$. □

Proposition 2. $\alpha\mathbf{0} = \mathbf{0} \quad \forall \alpha \in \mathbb{R}$.

[3] This property is equivalent to saying that every vector $\mathbf{a}$ can be represented in the form $\mathbf{a} = \alpha\mathbf{b}$ for some vector $\mathbf{b}$ and some number α. Indeed, if this property holds, one can take $\mathbf{b} = \mathbf{a}$ and $\alpha = 1$. On the other hand, suppose this property is not assumed to hold, but it is known that $\mathbf{a} = \alpha\mathbf{b}$. Then, using the axiom $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a}$, we obtain $1(\alpha\mathbf{b}) = (1 \cdot \alpha)\mathbf{b} = \alpha\mathbf{b} \Rightarrow 1\mathbf{a} = \mathbf{a}$.

Proof. $\alpha\mathbf{0} = \alpha(\mathbf{0}+\mathbf{0}) = \alpha\mathbf{0} + \alpha\mathbf{0} \Rightarrow \alpha\mathbf{0} = \mathbf{0}$. □

Proposition 3. $(-1)\mathbf{a} = -\mathbf{a} \quad \forall \mathbf{a} \in V$.

Proof. By Proposition 1 and distributivity,

$$\mathbf{0} = 0\mathbf{a} = (1+(-1))\mathbf{a} = (1\mathbf{a}) + ((-1)\mathbf{a}) = \mathbf{a} + ((-1)\mathbf{a}).$$ □

Proposition 4. If $\alpha\mathbf{a} = \mathbf{0}$, then either $\alpha = 0$ or $\mathbf{a} = \mathbf{0}$.

Proof. Suppose[4] $\alpha \neq 0$. Then

$$\mathbf{a} = 1\mathbf{a} = \left(\frac{1}{\alpha}\alpha\right)\mathbf{a} = \frac{1}{\alpha}(\alpha\mathbf{a}) = \frac{1}{\alpha}\mathbf{0} = \mathbf{0}.$$ □

As before, for any numbers $\alpha_1, ..., \alpha_n$, a vector $\mathbf{w}$ of the form

$$\mathbf{w} = \alpha_1\mathbf{a}_1 + ... + \alpha_n\mathbf{a}_n$$

is called a linear combination of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$, and the set of all linear combinations with all possible values of the coefficients $\alpha_1, ..., \alpha_n$ is called the linear span of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ and is denoted by $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$.

Note that the zero vector is denoted by $\mathbf{0}$, to distinguish it from the number 0.

3.2. Dot Product, Norm, Distance

Dot Product of Geometric Vectors

The length of a vector $\mathbf{a}$ is defined as the length of a directed segment generating it (directed segments generating one and the same vector are equal and therefore have the same length). Notation for the length: $\|\mathbf{a}\|$. The angle θ between nonzero vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$ is defined as the angle between the sides OA and OB in the triangle OAB.

The *dot product* (or *scalar product*) of vectors $\mathbf{a}$ and $\mathbf{b}$ is the quantity

$$\mathbf{a}\cdot\mathbf{b} = \begin{cases} \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta, & \mathbf{a}\neq\mathbf{0} \text{ and } \mathbf{b}\neq\mathbf{0}, \\ 0, & \mathbf{a}=\mathbf{0} \text{ or } \mathbf{b}=\mathbf{0}. \end{cases}$$

By virtue of the definition, it is obvious that

$$\mathbf{a}\cdot\mathbf{a} > 0 \quad \text{for } \mathbf{a}\neq\mathbf{0}; \quad \mathbf{a}\cdot\mathbf{a} = 0 \Leftrightarrow \mathbf{a} = \mathbf{0}. \tag{1}$$

It is also obvious that

$$\mathbf{a}\cdot\mathbf{b} = \mathbf{b}\cdot\mathbf{a} \quad \forall \mathbf{a}, \mathbf{b}. \tag{2}$$

If the vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$ are noncollinear, then in the plane passing through the points O, A, B one can introduce a Cartesian coordinate system

[4] This proposition cannot be obtained without the axiom $1\mathbf{a} = \mathbf{a}$. Indeed, take any abelian group V with zero element $\mathbf{0}$ and define multiplication by a number by the rule $\alpha\mathbf{a} = \mathbf{0}$ for all numbers α and vectors $\mathbf{a} \in V$. Then all the axioms of a vector space will be satisfied except this one.

with origin at the point O and first axis coinciding with the line OB and giving the point B a positive coordinate. Then the quantity $\|\mathbf{a}\| \cos\theta$ will be precisely the coordinate of the point A on this axis. From this there immediately follow the important properties of linearity of the dot product in the first argument:

$$(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c} \quad \forall \mathbf{a}, \mathbf{b}, \mathbf{c}, \tag{3}$$

$$(\alpha \mathbf{a}) \cdot \mathbf{b} = \alpha(\mathbf{a} \cdot \mathbf{b}) \quad \forall \mathbf{a}, \mathbf{b}, \quad \forall \alpha \in \mathbb{R}. \tag{4}$$

Property (2) immediately yields the analogous properties of linearity of the dot product in the second argument as well.

Vectors are called *orthogonal* if their dot product is equal to zero.

Dot Product and Coordinates

Suppose a Cartesian coordinate system with basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is given. Then

$$\mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 0, & i \neq j, \\ 1, & i = j. \end{cases} \tag{5}$$

Theorem. Suppose that in a given Cartesian coordinate system the vector $\mathbf{a}$ has coordinates a_1, a_2, a_3, and the vector $\mathbf{b}$ has coordinates b_1, b_2, b_3. Then the formula

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \tag{6}$$

holds.

Proof. Let us write

$$\mathbf{a} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3, \quad \mathbf{b} = b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2 + b_3 \mathbf{e}_3.$$

Relying on properties (2)–(4) of the dot product and relations (5), we find

$$\mathbf{a} \cdot \mathbf{b} = (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3) \cdot (b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2 + b_3 \mathbf{e}_3) =$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j (\mathbf{e}_i \cdot \mathbf{e}_j) = a_1 b_1 + a_2 b_2 + a_3 b_3. \qquad \square$$

Remark 1. If in some coordinate system the dot product of any vectors $\mathbf{a}$ and $\mathbf{b}$ is computed by formula (6), then that coordinate system is Cartesian.

Remark 2. In the case of a Cartesian coordinate system for vectors in a plane, formula (6) takes the form

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2.$$

On Generalizations

Formula (6) and properties (1)–(4) provide a basis for introducing an inner product in more general cases—for objects that are no longer vectors in geometric space. For example, if $\mathbf{a} = (a_1, ..., a_n)^T$, $\mathbf{b} = (b_1, ..., b_n)^T$ are column vectors in $\mathbb{R}^n$, one can define their dot product by analogy with formula (6):

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + ... + a_n b_n. \tag{7}$$

There is also another idea of a more general character—to take properties (1)–(4) as the foundation and call any function of column vectors $\mathbf{a}, \mathbf{b}$ satisfying axioms (1)–(4) an inner product.

For geometric vectors, the dot product was defined on the basis of concepts such as the length of a vector and the angle between vectors. In more general cases, it is simpler to introduce an inner product in some way and then use it to introduce the concepts of length (norm) and angle:

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}, \quad \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}.$$

For example, relying on (7), one can introduce length and angle in this way for vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. Here it is important that

$$|\mathbf{a} \cdot \mathbf{b}| \le \sqrt{\mathbf{a} \cdot \mathbf{a}}\sqrt{\mathbf{b} \cdot \mathbf{b}}.$$

This inequality (known as the Cauchy–Bunyakovsky–Schwarz inequality) is easily deduced from (7), but in fact it holds for all conceivable ways of specifying an inner product—we shall discuss this in detail later in this section.

Euclidean Space

Let V be a real vector space on which to each ordered pair of vectors $\mathbf{x}, \mathbf{y} \in V$ there is assigned a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ in such a way that:

(i) $\langle \mathbf{x}, \mathbf{x} \rangle \ge 0 \quad \forall \mathbf{x} \in V; \quad \langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$;
(ii) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle \quad \forall \mathbf{x}, \mathbf{y} \in V$;
(iii) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in V$;
(iv) $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle \quad \forall \alpha \in \mathbb{R}, \quad \forall \mathbf{x} \in V$.

The number $\langle \mathbf{x}, \mathbf{y} \rangle$ is called the *inner product* (or *scalar product*) of the vectors $\mathbf{x}$ and $\mathbf{y}$. A real vector space with an inner product is called *Euclidean*.

In $\mathbb{R}^n$, the inner product of vectors $\mathbf{x} = (x_1, ..., x_n)^T$ and $\mathbf{y} = (y_1, ..., y_n)^T$ is often introduced as the sum of pairwise products of their coordinates:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i = \mathbf{y}^T \mathbf{x}. \tag{8}$$

It is called the *natural* inner product on $\mathbb{R}^n$; this is the dot product.

However, on $\mathbb{R}^n$ an inner product can also be introduced in many other ways: for example, if we fix numbers $\lambda_1, ..., \lambda_n > 0$, then the expression

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} \lambda_i x_i y_i = \mathbf{y}^T \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \mathbf{x}$$

possesses properties (i)–(iv) and, consequently, defines an inner product.

Length of a Vector BEYOND MA1522

Let V be an arbitrary space with an inner product. The quantity

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$$

is called the *length* of the vector $\mathbf{x} \in V$.

Theorem (Cauchy–Bunyakovsky–Schwarz Inequality). For any vectors $\mathbf{x}, \mathbf{y} \in V$,

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|, \tag{9}$$

and equality is attained if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. Let us write the complex number $\langle \mathbf{x}, \mathbf{y} \rangle$ in trigonometric form:

$$\langle \mathbf{x}, \mathbf{y} \rangle = |\langle \mathbf{x}, \mathbf{y} \rangle| \xi, \quad \xi = \cos\varphi + i \sin\varphi.$$

If $\mathbf{y} = \mathbf{0}$, then equality holds in (9). Suppose $\mathbf{y} \neq \mathbf{0}$. For an arbitrary $t \in \mathbb{R}$, consider the expression

$$\langle \mathbf{x} + t\xi\mathbf{y}, \mathbf{x} + t\xi\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle + t\bar{\xi}\langle \mathbf{x}, \mathbf{y} \rangle + t\xi\overline{\langle \mathbf{x}, \mathbf{y} \rangle} + t^2 \xi\bar{\xi}\langle \mathbf{y}, \mathbf{y} \rangle =$$
$$= t^2 \|\mathbf{y}\|^2 + 2t|\langle \mathbf{x}, \mathbf{y} \rangle| + \|\mathbf{x}\|^2 \geq 0.$$

The nonnegativity of the quadratic trinomial in the variable t means that its discriminant is nonpositive:

$$D = |\langle \mathbf{x}, \mathbf{y} \rangle|^2 - \|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \leq 0 \quad \Rightarrow \quad |\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

Suppose that for $\mathbf{y} \neq \mathbf{0}$ equality holds in (9) $\Rightarrow D = 0 \Rightarrow$ for some real t we obtain $\langle \mathbf{x} + t\xi\mathbf{y}, \mathbf{x} + t\xi\mathbf{y} \rangle = 0 \Rightarrow \mathbf{x} + t\xi\mathbf{y} = \mathbf{0}$. It is also obvious that if $\mathbf{y} = \mathbf{0}$ or $\mathbf{x} = \alpha\mathbf{y}$, then (9) turns into an equality. □

Corollary. Length is a vector norm on V.

Proof. The first two properties of a norm are obvious, and the triangle inequality follows from the Cauchy–Bunyakovsky–Schwarz inequality:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle \leq$$
$$\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2.$$ □

A space with an inner product that is complete with respect to the norm $\|\cdot\|$ is usually called a *Hilbert* space.

Problem 1. For two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the equality $\|\mathbf{x} + \mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\|$ holds. Prove that $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent. Is this true in the case of the equality $\|\mathbf{x} + \mathbf{y}\|_p = \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$ for the Hölder norm with $p \neq 2$?

Problem 2. For matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$, the square of the sum of the diagonal entries of the matrix $\mathbf{A}^T\mathbf{B}$ is equal to the product of the sums of the diagonal entries of the matrices $\mathbf{A}^T\mathbf{A}$ and $\mathbf{B}^T\mathbf{B}$. Prove that $\mathbf{A}$ and $\mathbf{B}$ differ only by a scalar factor.

Problem 3. Let V be the vector space of real functions continuous on $[0, 1]$. Prove that the expression $\langle f, g \rangle = \int_0^1 f(t)g(t)\,\mathrm{d}t$ defines an inner product, but that the resulting Euclidean space V is not a Hilbert space.

3.3. Linear Combinations and Linear Spans

Linear Combinations is treated in Section 2.3.

3.4. Subspaces

Subspaces of a Vector Space

A nonempty set $W \subseteq V$ is called a *subspace* of the vector space V if it is itself a vector space with respect to the operations acting in V. Clearly, in order for W to be a subspace, it is necessary and sufficient that for any vectors $\mathbf{a}, \mathbf{b} \in W$ and any number α the inclusions $\mathbf{a} + \mathbf{b} \in W$ and $\alpha\mathbf{a} \in W$ hold.

If the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ belong to a subspace W, then

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\} \subseteq W.$$

Example. Consider the set V of all free vectors in the plane with a coordinate system centered at the point O. Since each free vector is generated by one and only one position vector, any subset of free vectors can be identified with a subset of position vectors $\overrightarrow{OA}$ or their endpoints—the points A.

The set V is obviously a vector space. Any line passing through the origin is a subspace in V. At the same time, if l is a line not passing through the origin, then it is not a subspace: let $A, B \in l$ and $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{OB}$; clearly, $C \notin l$.

Problem 1. Prove that the vector space $\mathbb{R}^n$ cannot be represented as the union of a finite number of sets, each of which does not coincide with $\mathbb{R}^n$ and is a subspace of it.

Sum and Intersection of Subspaces BEYOND MA1522

Let P and Q be subspaces of a vector space V. By the sum $P + Q$ we mean the set of all vectors of the form $\mathbf{w} = \mathbf{p} + \mathbf{q}$, where $\mathbf{p} \in P$, $\mathbf{q} \in Q$. By the intersection $P \cap Q$ we mean the usual intersection of sets.

Proposition. The sets $P + Q$ and $P \cap Q$ are subspaces in V.

Proof. 1) Consider an arbitrary linear combination of vectors $\mathbf{w}_1, \mathbf{w}_2 \in P + Q$. By the definition of the set $P + Q$ we have: $\mathbf{w}_1 = \mathbf{p}_1 + \mathbf{q}_1$ and $\mathbf{w}_2 = \mathbf{p}_2 + \mathbf{q}_2$, where $\mathbf{p}_1, \mathbf{p}_2 \in P$ and $\mathbf{q}_1, \mathbf{q}_2 \in Q$. Then

$$\alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 = (\alpha_1\mathbf{p}_1 + \alpha_2\mathbf{p}_2) + (\alpha_1\mathbf{q}_1 + \alpha_2\mathbf{q}_2) \in P + Q,$$

since the vector in the first parentheses belongs to P and the vector in the second parentheses belongs to Q (P and Q are subspaces, and therefore they contain all linear combinations of their vectors).

2) Similarly, consider a linear combination of vectors $\mathbf{w}_1, \mathbf{w}_2 \in P \cap Q$:

$$\alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 \in P \quad \text{and simultaneously} \quad \alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 \in Q$$
$$\Rightarrow \alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 \in P \cap Q.$$

□

Note that any two subspaces have a nonempty intersection: each of them contains at least the zero vector.

Theorem (Grassmann's Theorem)**.** Let W_1 and W_2 be finite-dimensional subspaces of a vector space V. Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof. Consider a basis $\{\mathbf{g}_1, ..., \mathbf{g}_r\}$ for the subspace $W_1 \cap W_2$ and extend it first to a basis for W_1:

$$\{\mathbf{g}_1, ..., \mathbf{g}_r, \mathbf{p}_1, ..., \mathbf{p}_k\}, \quad r + k = \dim(W_1),$$

and then to a basis for W_2:

$$\{\mathbf{g}_1, ..., \mathbf{g}_r, \mathbf{q}_1, ..., \mathbf{q}_m\}, \quad r + m = \dim(W_2).$$

Obviously,

$$W_1 + W_2 = \operatorname{span}\{\mathbf{g}_1, ..., \mathbf{g}_r, \mathbf{p}_1, ..., \mathbf{p}_k, \mathbf{q}_1, ..., \mathbf{q}_m\}.$$

Therefore, it remains to prove the linear independence of the vectors generating this linear span. Let

$$\alpha_1\mathbf{g}_1 + ... + \alpha_r\mathbf{g}_r + \beta_1\mathbf{p}_1 + ... + \beta_k\mathbf{p}_k + \gamma_1\mathbf{q}_1 + ... + \gamma_m\mathbf{q}_m = \mathbf{0}.$$

From this,

$$\alpha_1\mathbf{g}_1 + ... + \alpha_r\mathbf{g}_r + \beta_1\mathbf{p}_1 + ... + \beta_k\mathbf{p}_k = -(\gamma_1\mathbf{q}_1 + ... + \gamma_m\mathbf{q}_m) \in W_1 \cap W_2.$$

Since $W_1 \cap W_2 = \operatorname{span}\{\mathbf{g}_1, ..., \mathbf{g}_r\}$, for some coefficients $\delta_1, ..., \delta_r$ we have

$$\delta_1\mathbf{g}_1 + ... + \delta_r\mathbf{g}_r = -(\gamma_1\mathbf{q}_1 + ... + \gamma_m\mathbf{q}_m),$$

which is equivalent to the equality

$$\begin{gathered}\delta_1\mathbf{g}_1 + ... + \delta_r\mathbf{g}_r + \gamma_1\mathbf{q}_1 + ... + \gamma_m\mathbf{q}_m = \mathbf{0} \Rightarrow \\ \Rightarrow \delta_1 = ... = \delta_r = \gamma_1 = ... = \gamma_m = 0 \Rightarrow \\ \Rightarrow \alpha_1 = ... = \alpha_r = \beta_1 = ... = \beta_k = 0.\end{gathered}$$

□

Problem 2. Find the dimension of the sum of the subspace of $n \times n$ matrices with zero sum of entries in each row and the subspace of $n \times n$ matrices with zero sum of entries in each column.

Affine Sets (Linear Manifolds)

Let W be a subspace in a vector space V and $\mathbf{x}$ be some vector from V. A set of vectors of the form

$$M = \{\mathbf{v} = \mathbf{x} + \mathbf{w} \mid \mathbf{w} \in W\},$$

that is, the translate of the subspace W by the vector $\mathbf{x}$, is called an *affine set* (or *linear manifold*) in V. Notation: $M = \mathbf{x} + W$.

The subspace W is called the *direction subspace* for M and is uniquely determined by the set M.

Indeed, let

$$M = \mathbf{x}_1 + W_1 = \mathbf{x}_2 + W_2.$$

From this, $\mathbf{x}_1 - \mathbf{x}_2 \in W_1 \cap W_2$. Let $\mathbf{y} \in W_1$. Then $\mathbf{y} + (\mathbf{x}_1 - \mathbf{x}_2) \in W_2 \Rightarrow \mathbf{y} \in W_2$. Similarly, if $\mathbf{y} \in W_2$, then $\mathbf{y} + (\mathbf{x}_2 - \mathbf{x}_1) \in W_1 \Rightarrow \mathbf{y} \in W_1$. □

For $M = \mathbf{x} + W$, the vector $\mathbf{x}$ is called a *shift vector*. As the shift vector one can take any vector from M:

$$M = \mathbf{x} + W = \mathbf{y} + W \quad \forall \mathbf{y} \in M.$$

Indeed, let $\mathbf{y} = \mathbf{x} + \mathbf{w}_0$ for some $\mathbf{w}_0 \in W$. Then, if $\mathbf{z} = \mathbf{x} + \mathbf{w}$ for some $\mathbf{w} \in W$, we have $\mathbf{z} = \mathbf{y} + (\mathbf{x} - \mathbf{y}) + \mathbf{w} = \mathbf{y} + (-\mathbf{w}_0 + \mathbf{w})$. Hence, $\mathbf{x} + W \subseteq \mathbf{y} + W$. The reverse inclusion is proved similarly. □

If W is a finite-dimensional space, then its dimension is also called the *dimension of the affine set* $M = \mathbf{x} + W$.

Example 1. The set of solutions of a linear system $\mathbf{Ax} = \mathbf{b}$ with an $m \times n$ matrix $\mathbf{A}$ of rank r is an affine set $\mathbf{v} + W$, where $\mathbf{v}$ is a particular solution of this system and $W = \mathrm{Null}(\mathbf{A})$ is the solution space of the associated homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Example 2. A line in the plane or in three-dimensional space is an affine set of dimension 1. A plane in three-dimensional space is an affine set of dimension 2.

When studying affine sets, the elements of a vector space are usually called *points*. By analogy with geometric space, one can think of vectors as position vectors laid off from a common initial point identified with the zero vector.

Problem 3. Prove that two affine sets $\mathbf{a}_1 + L_1$ and $\mathbf{a}_2 + L_2$ with direction subspaces L_1 and L_2 intersect if and only if $\mathbf{a}_1 - \mathbf{a}_2 \in L_1 + L_2$.

3.5. Linear Independence

Linear Dependence

Vectors all of whose entries are equal to zero are called *zero vectors*, and sometimes simply *zeros*. We shall denote any zero vector by the symbol $\mathbf{0}$.

A linear combination of vectors is called *nontrivial* if at least one of its coefficients is nonzero. A system (in other words, a nonempty ordered collection of a finite number) of vectors is called *linearly dependent* if there exists for them a nontrivial linear combination equal to the zero vector. (MA1522 speaks of a *set* $\{\mathbf{a}_1, ..., \mathbf{a}_k\}$; a system lists its vectors in a fixed order and may contain repeated vectors.)

Lemma 1. If $\mathbf{a}_1, ..., \mathbf{a}_k$ is a linearly dependent system of $k > 1$ nonzero vectors, then there exists in it a vector $\mathbf{a}_m$, $m > 1$, that is a linear combination of the preceding vectors:

$$\mathbf{a}_m \in \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_{m-1}\}.$$

Proof. Consider a nontrivial linear combination equal to zero:

$$\alpha_1\mathbf{a}_1 + ... + \alpha_k\mathbf{a}_k = \mathbf{0},$$

and let m be the largest index such that $\alpha_m \neq 0$. If $m = 1$, then $\alpha_1\mathbf{a}_1 = \mathbf{0}$ and, since $\alpha_1 \neq 0$, we obtain $\mathbf{a}_1 = \mathbf{0}$, which contradicts the hypothesis of the lemma. Consequently, $m > 1$. Then

$$\alpha_1\mathbf{a}_1 + ... + \alpha_m\mathbf{a}_m = \mathbf{0} \quad \Rightarrow \quad \mathbf{a}_m = \left(-\frac{\alpha_1}{\alpha_m}\right)\mathbf{a}_1 + ... + \left(-\frac{\alpha_{m-1}}{\alpha_m}\right)\mathbf{a}_{m-1}. \ \square$$

Linear Independence

A system of vectors is called *linearly independent* if it is not linearly dependent. Thus, if the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent, then

$$\alpha_1\mathbf{a}_1 + ... + \alpha_k\mathbf{a}_k = \mathbf{0} \quad \Rightarrow \quad \alpha_1 = ... = \alpha_k = 0.$$

Lemma 2. Every subsystem of a linearly independent system is linearly independent.

Proof. Suppose that a subsystem is linearly dependent. This means that there exists a nontrivial linear combination of the vectors of this subsystem equal to zero. Then the linear combination of the vectors of the original system with the same coefficients for the vectors from the subsystem and zero coefficients for the other vectors is a nontrivial linear combination equal to zero. We obtain a contradiction with the linear independence of the original system. $\square$

Lemma 3. If a vector is a linear combination of linearly independent vectors, then the coefficients of this linear combination are uniquely determined.

Proof. Suppose the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent and

$$\mathbf{b} = \alpha_1\mathbf{a}_1 + ... + \alpha_k\mathbf{a}_k = \beta_1\mathbf{a}_1 + ... + \beta_k\mathbf{a}_k.$$

Hence

$$(\alpha_1 - \beta_1)\mathbf{a}_1 + ... + (\alpha_k - \beta_k)\mathbf{a}_k = \mathbf{0} \quad \Rightarrow \quad \alpha_1 - \beta_1 = ... = \alpha_k - \beta_k = 0. \ \square$$

Problem 1. For each n, find all values of the parameter a for which the columns of the tridiagonal matrix

$$\mathbf{A} = \begin{pmatrix} a & 1 & & & \\ -1 & a & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & a & 1 \\ & & & -1 & a \end{pmatrix}$$

of order n are linearly independent.

Problem 2. An $(n+1) \times n$ matrix has entries $a_{ij} > 0$ for $i = j$ and $a_{ij} < 0$ for $i \neq j$. Prove that for $n = 3$ its columns are linearly independent. Is this true for $n = 4$?

Transitivity of Linear Dependence

An important (though obvious) property: if

$$\operatorname{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\} \subseteq \operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\}$$
$$\text{and} \quad \operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\},$$

then

$$\operatorname{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}.$$

Monotonicity of the Number of Linearly Independent Vectors

Lemma 4. Suppose that each of the systems of vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ and $\mathbf{a}_1, ..., \mathbf{a}_k$ is linearly independent, and suppose that

$$\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \qquad (*)$$

Then $m \leq k$.

Proof. According to $(*)$, the system

$$\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_k$$

is linearly dependent. By Lemma 1, there exists a vector that is a linear combination of the preceding vectors; let this be the vector

$$\mathbf{a}_k \in \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\}.$$

From this it follows that

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\}.$$

By the transitivity of linear dependence,

$$\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\},$$

and therefore the system

$$\mathbf{b}_2, \mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}$$

is linearly dependent. By Lemma 1, in this system too there exists a vector that is linearly expressible in terms of the preceding ones, and this vector cannot be $\mathbf{b}_1$ (the vectors $\mathbf{b}_1, \mathbf{b}_2$ are linearly independent as a subsystem of a linearly independent system (Lemma 2)). Without loss of generality, we may assume that

$$\mathbf{a}_{k-1} \in \operatorname{span}\{\mathbf{b}_2, \mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-2}\}.$$

Suppose that $m > k$. Then, continuing the preceding constructions, at the k-th step we obtain

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \operatorname{span}\{\mathbf{b}_k, \mathbf{b}_{k-1}, ..., \mathbf{b}_1\}.$$

Consequently, $\mathbf{b}_{k+1} \in \operatorname{span}\{\mathbf{b}_k, \mathbf{b}_{k-1}, ..., \mathbf{b}_1\}$, which contradicts the assumption that the vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ are linearly independent. The resulting contradiction proves that $m \leq k$. □

Note that the lemma remains valid even without the assumption that the system $\mathbf{a}_1, ..., \mathbf{a}_k$ is linearly independent—prove this!

3.6. Basis and Coordinates

Basis and Dimension

A linearly independent set of vectors $\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is called a *basis* for the linear span V if $\text{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} = V$.

Theorem (Basis Theorem)**.** Any bases for a linear span V contain one and the same number of vectors.

Proof. Let $\{\mathbf{b}_1, ..., \mathbf{b}_m\}$ and $\{\mathbf{c}_1, ..., \mathbf{c}_r\}$ be two bases for the given linear span. Clearly,

$$\text{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} = \text{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\}.$$

Applying Lemma 4 (Section 3.5) twice, we obtain two inequalities: $m \leq r$ and $r \leq m$. Hence $m = r$. □

Definition. The number of vectors in the bases for a linear span V is called its *dimension* and is denoted by $\dim(V)$.

Theorem (Dimension of a Linear Span)**.**

$$\dim(\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}) \leq k.$$

Proof. It suffices to note that as a basis for the linear span of a given system of vectors one can choose a maximal linearly independent subsystem of it. □

As a basis for the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, one can always choose some subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$. A maximal linearly independent subsystem is called a *base* of the given system.

Proposition. For a subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ to be a basis for $V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, it is necessary and sufficient that it be a base.

Proof. Without loss of generality, we may assume that the subsystem is formed by the first k vectors $\mathbf{a}_1, ..., \mathbf{a}_k$. If this is a base, then each of the vectors $\mathbf{a}_{k+1}, ..., \mathbf{a}_n$ is linearly expressible in terms of $\mathbf{a}_1, ..., \mathbf{a}_k \Rightarrow V \subseteq \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq V \Rightarrow V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. Thus, $\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is a basis for V.

If the chosen subsystem is a basis for V, then by the preceding argument and the Basis Theorem, no base can have a larger number of vectors. □

Problem 1. The vectors $\mathbf{a}_1, ..., \mathbf{a}_{k+1}$ are linearly independent. Prove that in the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_{k+1}\}$ there exists a basis containing no vector from the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$.

Basis and Dimension of an Abstract Vector Space BEYOND MA1522

Let V be a finite-dimensional vector space. By definition, it is the linear span of a finite number of its vectors:

$$V = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}.$$

The concepts of linearly dependent and linearly independent systems of vectors in the abstract case do not differ in any way from the same concepts in the case of column vectors. The same holds with regard to basis and dimension:

- V can be represented as the linear span of some linearly independent subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$;
- a basis for the space V is defined as any linearly independent set of vectors for which V is the linear span; any two bases for V contain the same number of vectors; the number of vectors in a basis is called the dimension of the space V and is denoted by $\dim(V)$;
- any linearly independent set of vectors from V can be extended to a basis for V; moreover, this can be done using some of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$.

The proofs of these propositions repeat the proofs given in Sections 3.5–3.7 for the special case where vectors are understood to be column vectors.

Expansion in a Basis

Let V be a real vector space of dimension n and $T = \{\mathbf{f}_1, ..., \mathbf{f}_n\}$ be some basis for it. Then any vector $\mathbf{v} \in V$ has a unique expansion in this basis:

$$\mathbf{v} = x_1\mathbf{f}_1 + ... + x_n\mathbf{f}_n.$$

The coefficients $x_1, ..., x_n$ are called the coordinates of the vector $\mathbf{v}$ relative to the basis T. Clearly, between the elements of the vector space V and the set of columns $\mathbb{R}^n$ there is a one-to-one correspondence:

$$\mathbf{v} \leftrightarrow [\mathbf{v}]_T = \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Upon choosing another basis $S = \{\mathbf{g}_1, ..., \mathbf{g}_n\}$, there arises one more one-to-one correspondence between the same sets:

$$\mathbf{v} = y_1\mathbf{g}_1 + ... + y_n\mathbf{g}_n \leftrightarrow [\mathbf{v}]_S = \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

Consider the expansions

$$\begin{aligned} \mathbf{f}_1 &= p_{11}\mathbf{g}_1 + ... + p_{n1}\mathbf{g}_n, \\ &... \\ \mathbf{f}_n &= p_{1n}\mathbf{g}_1 + ... + p_{nn}\mathbf{g}_n \end{aligned} \qquad (*)$$

and introduce the $n \times n$ matrix $\mathbf{P} = \big(p_{ij}\big)$; its jth column is the coordinate vector $\big[\mathbf{f}_j\big]_S$, that is,

$$\mathbf{P} = \Big(\big[\mathbf{f}_1\big]_S \ \cdots \ \big[\mathbf{f}_n\big]_S\Big).$$

Substituting $(*)$ into the expansion of the vector $\mathbf{v}$ in the basis T, we find

$$[\mathbf{v}]_S = \mathbf{P}[\mathbf{v}]_T, \quad \text{i.e.} \quad \mathbf{y} = \mathbf{P}\mathbf{x}. \qquad (**)$$

This relation makes it possible to pass from the coordinates of a vector relative to the basis T to the coordinates of the same vector relative to the basis S.

By virtue of $(**)$, the matrix $\mathbf{P}$ is called the *transition matrix from T to S*. If $\mathbf{f}_i$ and $\mathbf{g}_i$ are the columns of coordinates of the corresponding vectors in some third basis, then according to $(*)$ we have

$$(\mathbf{f}_1 \ \cdots \ \mathbf{f}_n) = (\mathbf{g}_1 \ \cdots \ \mathbf{g}_n)\mathbf{P},$$

whence follows the invertibility of the matrix $\mathbf{P}$ and the fact that $\mathbf{P}^{-1}$ is the transition matrix from S to T. Since this equality expresses the vectors $\mathbf{f}_j$ through the vectors $\mathbf{g}_i$, some books (the full textbook among them) use the opposite name and call $\mathbf{P}$ the transition matrix from $\{\mathbf{g}_i\}$ to $\{\mathbf{f}_i\}$, that is, from S to T. However, the name is not what matters—what is important is that the matrix $\mathbf{P}$ be used properly when recalculating coordinates!

3.7. Dimensions

Completion to a Basis

Lemma (Basis Extension Lemma)**.** Every linearly independent set of vectors $\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is a subset of some basis for the given linear span.

Proof. It suffices to consider the case where the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent. The system of vectors $\mathbf{b}_1, ..., \mathbf{b}_m, \mathbf{a}_1, ..., \mathbf{a}_k$ is linearly dependent. By Lemma 1 (Section 3.5), there exists in it a vector that is linearly expressible in terms of the preceding ones. Let us remove this vector and consider the remaining subsystem. If it is linearly independent, then it is a basis for the linear span $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. If not, there is a vector in it that is linearly expressible in terms of the preceding ones. We remove it from the system as well, consider the remaining subsystem, and so on. Ultimately, the system of vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ will be completed to a basis by some of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$. □

Existence of a Basis

Does a basis exist for every linear span? By the Basis Extension Lemma, a basis exists if the linear span contains a linearly independent subsystem of vectors. This will be the case if there exists at least one nonzero vector.

Thus, there is no basis only in the case of the zero space $\{\mathbf{0}\}$, the linear span which contains a single vector—the zero vector. By definition, the dimension of the zero space is equal to zero. (MA1522 regards the empty set $\varnothing$ as the basis for the zero space; this agrees with $\dim(\{\mathbf{0}\}) = 0$.)

3.8. Transition Matrices

Expansion in a Basis is treated in Section 3.6.

Transformation of Coordinates

Suppose we have two affine coordinate systems: the first centered at the point O with basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and the second centered at the point O' with basis vectors $\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3$. Let us write

$$\begin{aligned}\mathbf{e}'_1 &= p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3,\\ \mathbf{e}'_2 &= p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3,\\ \mathbf{e}'_3 &= p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3\end{aligned}$$

and form the *transition matrix* from the basis $\{\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\}$ to the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ (named as in Section 3.6)

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & p_{13}\\ p_{21} & p_{22} & p_{23}\\ p_{31} & p_{32} & p_{33}\end{pmatrix}.$$

If there is some third system of basis vectors and the vectors $\mathbf{e}_i, \mathbf{e}'_i$ are regarded as columns of the coordinates of their expansions in this third system, then the equality

$$(\mathbf{e}'_1 \quad \mathbf{e}'_2 \quad \mathbf{e}'_3) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3)\mathbf{P}$$

holds. From this follows the invertibility of the matrix $\mathbf{P}$.

Suppose a point M has coordinates (x, y, z) in the first system and (x', y', z') in the second system. Obviously, $\overrightarrow{OM} = \overrightarrow{OO'} + \overrightarrow{O'M}$. Let the coordinates of the point O' in the first system be (x_0, y_0, z_0). Then

$$\begin{aligned}x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 &= (x_0\mathbf{e}_1 + y_0\mathbf{e}_2 + z_0\mathbf{e}_3) + x'(p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3) +\\ &+y'(p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3) + z'(p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3) =\\ &= (x_0 + p_{11}x' + p_{12}y' + p_{13}z')\mathbf{e}_1 +\\ &+(y_0 + p_{21}x' + p_{22}y' + p_{23}z')\mathbf{e}_2 +\\ &+(z_0 + p_{31}x' + p_{32}y' + p_{33}z')\mathbf{e}_3.\end{aligned}$$

Consequently, the coordinates of one and the same point in the first and second coordinate systems are related by the following formula:

$$\begin{pmatrix} x\\ y\\ z\end{pmatrix} = \begin{pmatrix} x_0\\ y_0\\ z_0\end{pmatrix} + \mathbf{P}\begin{pmatrix} x'\\ y'\\ z'\end{pmatrix}.$$

From this it is easy to understand, for example, how the general equations of one and the same plane in different affine coordinate systems are related. If in the first coordinate system we have the equation $Ax + By + Cz + D = 0$, then, writing it in matrix form, we find

$$(A \quad B \quad C)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = -D \quad \Leftrightarrow \quad (A \quad B \quad C)\left(\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + \mathbf{P}\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}\right) = -D.$$

Thus, the equation of the same plane in the second system takes the form

$$(A' \quad B' \quad C')\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = -D',$$

$$(A' \quad B' \quad C') = (A \quad B \quad C)\mathbf{P}, \quad D' = D + (Ax_0 + By_0 + Cz_0).$$

Chapter 4. Subspaces Associated to a Matrix

4.1. Column Space, Row Space, and Nullspace

Examples of Finite-Dimensional Vector Spaces

Vector spaces that are the span of some finite number of their vectors are called *finite-dimensional*.

1. *The set of polynomials of order n.*

   A polynomial in x of order n is an expression of the form

   $$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_1x + a_0.$$

   If $a_k \neq 0$ and $a_{k+1} = ... = a_{n-1} = 0$, then k is called the *degree* of the polynomial $f(x)$. Expressions of the form ax^i are called *monomials*.

   Let us regard $f(x)$ as a function of x. Then the sum of polynomials and the multiplication of a polynomial by a number are defined in the same way as in the case of general functions. In doing so, it is clear that the results of these operations remain polynomials. Obviously, the vector space of all polynomials of order n is the span of the monomials of the form

   $$x^{n-1}, x^{n-2}, ..., x^1, x^0 \equiv 1.$$

2. *The set of $m \times n$ matrices with fixed dimensions m and n.*

   In this case, vector addition is defined as matrix addition, and multiplication of a vector by a number as multiplication of a matrix by a number.

   Denote by $\mathbf{E}^{kl} = \left(\left(\mathbf{E}^{kl}\right)_{ij}\right)$ the $m \times n$ matrix with entries of the form

   $$\left(\mathbf{E}^{kl}\right)_{ij} = \begin{cases} 1, & i = k, j = l, \\ 0 & \text{otherwise.} \end{cases}$$

   There are exactly mn such matrices, and obviously the entire space of $m \times n$ matrices is their span.

3. *The set of all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$.*

   If the rank of the $m \times n$ matrix $\mathbf{A}$ equals r, then a basis for the solution space of this homogeneous system contains $n - r$ vectors, and the entire set of solutions coincides with their span. This vector space is called the *nullspace* of the matrix $\mathbf{A}$. Notation: $\text{Null}(\mathbf{A})$ (in some books $\ker \mathbf{A}$ or $\text{null } \mathbf{A}$). Its dimension is called the *nullity* of $\mathbf{A}$: $\text{nullity}(\mathbf{A}) = \dim(\text{Null}(\mathbf{A}))$.

4. *The set of all vectors of the form $\mathbf{y} = \mathbf{Ax}$ (for a given matrix $\mathbf{A}$).*

This is a set well familiar to us, coinciding with the span of the columns of the matrix $\mathbf{A}$. It is called the *column space* of the matrix $\mathbf{A}$ (in some books, the *image* of $\mathbf{A}$, im $\mathbf{A}$). Notation: Col($\mathbf{A}$).

The row space Row($\mathbf{A}$)*, the span of the rows of* $\mathbf{A}$*, is treated in Section 4.2.*

4.2. Rank

Separation of Variables and Matrices BEYOND MA1522

In the study of functions of two variables, a special role is played by functions with *separated variables*, $f(x,y) = u(x)v(y)$, or sums of such functions:[5]

$$f(x,y) = u_1(x)v_1(y) + ... + u_r(x)v_r(y).$$

Suppose we are given an $m \times n$ matrix $\mathbf{A}$. Its entry a_{ij} can be regarded as a function of the discrete variables $i \in \{1,...,m\}$, $j \in \{1,...,n\}$. In this case, separation of variables means that

$$a_{ij} = u_i v_j, \quad 1 \le i \le m, \quad 1 \le j \le n.$$

From this it is easy to deduce that $\mathbf{A}$ is the product of a column and a row:[6]

$$\mathbf{A} = \mathbf{u}\mathbf{v}^T, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

Skeleton Decomposition BEYOND MA1522

Now suppose that $a_{ij} = u_{i1}v_{j1} + ... + u_{ir}v_{jr}$, $1 \le i \le m$, $1 \le j \le n$. In this case $\mathbf{A}$ is a sum of r matrices of the "column-by-row" form:

$$\mathbf{A} = \sum_{k=1}^{r} \mathbf{u}_k \mathbf{v}_k^T, \quad \mathbf{u}_k = \begin{pmatrix} u_{1k} \\ \vdots \\ u_{mk} \end{pmatrix}, \quad \mathbf{v}_k = \begin{pmatrix} v_{1k} \\ \vdots \\ v_{nk} \end{pmatrix}.$$

This same equality, written in the form of a product of two matrices:

$$\mathbf{A} = \mathbf{U}\mathbf{V}^T = \begin{pmatrix} u_{11} & ... & u_{1r} \\ \vdots & \vdots & \vdots \\ u_{m1} & ... & u_{mr} \end{pmatrix} \begin{pmatrix} v_{11} & ... & v_{n1} \\ \vdots & \vdots & \vdots \\ v_{1r} & ... & v_{nr} \end{pmatrix}, \tag{1}$$

$$\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_r), \quad \mathbf{V} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_r),$$

is called a *skeleton decomposition* of the matrix $\mathbf{A}$. It means that every column of the matrix $\mathbf{A}$ is a linear combination of the columns of the matrix $\mathbf{U}$, and every row of the matrix $\mathbf{A}$ is a linear combination of the rows of the matrix $\mathbf{V}^T$. From this there immediately follows

[5] Separation of variables is applied with great success to the approximation of functions of general form.

[6] The matrix $\mathbf{u}\mathbf{v}^T$ is sometimes called the *outer product* $\mathbf{u} \otimes \mathbf{v} = \mathbf{u}\mathbf{v}^T$ of the vectors $\mathbf{u} \in \mathbb{C}^m$ and $\mathbf{v} \in \mathbb{C}^n$.

Theorem. The dimension of the column space of a matrix $\mathbf{A}$ coincides with the dimension of its *row space* $\mathrm{Row}(\mathbf{A})$, the span of its rows:

$$\dim(\mathrm{Col}(\mathbf{A})) = \dim(\mathrm{Row}(\mathbf{A})),$$

$$\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}, \quad \mathrm{Row}(\mathbf{A}) = \mathrm{span}\{\hat{\mathbf{a}}_1^T, ..., \hat{\mathbf{a}}_m^T\},$$

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{a}}_1^T \\ \vdots \\ \hat{\mathbf{a}}_m^T \end{pmatrix}.$$

Proof. Let the columns of the matrix $\mathbf{U}$ form a basis for the column space $\mathrm{Col}(\mathbf{A})$, and let the j-th column of the matrix $\mathbf{V}^T$ consist of the coefficients of their linear combination yielding the column $\mathbf{a}_j$. Then, obviously, $\mathbf{A} = \mathbf{U}\mathbf{V}^T$. By virtue of the remark preceding the theorem, the dimension of the row space of the matrix $\mathbf{A}$ does not exceed the number of rows of the matrix $\mathbf{V}^T$, which by construction equals the dimension of the column space of the matrix $\mathbf{A}$. The opposite inequality is proved similarly—the roles of columns and rows are interchanged by transposition. □

Matrix Rank

The dimension of the column space $\mathrm{Col}(\mathbf{A})$ (row space $\mathrm{Row}(\mathbf{A})$) of a matrix $\mathbf{A}$ is sometimes called its column (row) rank. Since the column and row ranks coincide, it is natural to call their common value simply the *rank* of the matrix; this is the definition used in MA1522:

$$\mathrm{rank}(\mathbf{A}) = \dim(\mathrm{Col}(\mathbf{A})) = \dim(\mathrm{Row}(\mathbf{A})).$$

From our proof of this fact it also follows that this value is equal to the smallest number of matrices of the "column-by-row" form whose sum yields the given matrix.

Nevertheless, another definition is often given: the *rank* of a matrix is the largest k for which the matrix has a nonzero $k \times k$ minor. Here and throughout this section, a $k \times k$ *minor* (or *minor of order* k) means the determinant of a $k \times k$ submatrix; it is a number, not to be confused with the (i, j) matrix minor $\mathbf{M}_{ij}$ of Section 2.8, which is a submatrix. A nonzero minor of this largest order and the corresponding submatrix are called a *basis minor* and a *basis submatrix*. Since a square matrix is invertible if and only if its determinant is nonzero (Section 2.6), the rank in this sense is equal to the largest order of an invertible square submatrix of the given matrix. Notation: $\mathrm{rank}(\mathbf{A})$.

Two obvious properties of the rank of an $m \times n$ matrix $\mathbf{A}$:

$$\mathrm{rank}(\mathbf{A}) \le \min\{m, n\}, \quad \mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^T).$$

It is less obvious that: *the largest order of a nonzero minor of a matrix coincides with its column and row rank.* Let us prove this.

Bordering of an Invertible Submatrix BEYOND MA1522

We begin with a useful auxiliary proposition. Suppose a square matrix $\mathbf{Q}$ of order $k+1$ has the block form:

$$\mathbf{Q} = \begin{pmatrix} \mathbf{P} & \mathbf{v} \\ \mathbf{u}^T & c \end{pmatrix}, \quad \mathbf{P} \in \mathbb{R}^{k\times k}, \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^{k\times 1}.$$

In this case $\mathbf{Q}$ is called a *bordering of the submatrix* $\mathbf{P}$.

Lemma (on noninvertible bordering)**.** If the submatrix $\mathbf{P}$ is invertible, while its bordering $\mathbf{Q}$ is a noninvertible matrix, then the last column of the matrix $\mathbf{Q}$ is a linear combination of the first k columns.

Proof. Using the rule for multiplying block matrices (see Section 2.3), it is easy to verify the validity of the equality

$$\begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{u}^T\mathbf{P}^{-1} & 1 \end{pmatrix} \begin{pmatrix} \mathbf{P} & \mathbf{v} \\ \mathbf{u}^T & c \end{pmatrix} = \begin{pmatrix} \mathbf{P} & \mathbf{v} \\ \mathbf{0} & \gamma \end{pmatrix}, \quad \gamma = c - \mathbf{u}^T\mathbf{P}^{-1}\mathbf{v}.$$

Denote the matrix on the right-hand side by $\mathbf{M}$. As the product of an invertible and a noninvertible matrix, $\mathbf{M}$ cannot be an invertible matrix. Yet it has block-triangular form, and if the blocks $\mathbf{P}$ and γ were both invertible, then $\mathbf{M}$ would have an inverse matrix of the form

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{P}^{-1} & -\mathbf{P}^{-1}\mathbf{v}\gamma^{-1} \\ \mathbf{0} & \gamma^{-1} \end{pmatrix}.$$

(The equality $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$ is verified directly.) Since $\mathbf{M}$ is not an invertible matrix, we necessarily have

$$\gamma = c - \mathbf{u}^T\mathbf{P}^{-1}\mathbf{v} = 0 \quad \Rightarrow \quad c = \mathbf{u}^T\mathbf{P}^{-1}\mathbf{v}.$$

Consequently,

$$\begin{pmatrix} \mathbf{P} \\ \mathbf{u}^T \end{pmatrix} (\mathbf{P}^{-1}\mathbf{v}) = \begin{pmatrix} \mathbf{v} \\ c \end{pmatrix}. \qquad \square$$

Basis Minor Theorem

Theorem. The columns (rows) containing a basis minor are linearly independent, and any column (any row) of the given matrix is a linear combination of them.

Proof. Without loss of generality, suppose that the basis submatrix $\mathbf{P}$ of order k is located in the upper left corner of the $m \times n$ matrix $\mathbf{A}$. Thus,

$$\mathbf{A} = \begin{pmatrix} \mathbf{P} & \mathbf{v}_{k+1} & \dots & \mathbf{v}_n \\ \mathbf{u}_{k+1}^T & a_{k+1,k+1} & \dots & a_{k+1,n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_m^T & a_{m,k+1} & \dots & a_{mn} \end{pmatrix},$$

$$\mathbf{u}_{k+1}, \dots, \mathbf{u}_m, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n \in \mathbb{R}^{k\times 1}.$$

By the hypothesis of the theorem, any submatrix of order $k+1$ of the form

$$\mathbf{M} = \begin{pmatrix} \mathbf{P} & \mathbf{v}_j \\ \mathbf{u}_i^T & a_{ij} \end{pmatrix}, \quad i, j > k,$$

is noninvertible. By the lemma on noninvertible bordering of an invertible submatrix, the last column in $\mathbf{M}$ is a linear combination of the first k columns. Moreover, the coefficients of this linear combination do not depend on i (since they are determined by the vector $\mathbf{P}^{-1}\mathbf{v}_j$). Hence, the j-th column of the matrix $\mathbf{A}$ for $j > k$ is a linear combination of the first k columns. The linear independence of the first k columns is proved as follows: suppose their linear combination with coefficients $\alpha_1, \dots, \alpha_k$ is equal to $\mathbf{0}$; then

$$\mathbf{P}\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{pmatrix} = \mathbf{0}.$$

The assertion of the theorem concerning the rows is proved by passing to the transposed matrix. □

Corollary 1. A matrix has rank r if and only if some $r \times r$ minor is nonzero, while all $(r+1) \times (r+1)$ minors bordering it are equal to zero.

Corollary 2. The largest order of a nonzero minor of a matrix $\mathbf{A}$ coincides with its column and row rank, that is, with $\operatorname{rank}(\mathbf{A}) = \dim(\operatorname{Col}(\mathbf{A})) = \dim(\operatorname{Row}(\mathbf{A}))$.

Remark. The basis minor theorem does not use the previously proved theorem on the equality of the column and row ranks. In essence, it provides yet another proof of this fact.

Thus the columns containing a basis minor form a basis for $\operatorname{Col}(\mathbf{A})$, just as the columns of $\mathbf{A}$ corresponding to the pivot columns of its reduced row-echelon form do; likewise, the rows containing a basis minor form a basis for $\operatorname{Row}(\mathbf{A})$.

Problem 1. Let $\mathbf{A}$ be an $n \times n$ matrix of rank k, and $\mathbf{B}$ any invertible $k \times k$ submatrix. Denote by $\mathbf{R}$ the $k \times n$ submatrix consisting of the rows of the matrix $\mathbf{A}$ containing the submatrix $\mathbf{B}$, and by $\mathbf{C}$ the $n \times k$ submatrix consisting of the columns containing $\mathbf{B}$. Prove that

$$\mathbf{A} = \mathbf{C}\mathbf{B}^{-1}\mathbf{R}.$$

Problem 2. Prove that the submatrix located at the intersection of r linearly independent rows and r linearly independent columns of a matrix of rank r is invertible.

Problem 3. It is known that $\mathbf{A}^T = -\mathbf{A}$. Prove that the rank of the matrix $\mathbf{A}$ is an even number.

Properties of Rank

Proposition 1. The rank of a sum of matrices does not exceed the sum of their ranks:

$$\operatorname{rank}(\mathbf{A}+\mathbf{B}) \le \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}).$$

Proof. Obviously, $\mathbf{A}$ and $\mathbf{B}$ must have the same number of columns:

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{pmatrix}.$$

Clearly, $\operatorname{span}\{\mathbf{a}_1+\mathbf{b}_1, ..., \mathbf{a}_n+\mathbf{b}_n\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}$. In the smaller span, let us choose some set of vectors forming a basis. According to the Basis Extension Lemma (Section 3.7), a basis for the larger span can be obtained by extending this set by some vectors from the larger span. Therefore,

$$\operatorname{rank}(\mathbf{A}+\mathbf{B}) = \dim(\operatorname{span}\{\mathbf{a}_1+\mathbf{b}_1, ..., \mathbf{a}_n+\mathbf{b}_n\}) \le$$
$$\le \dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}).$$

Let $p = \operatorname{rank}(\mathbf{A})$, $q = \operatorname{rank}(\mathbf{B})$, and suppose without loss of generality that a basis for $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$ is formed by the first p vectors, and a basis for $\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_n\}$ by the first q vectors. Then $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\} = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_p, \mathbf{b}_1, ..., \mathbf{b}_q\} \Rightarrow$

$$\dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}) \le p+q. \qquad \square$$

Proposition 2. The rank of a product of matrices does not exceed the rank of each of the factors:

$$\operatorname{rank}(\mathbf{AB}) \le \min\{\operatorname{rank}(\mathbf{A}), \operatorname{rank}(\mathbf{B})\}.$$

Proof. It suffices to note that each of the columns of the matrix $\mathbf{AB}$ is a linear combination of the columns of the matrix $\mathbf{A}$. Therefore, the column space of the matrix $\mathbf{AB}$ is contained in the column space of the matrix $\mathbf{A}$: $\operatorname{Col}(\mathbf{AB}) \subseteq \operatorname{Col}(\mathbf{A})$. Consequently, $\operatorname{rank}(\mathbf{AB}) \le \operatorname{rank}(\mathbf{A})$. Furthermore,

$$\operatorname{rank}(\mathbf{AB}) = \operatorname{rank}((\mathbf{AB})^T) = \operatorname{rank}(\mathbf{B}^T\mathbf{A}^T) \le \operatorname{rank}(\mathbf{B}^T) = \operatorname{rank}(\mathbf{B}). \qquad \square$$

Proposition 3. The rank of a matrix does not change when it is multiplied on the left or on the right by an invertible matrix.

Proof. Let $\mathbf{B} = \mathbf{PAQ}$, where $\mathbf{P}$ and $\mathbf{Q}$ are invertible matrices.[7] By virtue of the preceding proposition, $\operatorname{rank}(\mathbf{B}) \leq \operatorname{rank}(\mathbf{A})$. At the same time, $\mathbf{A} = \mathbf{P}^{-1}\mathbf{BQ}^{-1} \Rightarrow \operatorname{rank}(\mathbf{A}) \leq \operatorname{rank}(\mathbf{B})$. □

This proposition is useful when computing the rank. Usually this is done with the aid of *elementary row and column operations*, simplifying the form of the matrix by eliminating its entries; these operations reduce to multiplying the matrix on the left and on the right by certain invertible matrices (elementary matrices) and therefore preserve the rank.

Proposition 4. When k rows (columns) of a matrix are changed, its rank cannot change by more than k.

Proof. It suffices to consider the case $k = 1$. Let the rank be equal to r, and let the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}$ be basis columns. If the column $\mathbf{a}_j$ coincides with one of these columns, then under any change to it, this set of columns will have a linearly independent subset of $r - 1$ columns. Therefore, the rank of the new matrix is no less than $r - 1$ and, of course, no greater than $r + 1$. If the column $\mathbf{a}_j$ does not coincide with any of the chosen basis columns, then every column of the new matrix is a linear combination of the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}, \mathbf{a}_j$, so its rank is no greater than $r + 1$ and no less than r, since there is a set of r linearly independent columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}$. □

Corollary. When k columns are added (or removed), the rank of a matrix cannot change by more than k.

Problem 4. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of rank 1. Prove that if $\mathbf{AB} \neq \mathbf{0}$, then the rank of the matrix $\mathbf{A} + \mathbf{B}$ is at most 1.

Problem 5. Given vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, with $\mathbf{x} \neq \mathbf{0}$. Prove that there exists a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that $\operatorname{rank}(\mathbf{A}) \leq 2$ and $\mathbf{Ax} = \mathbf{y}$.

Problem 6. A matrix $\mathbf{A}$ has r columns, and a matrix $\mathbf{B}$ has r rows. Prove that $r \geq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}) - \operatorname{rank}(\mathbf{AB})$.

Problem 7. Prove that any matrix of rank r with nonnegative entries can be represented as the difference of matrices with nonnegative entries and rank not exceeding 2^r.

Homogeneous Linear Systems

A linear system with a zero right-hand side

$$\mathbf{Ax} = \mathbf{0} \qquad (*)$$

is called *homogeneous*. Suppose that the given system has m equations and n variables. Then the coefficient matrix $\mathbf{A}$ has size $m \times n$. Let us write $\mathbf{A}$

[7] Of course, the order of $\mathbf{P}$ is equal to the number of rows, and the order of $\mathbf{Q}$ to the number of columns of the matrix $\mathbf{A}$.

by its columns, $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$, and suppose that its rank is equal to r. Without loss of generality, we shall assume that a basis submatrix in $\mathbf{A}$ is located in the first r columns—we shall call them *basis* columns. The components of the solution $x_1, ..., x_r$ corresponding to the basis columns will also be called *basis* (or *basic*) variables, and the remaining components $x_{r+1}, ..., x_n$ will be called *free* variables. (Under this assumption the basis columns are exactly the columns of $\mathbf{A}$ corresponding to the pivot columns of its reduced row-echelon form, and the free variables are those to which parameters are assigned in the general solution.) Thus, the solution vector has the form

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_r \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix}.$$

The system $(*)$ is equivalent to the equality

$$x_1\mathbf{a}_1 + ... + x_r\mathbf{a}_r = -x_{r+1}\mathbf{a}_{r+1} - ... - x_n\mathbf{a}_n. \qquad (**)$$

By the basis minor theorem, the columns $\mathbf{a}_{r+1}, ..., \mathbf{a}_n$ belong to $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_r\}$. Therefore, for any choice of values of the free variables, the values of the basis variables satisfying equality $(**)$ exist and are uniquely determined. Thus, there exist $n - r$ vectors of the form

$$\mathbf{v}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{r1} \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} x_{12} \\ \vdots \\ x_{r2} \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad ..., \quad \mathbf{v}_{n-r} = \begin{pmatrix} x_{1,n-r} \\ \vdots \\ x_{r,n-r} \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

each of which is a solution of the system $(*)$:

$$\mathbf{A}\mathbf{v}_1 = \mathbf{0}, \quad ..., \quad \mathbf{A}\mathbf{v}_{n-r} = \mathbf{0}.$$

The vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ are linearly independent:

$$\alpha_1\mathbf{v}_1 + ... + \alpha_{n-r}\mathbf{v}_{n-r} = \begin{pmatrix} * \\ \vdots \\ * \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{n-r} \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \alpha_1 = ... = \alpha_{n-r} = 0.$$

Moreover, any linear combination of these vectors is a solution of the system $(*)$ and, furthermore, if $\mathbf{x}$ is an arbitrary solution of the system $(*)$, then

$$\mathbf{x} = x_{r+1}\mathbf{v}_1 + ... + x_n\mathbf{v}_{n-r}.$$

Thus, we have proved the following important statement.

Theorem. The set of solutions of the homogeneous linear system $(*)$ coincides with $\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}$, the span of the linearly independent vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$; that is, $\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}$ is a basis for the nullspace $\text{Null}(\mathbf{A})$.

Corollary (Rank–Nullity Theorem)**.** For the $m \times n$ matrix $\mathbf{A}$ of rank r, $\text{nullity}(\mathbf{A}) = \dim(\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}) = n - r$; in other words,

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

A linearly independent set of solutions $\{\mathbf{w}_1, ..., \mathbf{w}_k\}$ of the system $\mathbf{Ax} = \mathbf{0}$ is called a *basis for the solution space* (classically, a *fundamental system of solutions*) if $\text{span}\{\mathbf{w}_1, ..., \mathbf{w}_k\}$ coincides with the set of all solutions of the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Corollary. The number of vectors in any basis for the solution space of $\mathbf{Ax} = \mathbf{0}$ is equal to $n - r$, where $r = \text{rank}(\mathbf{A})$.

For the proof, it suffices to note that the span of the vectors of such a basis also has a basis consisting of the vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ constructed above.

Problem 8. Given square matrices $\mathbf{A}$ and $\mathbf{B}$ of order n such that $\mathbf{AB} = \mathbf{0}$ and the matrix $\mathbf{A} + \mathbf{B}$ is invertible. Prove that $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) = n$.

Rank and Consistency (Kronecker–Capelli Theorem)
Consider a linear system $\mathbf{Ax} = \mathbf{b}$ with an $m \times n$ matrix $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$. The matrix $(\mathbf{A} \mid \mathbf{b}) = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n \mid \mathbf{b})$ is called the *augmented matrix* of the given system.

Theorem. The linear system $\mathbf{Ax} = \mathbf{b}$ is consistent if and only if the rank of the coefficient matrix coincides with the rank of the augmented matrix:

$$\text{rank}(\mathbf{A}) = \text{rank}((\mathbf{A} \mid \mathbf{b})).$$

Proof. We already know (see Section 1.5) that the consistency of the system $\mathbf{Ax} = \mathbf{b}$ is equivalent to the equality of the spans $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\} = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\}$. It remains to note that $\text{rank}(\mathbf{A}) = \dim(\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\})$ and $\text{rank}((\mathbf{A} \mid \mathbf{b})) = \dim(\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\})$. □

General Solution of a Linear System

If U and V are two sets of vectors in $\mathbb{R}^n$, then the sum $U + V$ is defined as the set consisting of all possible sums of vectors of the form $\mathbf{u} + \mathbf{v}$, where $\mathbf{u} \in U$, $\mathbf{v} \in V$.

Theorem. Suppose that the system $\mathbf{Ax} = \mathbf{b}$ is consistent, and fix an arbitrary particular solution $\mathbf{u}$ ($\mathbf{Au} = \mathbf{b}$). Then the set of all solutions of the system $\mathbf{Ax} = \mathbf{b}$ has the form $\mathbf{u} + V$, where $V = \text{Null}(\mathbf{A})$ is the set of all solutions of the corresponding homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Proof. Let $\mathbf{x}$ be an arbitrary solution of the system $\mathbf{Ax} = \mathbf{b}$. Then, obviously, $\mathbf{A}(\mathbf{x} - \mathbf{u}) = \mathbf{0} \Rightarrow \mathbf{x} - \mathbf{u} \in V \Rightarrow \mathbf{x} \in \mathbf{u} + V$. Next, take an arbitrary vector $\mathbf{x} \in \mathbf{u} + V \Rightarrow \mathbf{x} = \mathbf{u} + \mathbf{v}$, $\mathbf{v} \in V \Rightarrow \mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{Au} + \mathbf{Av} = \mathbf{b} + \mathbf{0} = \mathbf{b}$. □

Corollary. The general solution of a consistent system $\mathbf{Ax} = \mathbf{b}$ has the form

$$\mathbf{x} = \mathbf{u} + c_1\mathbf{v}_1 + ... + c_{n-r}\mathbf{v}_{n-r},$$

where $\mathbf{u}$ is an arbitrary particular solution of the given system, $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ are linearly independent solutions of the corresponding homogeneous system (a basis for $\text{Null}(\mathbf{A})$), $r = \text{rank}(\mathbf{A})$, and the coefficients $c_1, ..., c_{n-r}$ are arbitrary real numbers (the parameters of the general solution).

Problem 9. What can be said about the matrix and right-hand side of a system $\mathbf{Ax} = \mathbf{b}$ with respect to a vector $\mathbf{x} \in \mathbb{R}^n$ if every vector in $\mathbb{R}^n$ is a solution of it?

Instability of Rank BEYOND MA1522

An $m \times n$ matrix $\mathbf{A}$ is said to be of *full rank* if its rank coincides with one of its dimensions, i.e., has the maximum possible value $\text{rank}(\mathbf{A}) = \min\{m, n\}$. Otherwise, one speaks of a matrix of *deficient rank* (or *rank-deficient matrix*).

It can be proved that if $\mathbf{A}$ is an $m \times n$ matrix of full rank, then for all sufficiently small $\varepsilon > 0$ the matrix $\mathbf{A} + \mathbf{F}$, where all entries of the perturbation matrix $\mathbf{F}$ are less than ε in modulus, will also be a matrix of full rank. For this it suffices to note that the determinant of any square submatrix (any minor) can be regarded as a function of its entries, and this function is continuous. If a continuous function is nonzero at some values of the variables, then it will also be nonzero in some sufficiently small neighborhood of those values.

At the same time, if $\mathbf{A}$ has deficient rank, then for any arbitrarily small ε there exists a perturbation matrix $\mathbf{F}$ with entries no greater than ε in modulus such that $\mathbf{A} + \mathbf{F}$ is a matrix of full rank. For example, the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

has rank 2, whereas for any $\varepsilon \neq 0$ the matrix

$$\mathbf{A} + \mathbf{F} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \varepsilon & 0 \end{pmatrix}$$

obviously has rank 3.

Chapter 5. Orthogonality, Projection, and Least Squares

5.1. Orthogonality

Orthogonality of Vectors

An inner product makes it possible to introduce the general notion of orthogonality of vectors: $\mathbf{x}$ and $\mathbf{y}$ are called *orthogonal* if $\mathbf{x} \cdot \mathbf{y} = 0$. Notation: $\mathbf{x} \perp \mathbf{y}$.

In Sections 5.1–5.3 the inner product is written as the dot product $\mathbf{x} \cdot \mathbf{y}$ of vectors in $\mathbb{R}^n$. Unless the text says otherwise, the definitions and results hold verbatim in any space with an inner product $\langle \mathbf{x}, \mathbf{y} \rangle$, real (a *Euclidean* space, see Section 3.2) or complex (a *unitary* space), once $\mathbf{x} \cdot \mathbf{y}$ is replaced by $\langle \mathbf{x}, \mathbf{y} \rangle$ and $\mathbb{R}^n$ by that space.

Note that in one and the same space an inner product can be introduced in many different ways (for $\mathbb{R}^n$, see Section 3.2), and vectors that are orthogonal with respect to one inner product may fail to be orthogonal with respect to another inner product.

In a Euclidean space, one can also introduce the general notion of the *angle* $\theta = \theta(\mathbf{x}, \mathbf{y})$ between vectors $\mathbf{x}$ and $\mathbf{y}$. By definition,

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

It should be noted that the right-hand side does not exceed 1 in modulus (by virtue of the Cauchy–Bunyakovsky–Schwarz inequality). For orthogonal vectors, $\theta = \frac{\pi}{2}$. For an obvious reason, this definition of angle *does not carry over* to the case of unitary spaces. However, the notion of orthogonality works, of course, there as well.

> **Theorem** (Pythagorean Theorem)**.** If $\mathbf{x} \perp \mathbf{y}$, then $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$.

Proof. Let $\mathbf{x} \cdot \mathbf{y} = 0$. Then $(\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} = \mathbf{x} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y}$. □

> **Remark.** In a Euclidean (but not in a unitary!) space, the Pythagorean theorem can be inverted: if $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$, then $\mathbf{x} \cdot \mathbf{y} = 0$—this is obvious since $\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} = 2\mathbf{x} \cdot \mathbf{y}$. However, in a unitary space the equality $\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle = 2\langle \mathbf{x}, \mathbf{y} \rangle$ is not true for arbitrary vectors.

Orthogonality of Sets

Let $L \subset \mathbb{R}^n$ be an arbitrary nonempty subset of vectors. A vector $\mathbf{x} \in \mathbb{R}^n$ is said to be *orthogonal* to the set L if $\mathbf{x} \cdot \mathbf{y} = 0$ for all $\mathbf{y} \in L$. Notation: $\mathbf{x} \perp L$. By definition, sets L and M are *orthogonal* if $\mathbf{x} \cdot \mathbf{y} = 0$ for any $\mathbf{x} \in L$ and $\mathbf{y} \in M$. Notation: $L \perp M$.

The set M of all vectors in $\mathbb{R}^n$, each of which is orthogonal to a given set L, is called its *orthogonal complement* in $\mathbb{R}^n$. Notation: $M = L^\perp$.

Proposition. For any set L, its orthogonal complement $L^\perp$ is a subspace. Moreover, $L \subset (L^\perp)^\perp$.

Proof. Let $\mathbf{x}, \mathbf{y} \in L^\perp$. Then $\mathbf{x} \cdot \mathbf{z} = \mathbf{y} \cdot \mathbf{z} = 0 \quad \forall \mathbf{z} \in L \Rightarrow (\alpha\mathbf{x} + \beta\mathbf{y}) \cdot \mathbf{z} = \alpha(\mathbf{x} \cdot \mathbf{z}) + \beta(\mathbf{y} \cdot \mathbf{z}) = 0 \quad \forall \mathbf{z} \in L \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in L^\perp$.

By definition, the set $(L^\perp)^\perp$ contains all vectors orthogonal to $L^\perp$, and hence all vectors from the set L. □

Orthogonal Sum of Subspaces

Recall that the *sum of subspaces* $L_1, L_2, ..., L_m$ is the set L of all vectors of the form $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 + ... + \mathbf{x}_m$, where $\mathbf{x}_i \in L_i$ for all i (see Section 3.4). It is elementary to verify that L is a subspace. Notation: $L = L_1 + ... + L_m$.

Recall also that L is called a *direct sum* (see Tyrtyshnikov, §12.4) if the subspaces L_i are nonzero and each vector $\mathbf{x} \in L$ has a unique decomposition of the form $\mathbf{x} = \mathbf{x}_1 + ... + \mathbf{x}_m$, where $\mathbf{x}_i \in L_i$ (if $\mathbf{x} = \mathbf{x}'_1 + ... + \mathbf{x}'_m$ and $\mathbf{x}'_i \in L_i \quad \forall i$, then necessarily $\mathbf{x}'_i = \mathbf{x}_i \quad \forall i$).

A sum $L = L_1 + ... + L_m$ of nonzero subspaces is called an *orthogonal sum* if $L_i \perp L_j$ for $i \neq j$. Notation: $L = L_1 \oplus ... \oplus L_m$.

Proposition. An orthogonal sum of subspaces $L = L_1 \oplus ... \oplus L_m$ is a direct sum. Furthermore, if $\mathbf{x}_i \in L_i$, then

$$\|\mathbf{x}_1 + ... + \mathbf{x}_m\|^2 = \|\mathbf{x}_1\|^2 + ... + \|\mathbf{x}_m\|^2. \qquad (*)$$

Proof. Let us first prove $(*)$. Taking into account that $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$, we find

$$\|\mathbf{x}_1 + ... + \mathbf{x}_m\|^2 = \sum_{i=1}^m \sum_{j=1}^m \mathbf{x}_i \cdot \mathbf{x}_j = \sum_{i=1}^m \mathbf{x}_i \cdot \mathbf{x}_i = \sum_{i=1}^m \|\mathbf{x}_i\|^2.$$

Next, suppose $\mathbf{x}_1 + ... + \mathbf{x}_m = \mathbf{x}'_1 + ... + \mathbf{x}'_m$, where $\mathbf{x}_i, \mathbf{x}'_i \in L_i \quad \forall i$. Then

$$0 = \|(\mathbf{x}_1 - \mathbf{x}'_1) + ... + (\mathbf{x}_m - \mathbf{x}'_m)\|^2 = \|\mathbf{x}_1 - \mathbf{x}'_1\|^2 + ... + \|\mathbf{x}_m - \mathbf{x}'_m\|^2$$
$$\Rightarrow \quad \mathbf{x}_i = \mathbf{x}'_i \quad \forall i.$$ □

Corollary 1. A finite set of nonzero pairwise orthogonal vectors is linearly independent.

Proof. Suppose the vectors $\mathbf{x}_1, ..., \mathbf{x}_m$ are pairwise orthogonal and nonzero. Then the sum of the spans $\text{span}\{\mathbf{x}_1\}, ..., \text{span}\{\mathbf{x}_m\}$ is an orthogonal sum and, if $\alpha_1\mathbf{x}_1 + ... + \alpha_m\mathbf{x}_m = \mathbf{0}$, then according to $(*)$

$$0 = \|\alpha_1\mathbf{x}_1 + ... + \alpha_m\mathbf{x}_m\|^2 = |\alpha_1|^2\|\mathbf{x}_1\|^2 + ... + |\alpha_m|^2\|\mathbf{x}_m\|^2$$
$$\Rightarrow \quad \alpha_1 = ... = \alpha_m = 0. \qquad \square$$

Corollary 2. If nonzero subspaces $L_1, ..., L_m$ are finite-dimensional and pairwise orthogonal, then

$$\dim(L_1 \oplus ... \oplus L_m) = \dim(L_1) + ... + \dim(L_m).$$

It suffices to recall that for a direct sum of finite-dimensional subspaces L_i, a basis is obtained as the union of bases for the subspaces L_i (see Tyrtyshnikov, §12.4).

5.2. Orthogonal and Orthonormal Bases

Orthogonal and Orthonormal Sets

A set $S = \{\mathbf{x}_1, ..., \mathbf{x}_n\}$ of nonzero vectors is called *orthogonal* if

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0, \quad i \neq j, \qquad (*)$$

and *orthonormal* if, in addition, $\|\mathbf{x}_1\| = ... = \|\mathbf{x}_n\| = 1$.[8] Thus, the Gram matrix (see Section 5.3) for an orthogonal set is diagonal with positive diagonal entries, and for an orthonormal set it is the identity matrix.

Consider the space $\mathbb{C}^n$ with the natural inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^*\mathbf{x}$ (real case: $\mathbb{R}^n$ with the dot product, and $\mathbf{X}^T$ in place of $\mathbf{X}^*$ below) and an orthonormal set of column vectors:

$$\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix}, \quad ..., \quad \mathbf{x}_n = \begin{pmatrix} x_{1n} \\ \vdots \\ x_{nn} \end{pmatrix} \in \mathbb{C}^n.$$

Let us form from them the $n \times n$ matrix

$$\mathbf{X} = (\mathbf{x}_1 \ \cdots \ \mathbf{x}_n) = \begin{pmatrix} x_{11} & ... & x_{1n} \\ ... & ... & ... \\ x_{n1} & ... & x_{nn} \end{pmatrix}$$

and note that the relations $(*)$ (together with normalization) are equivalent to the matrix equality

$$\mathbf{X}^*\mathbf{X} = \begin{pmatrix} \langle \mathbf{x}_1, \mathbf{x}_1 \rangle & ... & \langle \mathbf{x}_n, \mathbf{x}_1 \rangle \\ ... & ... & ... \\ \langle \mathbf{x}_1, \mathbf{x}_n \rangle & ... & \langle \mathbf{x}_n, \mathbf{x}_n \rangle \end{pmatrix} = \mathbf{I}.$$

[8] In MA1522 an orthogonal set may also contain the zero vector; here, as in the definition above, the vectors of an orthogonal set are nonzero.

A matrix $\mathbf{X} \in \mathbb{C}^{n\times n}$ with the property $\mathbf{X}^*\mathbf{X} = \mathbf{I}$ is called *unitary*. Thus, any square matrix whose columns form an orthonormal set is unitary, and the columns of any unitary matrix form an orthonormal set. It is also clear that for a matrix to be unitary it is necessary and sufficient that its rows form an orthonormal set (prove this!).

A real unitary matrix is called *orthogonal*. In the next subsection we note that the set of all orthogonal matrices of order n is a group under the operation of matrix multiplication. The same holds with respect to the set of all unitary matrices of order n.

Cartesian Systems and Orthogonal Matrices

Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $\tilde{\mathbf{e}}_1, \tilde{\mathbf{e}}_2, \tilde{\mathbf{e}}_3$ be the basis vectors of two Cartesian coordinate systems with a common origin. Let us express the vectors of the second system as linear combinations of the vectors of the first system:

$$\begin{aligned}\tilde{\mathbf{e}}_1 &= p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3,\\ \tilde{\mathbf{e}}_2 &= p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3,\\ \tilde{\mathbf{e}}_3 &= p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3,\end{aligned}$$

and observe that

$$\begin{pmatrix} p_{11} & p_{21} & p_{31}\\ p_{12} & p_{22} & p_{32}\\ p_{13} & p_{23} & p_{33}\end{pmatrix}\begin{pmatrix} p_{11} & p_{12} & p_{13}\\ p_{21} & p_{22} & p_{23}\\ p_{31} & p_{32} & p_{33}\end{pmatrix} =$$

$$= \begin{pmatrix} \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_3\\ \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_3\\ \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_3\end{pmatrix} = \begin{pmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

Thus, the transition matrix $\mathbf{P}$ for the bases of two Cartesian coordinate systems satisfies the matrix equality

$$\mathbf{P}^T\mathbf{P} = \mathbf{I}. \qquad (**)$$

It is also clear that if a transition matrix possesses property $(**)$, then a Cartesian system passes into a Cartesian system.

Definition. A square real matrix $\mathbf{P}$ satisfying equality $(**)$ is called *orthogonal*.

Orthogonal matrices of order 2 effect the transition between bases of Cartesian systems in the plane. Such, in particular, are the transition matrices implementing a rotation (see Tyrtyshnikov, §19.3).

The definition we have given is also applicable to matrices of order greater than 3. According to $(**)$, for orthogonal matrices of arbitrary order, inversion reduces to transposition:

$$\mathbf{P}^{-1} = \mathbf{P}^T.$$

Moreover, the product of two orthogonal matrices remains an orthogonal matrix: if $\mathbf{P}^T\mathbf{P} = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, then $(\mathbf{PQ})^T(\mathbf{PQ}) = \mathbf{Q}^T(\mathbf{P}^T\mathbf{P})\mathbf{Q} = \mathbf{Q}^T\mathbf{IQ} = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. Obviously, the identity matrix $\mathbf{I}$ is an orthogonal matrix. Consequently, *the set of all orthogonal matrices of a fixed order forms a group with respect to the operation of matrix multiplication.*

Proposition. Let $\mathbf{y} = \mathbf{Px}$, where $\mathbf{P}$ is an arbitrary orthogonal matrix of order n and $\mathbf{x} \in \mathbb{R}^n$. Then the sum of the squares of the entries of the column vector $\mathbf{y}$ equals the sum of the squares of the entries of the column vector $\mathbf{x}$.

Proof. $y_1^2 + ... + y_n^2 = \mathbf{y}^T\mathbf{y} = (\mathbf{Px})^T(\mathbf{Px}) = \mathbf{x}^T(\mathbf{P}^T\mathbf{P})\mathbf{x} = \mathbf{x}^T\mathbf{x} = x_1^2 + ... + x_n^2$. □

Corollary. Let $\mathbf{B} = \mathbf{PAQ}$, where $\mathbf{P}, \mathbf{Q}$ are arbitrary orthogonal matrices of order n and $\mathbf{A}$ is an arbitrary real matrix of order n. Then the sum of the squares of the entries of the matrix $\mathbf{B}$ equals the sum of the squares of the entries of the matrix $\mathbf{A}$.

5.3. Orthogonal Projection

Gram Matrix

Suppose we are given vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ in a space with an inner product $\langle \cdot, \cdot \rangle$, and let

$$\mathbf{x} = \alpha_1\mathbf{v}_1 + ... + \alpha_n\mathbf{v}_n, \quad \mathbf{y} = \beta_1\mathbf{v}_1 + ... + \beta_n\mathbf{v}_n.$$

Then a direct calculation yields

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} \overline{\beta_i} \left(\sum_{j=1}^{n} \langle \mathbf{v}_j, \mathbf{v}_i \rangle \alpha_j \right) = \mathbf{b}^*\mathbf{Ga}, \qquad (*)$$

where

$$\mathbf{G} = \mathbf{G}(\mathbf{v}_1, ..., \mathbf{v}_n) = \begin{pmatrix} \langle \mathbf{v}_1, \mathbf{v}_1 \rangle & ... & \langle \mathbf{v}_n, \mathbf{v}_1 \rangle \\ ... & ... & ... \\ \langle \mathbf{v}_1, \mathbf{v}_n \rangle & ... & \langle \mathbf{v}_n, \mathbf{v}_n \rangle \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}.$$

The matrix $\mathbf{G}$ formed from the inner products of the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ is called their *Gram matrix*.[9] In the real case (in particular, in $\mathbb{R}^n$ with the dot product), $(*)$ reads $\mathbf{x} \cdot \mathbf{y} = \mathbf{b}^T\mathbf{Ga}$ with $\mathbf{G} = \left(\mathbf{v}_j \cdot \mathbf{v}_i\right)$, and $\mathbf{a}^*$ below becomes $\mathbf{a}^T$.

Theorem (Gram Matrix Theorem)**.** Vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent if and only if their Gram matrix is not invertible.

[9] Note that the entry in position i, j has the form $\langle \mathbf{v}_j, \mathbf{v}_i \rangle$. Often $\mathbf{G}^T$ is called the Gram matrix (in the real case, of course, $\mathbf{G}^T = \mathbf{G}$).

Proof. Let $\mathbf{x} = \alpha_1\mathbf{v}_1 + ... + \alpha_n\mathbf{v}_n$. Using $(*)$ for $\mathbf{x} = \mathbf{y}$, we find

$$\langle \mathbf{x}, \mathbf{x} \rangle = \mathbf{a}^*\mathbf{G}\mathbf{a}, \quad \mathbf{a} = (\alpha_1, ..., \alpha_n)^T. \qquad (\#)$$

If $\mathbf{G}$ is not invertible, then there exists a column $\mathbf{a} \neq \mathbf{0}$ such that $\mathbf{G}\mathbf{a} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0} \Rightarrow$ the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent.

Conversely, if these vectors are linearly dependent, then $\mathbf{x} = \mathbf{0}$ for some $\mathbf{a} \neq \mathbf{0}$. It is easy to see that $\mathbf{G}\mathbf{a} = (\langle \mathbf{x}, \mathbf{v}_1 \rangle, ..., \langle \mathbf{x}, \mathbf{v}_n \rangle)^T = \mathbf{0}$ is a nontrivial linear combination of the columns of the matrix $\mathbf{G}$ that is equal to zero $\Rightarrow$ the columns of $\mathbf{G}$ are linearly dependent $\Rightarrow$ $\mathbf{G}$ is not invertible. □

Problem 1. In a space with an inner product, two sets of vectors are given: $\mathbf{u}_1, ..., \mathbf{u}_m$ and $\mathbf{v}_1, ..., \mathbf{v}_m$. Suppose that $L^\perp \cap M = \{\mathbf{0}\}$, where L and M are the spans of the vectors of the first and second sets. Prove that at least one of these sets is linearly dependent if and only if the $m \times m$ matrix $\mathbf{A}$ with entries $a_{ij} = \langle \mathbf{v}_j, \mathbf{u}_i \rangle$ is not invertible.

Orthogonal Projection and Best Approximation

Let L be a subspace of $\mathbb{R}^n$ of dimension m. We already know (see Tyrtyshnikov, §23.6) that for any $\mathbf{x} \in \mathbb{R}^n$ there exists an element of best approximation $\mathbf{z}_0 \in L$—such that $\|\mathbf{x} - \mathbf{z}_0\| \leq \|\mathbf{x} - \mathbf{z}\|$ for all $\mathbf{z} \in L$. In this special case—for a norm induced by an inner product—uniqueness of $\mathbf{z}_0$ holds, and there is a very simple way of obtaining it.

We proceed from the assumption that a basis $\{\mathbf{z}_1, ..., \mathbf{z}_m\}$ for L is given. Then

$$\mathbf{z}_0 = \alpha_1\mathbf{z}_1 + ... + \alpha_m\mathbf{z}_m.$$

Let us find the coefficients $\alpha_1, ..., \alpha_m$ from the condition

$$\mathbf{x} - \mathbf{z}_0 \perp L \quad \Leftrightarrow \quad (\mathbf{x} - \mathbf{z}_0) \cdot \mathbf{z}_1 = 0, ..., (\mathbf{x} - \mathbf{z}_0) \cdot \mathbf{z}_m = 0 \quad \Leftrightarrow$$

$$\Leftrightarrow \quad \begin{cases} \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_1) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_1) = \mathbf{x} \cdot \mathbf{z}_1, \\ \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_2) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_2) = \mathbf{x} \cdot \mathbf{z}_2, \\ ... \\ \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_m) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_m) = \mathbf{x} \cdot \mathbf{z}_m. \end{cases}$$

Obviously, we have a linear system whose coefficient matrix coincides with the Gram matrix $\mathbf{G} = \mathbf{G}(\mathbf{z}_1, ..., \mathbf{z}_m)$ of the vectors $\mathbf{z}_1, ..., \mathbf{z}_m$. By the Gram matrix theorem for linearly independent vectors, it is invertible $\Rightarrow$ the linear system in $\alpha_1, ..., \alpha_m$ has a unique solution $\Rightarrow$ the vector $\mathbf{z}_0$ subject to the condition $\mathbf{x} - \mathbf{z}_0 \perp L$ exists and is unique.

The vector $\mathbf{x}_n \equiv \mathbf{x} - \mathbf{z}_0$ in the case $\mathbf{x}_n \perp L$, $\mathbf{z}_0 \in L$ is called the *perpendicular* dropped from $\mathbf{x}$ onto L, and $\mathbf{x}_p \equiv \mathbf{z}_0$ is called the *orthogonal projection* (or just *projection*) of the vector $\mathbf{x}$ onto L. Thus, $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ with $\mathbf{x}_p \in L$ and $\mathbf{x}_n \perp L$.

Theorem (Orthogonal Projection Theorem)**.** For any vector $\mathbf{x}$ and any finite-dimensional subspace L, there exist a unique perpendicular $\mathbf{x}_n \perp L$ and a unique projection $\mathbf{x}_p \in L$ such that $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$. Moreover,

$$\|\mathbf{x}_n\| = \|\mathbf{x} - \mathbf{x}_p\| < \|\mathbf{x} - \mathbf{z}\| \quad \forall \mathbf{z} \in L, \quad \mathbf{z} \neq \mathbf{x}_p.$$

Proof. It remains to prove only that $\mathbf{x}_p$ is the uniquely determined element of best approximation in L for the vector $\mathbf{x}$. Let $\mathbf{z}$ be an arbitrary vector in L. Then $\mathbf{x} - \mathbf{z} = (\mathbf{x} - \mathbf{x}_p) + (\mathbf{x}_p - \mathbf{z})$, where $\mathbf{x} - \mathbf{x}_p \perp L$ and $\mathbf{x}_p - \mathbf{z} \in L$. From this it follows that $\mathbf{x} - \mathbf{x}_p$ and $\mathbf{x}_p - \mathbf{z}$ are the perpendicular and the orthogonal projection onto L for the vector $\mathbf{x} - \mathbf{z}$. By the Pythagorean theorem,

$$\|\mathbf{x} - \mathbf{z}\|^2 = \|\mathbf{x} - \mathbf{x}_p\|^2 + \|\mathbf{x}_p - \mathbf{z}\|^2 \quad \Rightarrow \quad \|\mathbf{x} - \mathbf{x}_p\| < \|\mathbf{x} - \mathbf{z}\| \quad \forall \mathbf{z} \neq \mathbf{x}_p. \ \square$$

Corollary. If L is a finite-dimensional subspace, then $L = (L^\perp)^\perp$.

Proof. We already know that $L \subset (L^\perp)^\perp$ (see Section 5.1). Take $\mathbf{x} \in (L^\perp)^\perp$ and drop the perpendicular $\mathbf{x}_n$ from it onto L. By the definition of the orthogonal complement, $\mathbf{x}_n \in L^\perp$. At the same time, $\mathbf{x}_n \perp L^\perp \Rightarrow \mathbf{x}_n \cdot \mathbf{x}_n = 0 \Rightarrow \mathbf{x}_n = \mathbf{0}$. Consequently, $\mathbf{x} \in L \Rightarrow (L^\perp)^\perp \subset L$. $\square$

Problem 2. Let L and M be subspaces in a finite-dimensional space V with an inner product. Are the equalities $L^\perp \cap M = \{\mathbf{0}\}$ and $L \cap M^\perp = \{\mathbf{0}\}$ equivalent?

Problem 3. In $\mathbb{R}^n$, vectors $\mathbf{a}_1, ..., \mathbf{a}_{n+1}$ are such that $\mathbf{a}_i \cdot \mathbf{a}_j < 0$ for $i \neq j$. Prove that any n of them are linearly independent.

Problem 4. Prove that in $\mathbb{R}^n$, any set of $n+2$ vectors contains a pair of vectors whose inner product is nonnegative.

Problem 5. Let P_n be the vector space of all real polynomials of degree at most n with the inner product $\langle f, g \rangle = \int_{-1}^{1} f(t)g(t)\,\mathrm{d}t$. Prove that the distance from the polynomial x^n to the subspace P_{n-1} does not exceed $\frac{\sqrt{2}}{2^n}$.

Gram–Schmidt Process

From the orthogonal projection theorem it immediately follows that in any finite-dimensional space V with an inner product (for example, in any subspace of $\mathbb{R}^m$) there exists an orthonormal basis.

Indeed, let us take an arbitrary basis $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ for V and assume that for the span $L_{n-1} = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-1}\}$ an orthonormal basis $\{\mathbf{q}_1, ..., \mathbf{q}_{n-1}\}$ has already been constructed (of course, $L_{n-1} = \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_{n-1}\}$). Let $\mathbf{h}_n$ be the perpendicular dropped from the vector $\mathbf{v}_n$ onto L_{n-1}. Clearly, $\mathbf{h}_n \neq \mathbf{0}$ (otherwise $\mathbf{v}_n \in L_{n-1} \Rightarrow$ the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent). Set $\mathbf{q}_n = \frac{\mathbf{h}_n}{\|\mathbf{h}_n\|}$. Then $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$ will be the desired orthonormal basis for V.

Note that in the constructed basis, for any $k = 1, ..., n$, the first k vectors $\mathbf{q}_1, ..., \mathbf{q}_k$ form an orthonormal basis for the span $L_k = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_k\}$. Thus,

$$\operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_k\} = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_k\}, \quad k = 1, ..., n.$$

Actual computations begin with obtaining the vector $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$. Then from the vector $\mathbf{v}_2$ the perpendicular $\mathbf{h}_2$ is dropped onto L_1 and normalized: $\mathbf{q}_2 = \frac{\mathbf{h}_2}{\|\mathbf{h}_2\|}$. And so on. When dropping the perpendicular onto L_k, it is reasonable to seek the expansion of the corresponding projection not in terms of the original vectors $\mathbf{v}_1, ..., \mathbf{v}_k$, but in terms of the already constructed orthonormal set $\mathbf{q}_1, ..., \mathbf{q}_k$. The advantage is obvious: the Gram matrix for $\mathbf{q}_1, ..., \mathbf{q}_k$ is the identity matrix!

This algorithm is called the *Gram–Schmidt process*. Here is its formal description:

$$\mathbf{h}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} (\mathbf{v}_k \cdot \mathbf{q}_i)\mathbf{q}_i, \quad \mathbf{q}_k = \frac{\mathbf{h}_k}{\|\mathbf{h}_k\|}, \quad k = 1, ..., n.$$

Problem 6. In the space of real polynomials, an inner product $\langle f, g\rangle$ is defined in an arbitrary manner, but so that for any polynomials $f(x)$ and $g(x)$ the equality[10]

$$\langle xf(x), g(x)\rangle = \langle f(x), xg(x)\rangle$$

holds, and suppose that applying the Gram–Schmidt process to the polynomials $1, x, x^2, ..., x^n$ yields polynomials $L_0(x), L_1(x), ..., L_n(x)$. Prove that the three-term recurrence relations

$$L_k(x) = a_k x L_{k-1}(x) + b_k L_{k-1}(x) + c_k L_{k-2}(x),$$
$$2 \le k \le n, \quad a_k, b_k, c_k \in \mathbb{R},$$

hold.

Completion to an Orthogonal Basis

Let V be a space of dimension n with an inner product (for example, $V = \mathbb{R}^n$ with the dot product).

Lemma (Completion to an Orthogonal Basis)**.** Any orthogonal (orthonormal) set of vectors $\mathbf{v}_1, ..., \mathbf{v}_k \in V$ can be completed by some vectors from V to an orthogonal (orthonormal) basis for V.

Proof. Complete $\mathbf{v}_1, ..., \mathbf{v}_k$ by some vectors to a basis for V, and then apply the Gram–Schmidt process to the resulting basis. □

Corollary. If L_k is a subspace of dimension k, then $\dim\big(L_k^\perp\big) = n - k$. Moreover,

$$V = L_k \oplus L_k^\perp.$$

[10] Verify that it holds, for example, if the inner product is given by the formula $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x$. In this case, the resulting orthogonal polynomials are called *Legendre polynomials*.

Proof. In V there exists an orthonormal basis $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$ such that $L_k = \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_k\}$. It is then obvious that any vector orthogonal to L_k is a linear combination of the vectors $\mathbf{q}_{k+1}, ..., \mathbf{q}_n$. □

5.4. QR Factorization

QR Factorization of a Matrix

Suppose $\mathbf{A} \in \mathbb{C}^{n\times m}$ has linearly independent columns $\mathbf{a}_1, ..., \mathbf{a}_m \in \mathbb{C}^n$, and the Gram–Schmidt process is applied to them using the natural inner product (real case: $\mathbf{A} \in \mathbb{R}^{n\times m}$ and the dot product in $\mathbb{R}^n$). Suppose that as a result we obtain orthonormal vectors $\mathbf{q}_1, ..., \mathbf{q}_m \in \mathbb{C}^n$.

The relations $\mathbf{a}_k \in \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_k\}$ hold for $k = 1, ..., m$ and mean that for some numbers r_{ik} we have the equalities

$$\mathbf{a}_k = \sum_{i=1}^{k} r_{ik}\mathbf{q}_i, \quad k = 1, ..., m,$$

or, in matrix form,

$$\mathbf{A} = \mathbf{QR}, \quad \mathbf{Q} = (\mathbf{q}_1 \ \cdots \ \mathbf{q}_m), \quad \mathbf{R} = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1m} \\ & r_{22} & \cdots & r_{2m} \\ & & \ddots & \vdots \\ 0 & & & r_{mm} \end{pmatrix}.$$

Definition. A factorization $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ has orthonormal columns and $\mathbf{R}$ is an upper triangular matrix, is called a *QR factorization* of the matrix $\mathbf{A}$.[11]

Thus, we have just proved that *for any rectangular matrix with linearly independent columns there exists a QR factorization.* In particular, it exists for any invertible matrix. In fact, a more general result holds.

Theorem. Any rectangular matrix in which the number of rows is not less than the number of columns possesses a QR factorization with a matrix $\mathbf{R}$ in row-echelon form.

Proof. Let $\mathbf{a}_{i_1}$ be the first nonzero column of the matrix $\mathbf{A}$, $\mathbf{a}_{i_2}$ be the first column such that $\mathbf{a}_{i_2} \notin \operatorname{span}\{\mathbf{a}_{i_1}\}$, $\mathbf{a}_{i_3}$ be the first column such that $\mathbf{a}_{i_3} \notin \operatorname{span}\{\mathbf{a}_{i_1}, \mathbf{a}_{i_2}\}$, and so on. As a result, we obtain in $\mathbf{A}$ columns

$$\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_r}, \quad i_1 < i_2 < ... < i_r,$$

forming a basis for $\operatorname{Col}(\mathbf{A})$ and possessing the following properties:

[11] MA1522 defines the QR factorization only for a matrix with linearly independent columns, and requires in addition that $\mathbf{R}$ be invertible with positive diagonal entries; the definition here is broader.

$$\begin{aligned}
&\mathbf{a}_j = \mathbf{0} && \text{for } \ j < i_1;\\
&\mathbf{a}_j \in \operatorname{span}\{\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_l}\} && \text{for } \ i_l < j < i_{l+1}, \quad l = 1, ..., r-1;\\
&\mathbf{a}_j \in \operatorname{span}\{\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_r}\} && \text{for } \ i_r < j.
\end{aligned}$$

Let us find the QR factorization:

$$\begin{pmatrix}\mathbf{a}_{i_1} & \cdots & \mathbf{a}_{i_r}\end{pmatrix} = \begin{pmatrix}\mathbf{q}_{i_1} & \cdots & \mathbf{q}_{i_r}\end{pmatrix}\mathbf{R}_r.$$

Let us complete the columns $\mathbf{q}_{i_1}, ..., \mathbf{q}_{i_r}$ to an orthonormal basis for the n-dimensional space of columns, and from the resulting columns form the matrix $\mathbf{Q}$, keeping the initial columns in positions $i_1, ..., i_r$.

Writing $\mathbf{A} = \mathbf{QR}$, we see that in the matrix $\mathbf{R}$ the first r entries of the i_l-th column are the same as in the l-th column of the matrix $\mathbf{R}_r$. At the same time, the j-th column for $i_l < j < i_{l+1}$ has zeros in positions below the i_l-th. □

Problem 1. Suppose $\mathbf{A} \in \mathbb{C}^{n\times n}$ has columns $\mathbf{a}_1, ..., \mathbf{a}_n \in \mathbb{C}^n$. Prove the inequality

$$|\det(\mathbf{A})| \le \prod_{j=1}^{n} \|\mathbf{a}_j\|.$$

Problem 2. Let $\mathbf{A}$ be a matrix of order n with entries $a_{ij} = \pm 1$. Prove that if $|\det(\mathbf{A})| = n^{\frac{n}{2}}$ (such matrices are called *Hadamard matrices*) and $n \ge 3$, then n is divisible by 4.

5.5. Least Squares Approximation

Method of Least Squares

If a linear system $\mathbf{Ax} = \mathbf{b}$ is inconsistent, then the equality $\mathbf{Ax} = \mathbf{b}$ does not hold for any vector $\mathbf{x}$. In this case, nevertheless, one tries to consider those $\mathbf{x}$ for which the vector $\mathbf{b} - \mathbf{Ax}$ (it is called the *residual* for $\mathbf{x}$) has the minimum possible norm. A vector $\mathbf{x}$ is called a *least squares solution* of the system $\mathbf{Ax} = \mathbf{b}$ if

$$\|\mathbf{b} - \mathbf{Ax}\| = \min_{\mathbf{z}} \|\mathbf{b} - \mathbf{Az}\|.$$

In this method of defining a "generalized solution," in the real case one is indeed dealing with the minimum value of the sum of squares (whence the name of the method)

$$\|\mathbf{b} - \mathbf{Ax}\|^2 = \sum_{i=1}^{m} (b_i - a_{i1}x_1 - ... - a_{in}x_n)^2.$$

Below, the formulas are written for the real case; for a complex matrix $\mathbf{A}$, every transpose ($\mathbf{A}^T$, $\mathbf{V}^T$, $\mathbf{u}_k^T$) is to be replaced by the conjugate transpose ($\mathbf{A}^*$, $\mathbf{V}^*$, $\mathbf{u}_k^*$).

Proposition. Let $\mathbf{A}$ be an $m \times n$ matrix of rank r. The set of least squares solutions of the system $\mathbf{Ax} = \mathbf{b}$ is an affine set of dimension $n - r$, that is, a translate $\mathbf{x}_0 + W$ of a subspace W of dimension $n - r$.

Proof. Let $\mathbf{b}_n$ be the perpendicular dropped from the vector $\mathbf{b}$ onto the subspace $\mathrm{Col}(\mathbf{A})$, and let $\mathbf{b}_p \in \mathrm{Col}(\mathbf{A})$ be the corresponding orthogonal projection. Then the system $\mathbf{Az} = \mathbf{b}_p$ is consistent, and if $\mathbf{z}$ is an arbitrary solution of it, then $\|\mathbf{b}_n\| = \|\mathbf{b} - \mathbf{Az}\| < \|\mathbf{b} - \mathbf{Ax}\|$ for all $\mathbf{x}$ such that $\mathbf{Ax} \neq \mathbf{b}_p$. Hence, the set of least squares solutions coincides with the set of solutions of the consistent system $\mathbf{Az} = \mathbf{b}_p$. □

Corollary. The set of least squares solutions of the system $\mathbf{Ax} = \mathbf{b}$ coincides with the set of solutions of the system $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$ (the *normal equation*).

Proof. $\mathbf{b} - \mathbf{Az} \perp \mathrm{Col}(\mathbf{A}) \Leftrightarrow \mathbf{A}^T(\mathbf{b} - \mathbf{Az}) = \mathbf{0}$. □

Among all least squares solutions, the one of minimal norm, $\hat{\mathbf{x}}$, stands out—it is called the *least squares solution of minimum norm*. Geometrically, it is clear that $\hat{\mathbf{x}}$ is the perpendicular dropped onto $\mathrm{Null}(\mathbf{A})$ from any particular solution $\mathbf{z}$ of the consistent system $\mathbf{Az} = \mathbf{b}_p$ (the vector $\mathbf{b}_p$ is the orthogonal projection of the vector $\mathbf{b}$ onto $\mathrm{Col}(\mathbf{A})$). Thus, *the least squares solution of minimum norm exists and is unique.*

The singular value decomposition (see Section 6.5) makes it possible to give an explicit form of the least squares solution of minimum norm. Let $\mathbf{A} = \mathbf{U\Sigma V}^T$ be a singular value decomposition, where $\sigma_1 \geq \ldots \geq \sigma_r > 0$ are the nonzero singular values of $\mathbf{A}$, and $\mathbf{u}_k$ and $\mathbf{v}_k$ are the columns of $\mathbf{U}$ and $\mathbf{V}$ (the left and right singular vectors). Then

$$\hat{\mathbf{x}} = \sum_{k=1}^{r} \frac{\mathbf{u}_k^T \mathbf{b}}{\sigma_k} \mathbf{v}_k. \qquad (*)$$

For the proof it suffices to verify that $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}} \perp \mathrm{Col}(\mathbf{A})$ and $\hat{\mathbf{x}} \perp \mathrm{Null}(\mathbf{A})$.

The simplicity of the formula should not create the impression that there are no problems in computing $\hat{\mathbf{x}}$. The main problem, properly speaking, is that in the case $r < \min(m, n)$ the rank r can be increased by an arbitrarily small perturbation of the entries of the matrix, which means that the least squares solution of minimum norm, despite the fact of its existence and uniqueness, is not a continuous function of the entries of the matrix $\mathbf{A}$. For example, let $m = n = 1$ and consider the system $0 \cdot x = 1$. Its least squares solution of minimum norm is obviously $\hat{x} = 0$, whereas the least squares solution of minimum norm of the perturbed system $\varepsilon \cdot x = 1$ is $\hat{x}(\varepsilon) = \frac{1}{\varepsilon}$. As we see, $\hat{x}(\varepsilon)$ does not tend to $\hat{x}$ as $\varepsilon \to 0$. The very problem of computing such an unstable object does not seem terribly meaningful.

At the same time, problems of this kind constantly arise in applications, and some methods for solving them are required of us. When constructing such methods, one should bear in mind that they must be, above all, *methods for modifying the very formulation of the problem.* Such questions are connected with the so-called *regularization methods.*[12]

Problem 1. Find the least squares solution of minimum norm of the inconsistent system

$$x_1 + x_2 + ... + x_n = 1,$$
$$x_1 + x_2 + ... + x_n = 0.$$

Pseudoinverse Matrix BEYOND MA1522

Formula $(*)$ for the least squares solution of minimum norm can also be written in the form

$$\hat{\mathbf{x}} = \mathbf{M}\mathbf{b}, \quad \mathbf{M} = \mathbf{V}\begin{pmatrix} \frac{1}{\sigma_1} & & & \\ & \ddots & & \\ & & \frac{1}{\sigma_r} & \\ & & & 0 \end{pmatrix}\mathbf{U}^T.$$

The matrix $\mathbf{M}$ is called the *pseudoinverse* (or *Moore–Penrose pseudoinverse*) of the matrix $\mathbf{A}$. By virtue of the uniqueness of the least squares solution of minimum norm, the pseudoinverse matrix is uniquely determined by the matrix $\mathbf{A}$. Notation: $\mathbf{M} = \mathbf{A}^+$.

Problem 2. Let $\mathbf{A}$ be an arbitrary rectangular matrix and $\mathbf{A}^+$ its pseudoinverse matrix. Prove that the following relations hold:

$$(\mathbf{A}\mathbf{A}^+)^T = \mathbf{A}\mathbf{A}^+, \quad (\mathbf{A}^+\mathbf{A})^T = \mathbf{A}^+\mathbf{A}, \quad \mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+.$$

Prove also that $\mathbf{A}^+$ is the unique matrix satisfying this system of equations.

[12]The general theory of regularization methods was created by the founder of the Faculty of Computational Mathematics and Cybernetics at Moscow State University, Academician Andrei Nikolaevich Tikhonov.

Chapter 6. Eigenanalysis

6.1. Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be a matrix of order n and let $\mathbf{P}$ be an invertible matrix with columns $\mathbf{p}_1, ..., \mathbf{p}_n$. It is easy to see that the equality $\mathbf{AP} = \mathbf{PD}$, where $\mathbf{D} = \text{diag}(\lambda_1, ..., \lambda_n)$, is equivalent to the system of equalities

$$\mathbf{A}\mathbf{p}_j = \lambda_j \mathbf{p}_j, \quad j = 1, ..., n.$$

These equalities lead us to the important concepts of an eigenvalue of a matrix and an eigenvector.

Definition. Let $\mathbf{A}$ be a matrix of order n. A scalar $\lambda \in \mathbb{C}$ and a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ related by $\mathbf{Av} = \lambda\mathbf{v}$ are called an *eigenvalue* and an *eigenvector* of the matrix $\mathbf{A}$ (an eigenvector *associated to* λ). The pair $\lambda, \mathbf{v}$ is sometimes called an *eigenpair* of the matrix $\mathbf{A}$.

In MA1522 the matrix $\mathbf{A}$ is real, and only real eigenvalues $\lambda \in \mathbb{R}$, with eigenvectors $\mathbf{v} \in \mathbb{R}^n$, are considered.

Theorem. A matrix $\mathbf{A}$ of order n is diagonalizable (see Section 6.2) if and only if it has n linearly independent eigenvectors.

Proof. Let $\mathbf{p}_1, ..., \mathbf{p}_n$ be linearly independent eigenvectors of the matrix $\mathbf{A}$ associated to eigenvalues $\lambda_1, ..., \lambda_n$:

$$\mathbf{A}\mathbf{p}_j = \lambda_j \mathbf{p}_j, \quad j = 1, ..., n \quad \Leftrightarrow \quad \mathbf{AP} = \mathbf{P}\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \quad \mathbf{P} = (\mathbf{p}_1 \ \cdots \ \mathbf{p}_n).$$

The matrix $\mathbf{P}$ is invertible—as a matrix with linearly independent columns. □

An example of a non-diagonalizable matrix: $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Suppose that

$$\begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \Rightarrow$$

$$\Rightarrow \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

From this,

$$\begin{pmatrix} p_{21} & p_{22} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} p_{11}\lambda_1 & p_{12}\lambda_2 \\ p_{21}\lambda_1 & p_{22}\lambda_2 \end{pmatrix}.$$

At least one of the numbers λ_1, λ_2 must be nonzero. Suppose for definiteness that $\lambda_1 \neq 0 \Rightarrow p_{21} = 0 \Rightarrow p_{11} = 0$. We obtain a contradiction, since a matrix with a zero column cannot be invertible. □

The Characteristic Equation

Let λ be an arbitrary eigenvalue of a matrix $\mathbf{A}$. For a fixed λ, all eigenvectors $\mathbf{v}$ associated to it satisfy the homogeneous linear system

$$(\lambda\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}.$$

A scalar λ is an eigenvalue of the matrix $\mathbf{A} \Leftrightarrow$ this system has a nontrivial solution $\Leftrightarrow \det(\lambda\mathbf{I} - \mathbf{A}) = 0$.

Definition. The equation $\det(x\mathbf{I} - \mathbf{A}) = 0$ in x is called the *characteristic equation* of the matrix $\mathbf{A}$. The left-hand side of this equation is a polynomial of degree n in x, called the *characteristic polynomial* of the matrix $\mathbf{A}$ and denoted $\mathrm{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})$.

The full textbook uses $\det(\mathbf{A} - \lambda\mathbf{I})$ instead; the two polynomials differ by the factor $(-1)^n$ and have the same roots.

Proposition. The characteristic polynomial of a matrix $\mathbf{A}$ has the form

$$\mathrm{char}(\mathbf{A}) = x^n - s_{n-1}x^{n-1} + s_{n-2}x^{n-2} - \ldots + (-1)^n s_0,$$

where s_k is the sum of all $(n-k) \times (n-k)$ minors of the matrix $\mathbf{A}$ (determinants of $(n-k) \times (n-k)$ submatrices) located at the intersection of columns and rows with identical indices.

Proof. To obtain the coefficient of x^k, one must select among the $n!$ terms of the determinant

$$\det(x\mathbf{I} - \mathbf{A}) = \sum_{\sigma \in S_n} d_\sigma$$

those and only those terms d_σ that contain a product of exactly k diagonal entries of the form $x - a_{ii}$ (they and only they are polynomials of degree k in x), extract in each of them the leading-degree term of the form $x^k c_\sigma$, and sum the resulting coefficients c_σ. Obviously, the sum of all c_σ corresponding to k diagonal entries in fixed positions $i_1 < \ldots < i_k$ is equal to the determinant of the submatrix of $-\mathbf{A}$ located in the rows and columns complementary to the rows and columns with indices $i_1, \ldots, i_k$, that is, to $(-1)^{n-k}$ times the corresponding $(n-k) \times (n-k)$ minor of $\mathbf{A}$. Hence the coefficient of x^k is $(-1)^{n-k} s_k$. □

In particular, $s_{n-1} = a_{11} + ... + a_{nn}$ is the quantity called the *trace* of the matrix $\mathbf{A}$. Notation: $\operatorname{tr}(\mathbf{A})$. By Vieta's formulas, the trace equals the sum of all eigenvalues counting multiplicities. Note also that $s_0 = \det(\mathbf{A})$.

For $n \leq 4$, the eigenvalues (as roots of a polynomial of degree $n \leq 4$) can be expressed in radicals in terms of the coefficients of the characteristic polynomial and, consequently, in terms of the entries of the matrix. For $n \geq 5$, such formulas no longer exist (the celebrated result of Abel, Ruffini, and Galois).

Algebraic Multiplicity of an Eigenvalue

The multiplicity of an eigenvalue λ as a root of the characteristic polynomial is called its *algebraic multiplicity* and is denoted r_λ: it is the largest integer r_λ such that $\operatorname{char}(\mathbf{A}) = (x - \lambda)^{r_\lambda} p(x)$ for some polynomial $p(x)$.

From the fundamental theorem of algebra we immediately obtain the following theorem.

Theorem. Every complex matrix $\mathbf{A}$ of order n has n complex eigenvalues counting algebraic multiplicities.

Characteristic Polynomial and Similarity

Theorem. The characteristic polynomials of similar matrices coincide.

Proof. Let $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, where $\mathbf{P}$ is an invertible matrix. Then

$$\begin{aligned}\det(x\mathbf{I} - \mathbf{B}) &= \det(x\mathbf{P}^{-1}\mathbf{P} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \det(\mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P}) = \\ &= \det(\mathbf{P}^{-1})\det(\mathbf{P})\det(x\mathbf{I} - \mathbf{A}) = \det(\mathbf{P}^{-1}\mathbf{P})\det(x\mathbf{I} - \mathbf{A}) = \\ &= \det(x\mathbf{I} - \mathbf{A}).\end{aligned}$$

□

Corollary. The eigenvalues and their algebraic multiplicities for similar matrices coincide.

Problem 1. Find the characteristic polynomial of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 1 & & & 0 \end{pmatrix}_{n\times n}.$$

Problem 2. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same order. Prove that $\mathbf{AB}$ and $\mathbf{BA}$ have the same characteristic polynomials.

Geometric Multiplicity of an Eigenvalue

Fix an eigenvalue λ of a linear transformation $T : V \to V$ (that is, $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$) and consider the set E_λ of all vectors $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$.

Proposition. The set E_λ is a subspace invariant under T.

Proof. Let $\mathbf{u}, \mathbf{v} \in E_\lambda \Rightarrow T(\mathbf{u}) = \lambda\mathbf{u}, T(\mathbf{v}) = \lambda\mathbf{v} \Rightarrow T(\alpha\mathbf{u} + \beta\mathbf{v}) = \lambda(\alpha\mathbf{u} + \beta\mathbf{v}) \Rightarrow \alpha\mathbf{u} + \beta\mathbf{v} \in E_\lambda$. The invariance of E_λ is obvious: if $\mathbf{v} \in E_\lambda$, then $T(\mathbf{v}) = \lambda\mathbf{v} \in E_\lambda$. □

Definition. The subspace E_λ is called the *eigenspace* associated to λ, and its dimension $\dim(E_\lambda)$ is called the *geometric multiplicity* of the eigenvalue λ.

For a square matrix $\mathbf{A}$ of order n, regarded as the linear transformation $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ of $\mathbb{R}^n$, the eigenspace is $E_\lambda = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\} = \text{Null}(\lambda\mathbf{I} - \mathbf{A})$, and the geometric multiplicity of λ is $\dim(E_\lambda) = \text{nullity}(\lambda\mathbf{I} - \mathbf{A})$.

6.2. Diagonalization

Similarity Transformation

Let $T : V \to V$ be a linear transformation of an n-dimensional vector space V and let $\mathbf{A}$ be its matrix when one and the same basis of V is chosen for the images and the preimages. In this case, the study of the transformation T reduces completely to the study of its matrix $\mathbf{A}$.

It is natural to try to choose a basis in such a way that the matrix $\mathbf{A}$ acquires the "simplest possible" form. If a linear transformation T is given by its matrix $\mathbf{A}$ in some basis, then choosing a new basis gives for the same transformation another matrix $\mathbf{B}$, which will be similar to the given matrix (see Section 7.1). The transition from $\mathbf{A}$ to a matrix $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ similar to it is called a *similarity transformation*.

The following question arises: to what "simplest possible" form can a given matrix be reduced by means of a similarity transformation? In effect, we are now beginning a not-so-simple journey toward a complete answer to this question.

Diagonalizable Matrices

Matrices similar to diagonal matrices are called *diagonalizable* (the full textbook also calls them *matrices of simple structure*). Thus a square matrix $\mathbf{A}$ is diagonalizable if there exists an invertible matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ is a diagonal matrix, or, equivalently, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$.

Consider the problem of diagonalizing a (3×3)-matrix:

$$\mathbf{AP}=\mathbf{PD} \quad \Leftrightarrow \quad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix}.$$

This equality is equivalent to the three equalities:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix} = \lambda_j \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix}, \quad j=1,2,3.$$

Suppose that the value λ_j is known. Then the entries of the j-th column of the matrix $\mathbf{P}$ satisfy the homogeneous system

$$\begin{pmatrix} \lambda_j - a_{11} & -a_{12} & -a_{13} \\ -a_{21} & \lambda_j - a_{22} & -a_{23} \\ -a_{31} & -a_{32} & \lambda_j - a_{33} \end{pmatrix} \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \qquad (*)$$

This system must have a nontrivial solution $\Leftrightarrow$ the determinant of the coefficient matrix is equal to zero. Thus, λ_j satisfies the following equation in x:

$$\det \begin{pmatrix} x - a_{11} & -a_{12} & -a_{13} \\ -a_{21} & x - a_{22} & -a_{23} \\ -a_{31} & -a_{32} & x - a_{33} \end{pmatrix} = 0. \qquad (\#)$$

This is a cubic equation of the form $x^3 - s_2 x^2 + s_1 x - s_0 = 0$, where, as is easy to see,

$$s_2 = a_{11} + a_{22} + a_{33},$$

$$s_1 = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \det \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix} + \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix},$$

$$s_0 = \det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

The question of diagonalizing matrices of order 3 also arises when studying quadric surfaces (see Section 6.3)—when searching for a coordinate system in which the matrix of the quadratic part (a real symmetric matrix of order 3) becomes diagonal.[13]

[13] In Section 6.3 it is proved that *every real symmetric matrix* $\mathbf{A}$ *is orthogonally similar to a diagonal matrix* $\mathbf{D}$—this means that $\mathbf{A}=\mathbf{PDP}^{-1}=\mathbf{PDP}^T$, where $\mathbf{P}$ is an orthogonal matrix, that is, $\mathbf{A}$ is orthogonally diagonalizable. This same statement will soon appear as a consequence of more general results.

Eigenvectors for Distinct Eigenvalues

Theorem. Eigenvectors associated to pairwise distinct eigenvalues of a matrix are linearly independent.

Proof. Let $\mathbf{v}_1, ..., \mathbf{v}_m$ be eigenvectors for pairwise distinct eigenvalues $\lambda_1, ..., \lambda_m$ of a matrix $\mathbf{A}$. Suppose $\alpha_1\mathbf{v}_1 + ... + \alpha_m\mathbf{v}_m = \mathbf{0}$. Multiply both sides on the left by the matrix $\mathbf{A}$:

$$\alpha_1\lambda_1\mathbf{v}_1 + ... + \alpha_m\lambda_m\mathbf{v}_m = \mathbf{0}.$$

From this equality subtract the preceding one multiplied by λ_m:

$$\alpha_1(\lambda_1 - \lambda_m)\mathbf{v}_1 + ... + \alpha_{m-1}(\lambda_{m-1} - \lambda_m)\mathbf{v}_{m-1} = \mathbf{0}.$$

From this it is clear that the linear independence of the vectors $\mathbf{v}_1, ..., \mathbf{v}_{m-1}$ would imply the linear independence of the vectors $\mathbf{v}_1, ..., \mathbf{v}_m$. The proof is completed by applying induction. □

Corollary. If a matrix of order n has n distinct eigenvalues, then it is diagonalizable.

6.3. Orthogonally Diagonalizable Matrices

The Method of Rotations BEYOND MA1522

Let us try to simplify the quadratic part $f_2(x, y, z)$ (see Tyrtyshnikov, §20.1) using the same idea of rotating the coordinate system as in the planar case (see Tyrtyshnikov, §19.3). However, we now have three coordinate planes generated by the three pairs of coordinate axes. The goal of a rotation is to obtain zero in place of some single pair of entries $a_{ij} = a_{ji}$ for $i \neq j$. Consider the three possibilities:

$$\begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} \tilde{a}_{11} & 0 & \tilde{a}_{13} \\ 0 & \tilde{a}_{22} & \tilde{a}_{23} \\ \tilde{a}_{13} & \tilde{a}_{23} & \tilde{a}_{33} \end{pmatrix}, \tag{1}$$

$$\begin{pmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} \cos\varphi & 0 & -\sin\varphi \\ 0 & 1 & 0 \\ \sin\varphi & 0 & \cos\varphi \end{pmatrix} = \\ = \begin{pmatrix} \tilde{a}_{11} & \tilde{a}_{12} & 0 \\ \tilde{a}_{12} & \tilde{a}_{22} & \tilde{a}_{23} \\ 0 & \tilde{a}_{23} & \tilde{a}_{33} \end{pmatrix}, \tag{2}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{pmatrix} = \\ = \begin{pmatrix} \tilde{a}_{11} & \tilde{a}_{12} & \tilde{a}_{13} \\ \tilde{a}_{12} & \tilde{a}_{22} & 0 \\ \tilde{a}_{13} & 0 & \tilde{a}_{33} \end{pmatrix}. \tag{3}$$

Denote by d_0, h_0 and d_1, h_1 the sums of the squares of the diagonal and off-diagonal entries of the original and new matrices in each of the three cases:

$$d_0 = a_{11}^2 + a_{22}^2 + a_{33}^2, \quad h_0 = 2a_{12}^2 + 2a_{13}^2 + 2a_{23}^2,$$
$$d_1 = \tilde{a}_{11}^2 + \tilde{a}_{22}^2 + \tilde{a}_{33}^2, \quad h_1 = 2\tilde{a}_{12}^2 + 2\tilde{a}_{13}^2 + 2\tilde{a}_{23}^2.$$

According to the properties of orthogonal matrices (see Section 5.2),

$$d_1 + h_1 = d_0 + h_0 \quad \Rightarrow \quad h_1 = h_0 - (d_1 - d_0).$$

For the same reason, in case (1) we have: $a_{11}^2 + a_{22}^2 + 2a_{12}^2 = \tilde{a}_{11}^2 + \tilde{a}_{22}^2$, and since $a_{33} = \tilde{a}_{33}$, we get $d_1 - d_0 = 2a_{12}^2$. In case (2), $d_1 - d_0 = 2a_{13}^2$, and in case (3), $d_1 - d_0 = 2a_{23}^2$.

Let the indices i, j determine the coordinate plane in which the rotation is performed (and indicate which of the relations (1), (2), or (3) holds). Choose them in such a way that the eliminated entry a_{ij} is maximal in modulus. Then, obviously,

$$d_1 - d_0 = 2a_{ij}^2 \geq 2\left(\frac{h_0}{6}\right) = \frac{h_0}{3} \quad \Rightarrow \quad h_1 \leq \frac{2}{3}h_0.$$

Let $\mathbf{A}_0 = \left(a_{ij}\right)$ and $\mathbf{A}_1 = \left(\tilde{a}_{ij}\right)$. Regarding $\mathbf{A}_1$ as a new initial matrix, we choose in it the off-diagonal entry of maximal modulus and, annihilating it by means of a rotation, obtain a matrix $\mathbf{A}_2$. Continuing to act in the same way, we construct a sequence of matrices $\mathbf{A}_k = \left(a_{ij}^{(k)}\right)$, $k = 0, 1, ...$

Let h_k denote the sum of the squares of the off-diagonal entries of the matrix $\mathbf{A}_k$. Then

$$h_k \leq \left(\frac{2}{3}\right)^k h_0 \to 0 \quad \text{as } k \to \infty.$$

Consequently, for any fixed $i \neq j$, the sequence of off-diagonal entries $a_{ij}^{(k)}$ converges to zero as $k \to \infty$.

Nested Subsequences BEYOND MA1522

Lemma (on bounded sequences). Suppose we are given a finite number of bounded sequences $\left\{s_1^{(k)}\right\}, ..., \left\{s_m^{(k)}\right\}$, $k = 1, 2, ...$ Then one can choose a sequence of indices $k_1 < k_2 < ...$ in such a way that each of the subsequences $\left\{s_1^{(k_l)}\right\}, ..., \left\{s_m^{(k_l)}\right\}$, $l = 1, 2, ...$, is convergent.

Proof. From the bounded sequence $\left\{s_1^{(k)}\right\}$ we select a convergent subsequence $\left\{s_1^{(k_l)}\right\}$ and instead of the original sequences consider the subsequences $\left\{s_1^{(k_l)}\right\}, ..., \left\{s_m^{(k_l)}\right\}$, $l = 1, 2, ...$ They remain bounded, of course, and at the same time the first of them will be convergent. Next, from the bounded sequence $\left\{s_2^{(k_l)}\right\}$ we select a convergent subsequence (a subsequence of a subsequence with respect to the original sequence) and pass to the subsequences $\left\{s_1^{\left(k_{l_i}\right)}\right\}, ..., \left\{s_m^{\left(k_{l_i}\right)}\right\}$, $i = 1, 2, ...$ The resulting nested subsequences will still be bounded, while the first two are now convergent. And so on. □

Diagonalization in the Limit BEYOND MA1522

Let us return to the method of rotations. Whether the sequences of diagonal entries $a_{ii}^{(k)}$ converge to finite limits is not very important for our immediate goal. Each of them is bounded and therefore possesses a convergent subsequence. Moreover, by the lemma on bounded sequences, there is a subsequence of matrices $\mathbf{A}_k$ in which each of the sequences of diagonal entries converges to some limit.

In order not to clutter the notation, let us assume that $\mathbf{A}_k$ is that very subsequence for which all the sequences $a_{ij}^{(k)}$ are convergent (as we know, to zero when $i \neq j$). Let

$$\lim_{k\to\infty} a_{ii}^{(k)} = \lambda_i, \quad i = 1, 2, 3.$$

It is clear that

$$\mathbf{A}_k = \mathbf{P}_k^T \mathbf{A}_0 \mathbf{P}_k, \quad k = 1, 2, ..., \qquad (\#)$$

where the matrices $\mathbf{P}_k = \left(p_{ij}^{(k)}\right)$ are products of the rotation matrices used (from relations (1), (2), or (3)). Therefore, for every k, the matrix $\mathbf{P}_k$ is orthogonal (as a product of orthogonal matrices). Consequently, the sum of the squares of all entries of the matrix $\mathbf{P}_k$ is the same for every k (and equal to 3). Hence each sequence $p_{ij}^{(k)}$ is bounded as $k \to \infty$ and therefore possesses a convergent subsequence.

By the lemma on bounded sequences, there exists a subsequence of matrices $\mathbf{P}_k$ in which each sequence $p_{ij}^{(k)}$ is convergent. To simplify the notation, let us assume that $\mathbf{P}_k$ is precisely such a subsequence. Let

$$\lim_{k\to\infty} p_{ij}^{(k)} = p_{ij}, \quad i, j = 1, 2, 3.$$

For each k, equality (#) holds. Passing to the limit in the corresponding entrywise equalities, we obtain

$$\mathbf{D} \equiv \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = \mathbf{P}^T \mathbf{A}_0 \mathbf{P}.$$

Moreover, for each k we have: $\mathbf{P}_k^T\mathbf{P}_k = \mathbf{I} \Rightarrow \mathbf{P}^T\mathbf{P} = \mathbf{I}$. As a result, we have proved the following important theorem:

Theorem. For any real symmetric matrix $\mathbf{A}$ of order 3, there exist an orthogonal matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{D} = \mathbf{P}^T\mathbf{A}\mathbf{P}, \quad \text{that is,} \quad \mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T.$$

Corollary. There exists a Cartesian coordinate system in which the equation of a quadric surface has the form

$$f(\tilde{x}, \tilde{y}, \tilde{z}) = \lambda_1\tilde{x}^2 + \lambda_2\tilde{y}^2 + \lambda_3\tilde{z}^2 + 2b_1\tilde{x} + 2b_2\tilde{y} + 2b_3\tilde{z} + a = 0.$$

Remark. By the orthogonality of the matrix $\mathbf{P}$, the equality $\mathbf{D} = \mathbf{P}^T\mathbf{A}\mathbf{P}$ holds if and only if $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D}$. The latter means that the j-th column $\mathbf{p}_j$ of the matrix $\mathbf{P}$ satisfies the equation $\mathbf{A}\mathbf{p}_j = \lambda_j\mathbf{p}_j \Leftrightarrow \big(\lambda_j\mathbf{I} - \mathbf{A}\big)\mathbf{p}_j = \mathbf{0}$. Taking into account that $\mathbf{p}_j \neq \mathbf{0}$, we find: $\det\big(\lambda_j\mathbf{I} - \mathbf{A}\big) = 0$. Thus, the numbers $\lambda_1, \lambda_2, \lambda_3$ are the roots of the cubic polynomial $\operatorname{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})$. Once they have been found, the column $\mathbf{p}_j$ can be obtained as a solution of the homogeneous linear system $\big(\lambda_j\mathbf{I} - \mathbf{A}\big)\mathbf{p}_j = \mathbf{0}$.

Orthogonal Diagonalization of Real Symmetric Matrices

In fact, the very same method of rotations allows us to obtain a more general theorem.

Theorem (on the diagonalization of real symmetric matrices)**.** A real symmetric matrix $\mathbf{A}$ of arbitrary order n can be reduced to a diagonal matrix $\mathbf{D}$ by means of some orthogonal matrix $\mathbf{P}$:

$$\mathbf{D} = \mathbf{P}^T\mathbf{A}\mathbf{P}, \quad \text{that is,} \quad \mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T.$$

In other words, every real symmetric matrix is orthogonally diagonalizable.

Proof. Starting with $\mathbf{A}_0 = \mathbf{A}$, let us construct a sequence of matrices $\mathbf{A}_k = \left(a_{ij}^{(k)}\right)$, $k = 0, 1, ...$, in which $\mathbf{A}_k$ is obtained from $\mathbf{A}_{k-1}$ by multiplying on the left and on the right by rotation matrices:

$$\mathbf{A}_k = \mathbf{R}_k^T\mathbf{A}_{k-1}\mathbf{R}_k, \qquad (*)$$

where $\mathbf{R}_k$ differs from the identity matrix $\mathbf{I}$ only in the four entries of the (2×2)-submatrix located at the intersection of the rows and columns with indices $i < j$ and equal to

$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}.$$

The matrix $\mathbf{R}_k$ effects a rotation through the angle φ in the coordinate plane determined by the indices $i \neq j$. Every rotation matrix of this form is obviously orthogonal.

Clearly, the symmetry of the matrix $\mathbf{A}_0 = \mathbf{A}$ is inherited by all the matrices $\mathbf{A}_k$. From our previous investigations we already know that φ can be chosen in such a way that $a_{ij}^{(k)} = a_{ji}^{(k)} = 0$. Denote by d_k and h_k the sums of the squares of the diagonal and off-diagonal entries of the matrix $\mathbf{A}_k$. Then

$$\left(a_{ii}^{(k)}\right)^2 + \left(a_{jj}^{(k)}\right)^2 = \left(a_{ii}^{(k-1)}\right)^2 + \left(a_{jj}^{(k-1)}\right)^2 + 2\left(a_{ij}^{(k-1)}\right)^2 \quad \Rightarrow$$

$$\Rightarrow \quad d_k - d_{k-1} = 2\left(a_{ij}^{(k-1)}\right)^2.$$

For each k, we shall choose the plane of rotation (the indices $i < j$) in such a way that the eliminated entry $a_{ij}^{(k-1)}$ is maximal in modulus among all off-diagonal entries of the matrix $\mathbf{A}_{k-1}$. The total number of off-diagonal entries is $n^2 - n$. Therefore

$$\left(a_{ij}^{(k-1)}\right)^2 \geq \frac{h_{k-1}}{n^2 - n}.$$

Hence, taking into account the equality $d_k + h_k = d_{k-1} + h_{k-1}$, we obtain

$$h_k \leq h_{k-1} - \frac{2}{n^2 - n} h_{k-1} \leq \left(1 - \frac{2}{n^2 - n}\right)^k h_0 \to 0 \quad \text{as } k \to \infty.$$

From relations $(*)$ it follows that

$$\mathbf{A}_k = \mathbf{P}_k^T \mathbf{A} \mathbf{P}_k, \quad k = 1, 2, ...,$$

where for all k the matrices $\mathbf{P}_k = \left(p_{ij}^{(k)}\right)$ are orthogonal (as products of orthogonal matrices).

For any fixed i, j, the sequences $a_{ij}^{(k)}, p_{ij}^{(k)}$ are bounded. By the lemma on bounded sequences, there exists a sequence of indices $k_1 < k_2 < ...$ such that each of the subsequences $a_{ij}^{(k_l)}, p_{ij}^{(k_l)}$ is convergent. Note that $a_{ij}^{(k_l)} \to 0$ for $i \neq j$. Let

$$\lim_{l \to \infty} a_{ii}^{(k_l)} = \lambda_i, \quad i = 1, ..., n, \quad \lim_{l \to \infty} p_{ij}^{(k_l)} = p_{ij}, \quad i, j = 1, ..., n.$$

Introduce the matrices

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \quad \mathbf{P} = (p_{ij}).$$

For all $l = 1, 2, ...$ we have: $\mathbf{A}_{k_l} = \mathbf{P}_{k_l}^T \mathbf{A} \mathbf{P}_{k_l}$. Passing to the limit in the entrywise equalities, we obtain

$$\mathbf{D} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

From the orthogonality conditions $\mathbf{P}_{k_l}^T \mathbf{P}_{k_l} = \mathbf{I}$ it follows that in the limit $\mathbf{P}^T \mathbf{P} = \mathbf{I}$. Hence the matrix $\mathbf{P}$ is orthogonal. $\square$

We have just obtained one of the most important results—both for matrix theory itself and for its numerous applications. In our course we shall return to discussing it in connection with a number of fundamental concepts of linear algebra. Our proof is remarkable for its constructiveness: it simultaneously provides a method for the approximate computation of the matrices $\mathbf{D}$ and $\mathbf{P}$. This is one of the earliest practical methods of computational algebra, proposed by C. Jacobi in 1846.[14]

> **Problem 1.** Given a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ with a nonzero sum of the entries on the main diagonal, prove the existence of an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n\times n}$ such that in the matrix $\mathbf{Q}^T\mathbf{A}\mathbf{Q}$ all entries on the main diagonal are equal.

Schur's Theorem BEYOND MA1522

Let $\lambda_1, ..., \lambda_n$ be a complete set of n eigenvalues of a matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ counting multiplicities. Suppose an arbitrary numbering of the eigenvalues is fixed.

> **Theorem** (Schur)**.** For any matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ with an arbitrary prescribed numbering of its eigenvalues $\lambda_1, ..., \lambda_n$, there exists a unitary matrix $\mathbf{X} \in \mathbb{C}^{n\times n}$ such that $\mathbf{B} = (b_{ij}) = \mathbf{X}^*\mathbf{A}\mathbf{X}$ is an upper triangular matrix with diagonal entries $b_{ii} = \lambda_i$, $i = 1, ..., n$.

Proof. Let $\mathbf{A}\mathbf{x}_1 = \lambda_1\mathbf{x}_1$, $\|\mathbf{x}_1\| = 1$ (the length is defined by the standard inner product). Construct an orthonormal basis $\mathbf{x}_1, ..., \mathbf{x}_n$ starting with the vector $\mathbf{x}_1$, and let $\mathbf{X} = (\mathbf{x}_1 \ \cdots \ \mathbf{x}_n)$. It is easy to verify that

$$\mathbf{A}\mathbf{X} = \mathbf{X}\begin{pmatrix} \lambda_1 & \mathbf{u}^T \\ \mathbf{0} & \mathbf{B} \end{pmatrix}, \quad \mathbf{B} \in \mathbb{C}^{(n-1)\times(n-1)}, \quad \mathbf{u} \in \mathbb{C}^{n-1}.$$

Note that $\operatorname{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A}) = (x - \lambda_1)(x - \lambda_2) \cdot ... \cdot (x - \lambda_n) = (x - \lambda_1)\det(x\mathbf{I}_{n-1} - \mathbf{B})$. Hence, $\mathbf{B}$ has the eigenvalues $\lambda_2, ..., \lambda_n$.

Arguing by induction, suppose that $\mathbf{Y}^*\mathbf{B}\mathbf{Y} = \mathbf{T}$, where $\mathbf{Y}$ is a unitary matrix of order $n-1$ and $\mathbf{T}$ is an upper triangular matrix of order $n-1$ with diagonal entries $\lambda_2, ..., \lambda_n$. As a result,

$$(\tilde{\mathbf{Y}}^*\mathbf{X}^*)\mathbf{A}(\mathbf{X}\tilde{\mathbf{Y}}) = \begin{pmatrix} \lambda_1 & \mathbf{u}^T\mathbf{Y} \\ \mathbf{0} & \mathbf{T} \end{pmatrix}, \quad \tilde{\mathbf{Y}} = \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{Y} \end{pmatrix}.$$

From the unitarity of the matrix $\mathbf{Y}$ it follows that $\tilde{\mathbf{Y}}$ is a unitary matrix. The matrix $\mathbf{X}\tilde{\mathbf{Y}}$ is unitary as a product of unitary matrices. □

The theorem on the triangular form of a matrix (see Tyrtyshnikov, §30.6) is, of course, a corollary of Schur's theorem. Moreover, Schur's theorem asserts more: a triangular form with a prescribed ordering of the eigenvalues on the diagonal is achieved by a similarity transformation using a unitary matrix.

[14] The latest results in the study of the method of rotations belong to the very recent past: in the 1990s its special capabilities associated with the high-accuracy computation of entries of the matrix $\mathbf{D}$ that are small in modulus were discovered.

Let us note the constructive character of the proof of Schur's theorem given above. Once an eigenvalue λ_1 and an associated eigenvector $\mathbf{x}_1$ have been found, the problem of determining the remaining eigenvalues reduces to a similar problem of order $n-1$.[15] This kind of dimension-reduction device is sometimes called *deflation*.

Problem 2. Prove that for any complex matrix $\mathbf{A}$ of order 3 there exists a unitary matrix $\mathbf{Q}$ such that the matrix $\mathbf{B} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$ is *tridiagonal*. (A matrix $\mathbf{B}$ is called tridiagonal if $b_{ij} = 0$ for $|i-j| > 1$.)[16]

Normal Matrices BEYOND MA1522

The foundation of matrix techniques consists of transformations and decompositions of general matrices obtained with the help of special classes of matrices.

A square complex matrix $\mathbf{A}$ is called *normal* if $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$.

Theorem. A matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is normal if and only if for some unitary matrix $\mathbf{Q} \in \mathbb{C}^{n\times n}$ the matrix $\mathbf{Q}^*\mathbf{A}\mathbf{Q}$ is diagonal.

Proof. By Schur's theorem, there exists a unitary matrix $\mathbf{Q}$ reducing $\mathbf{A}$ to upper triangular form $\mathbf{B} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$. The equality $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$ is equivalent to the equality $\mathbf{B}^*\mathbf{B} = \mathbf{B}\mathbf{B}^*$. It remains to see what this means in the case of an upper triangular matrix $\mathbf{B}$:

$$\begin{pmatrix} \overline{b}_{11} & & & \\ \overline{b}_{12} & \overline{b}_{22} & & \\ \vdots & \vdots & \ddots & \\ \overline{b}_{1n} & \overline{b}_{2n} & \dots & \overline{b}_{nn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ & b_{22} & \dots & b_{2n} \\ & & \ddots & \vdots \\ & & & b_{nn} \end{pmatrix} =$$

$$= \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ & b_{22} & \dots & b_{2n} \\ & & \ddots & \vdots \\ & & & b_{nn} \end{pmatrix} \begin{pmatrix} \overline{b}_{11} & & & \\ \overline{b}_{12} & \overline{b}_{22} & & \\ \vdots & \vdots & \ddots & \\ \overline{b}_{1n} & \overline{b}_{2n} & \dots & \overline{b}_{nn} \end{pmatrix}.$$

Equating the entries in position $(1,1)$, we obtain

$$|b_{11}|^2 = |b_{11}|^2 + |b_{12}|^2 + \dots + |b_{1n}|^2 \Rightarrow b_{12} = \dots = b_{1n} = 0.$$

Taking this into account, we equate the entries in position $(2,2)$:

$$|b_{22}|^2 = |b_{22}|^2 + |b_{23}|^2 + \dots + |b_{2n}|^2 \Rightarrow b_{23} = \dots = b_{2n} = 0.$$

[15] It does not follow from anywhere, however, that an eigenvector of the matrix $\mathbf{B}$ automatically corresponds to some eigenvector of the matrix $\mathbf{A}$.

[16] It was proved recently that the same holds for any complex matrix of order 4 (V. Pati, 2001) and that there exist matrices of order 5 that cannot be reduced to tridiagonal form by a similarity transformation using a unitary matrix.

And so on. Conclusion: an upper triangular matrix is normal if and only if it is diagonal. Hence, the equality $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$ holds if and only if $\mathbf{B}$ is a diagonal matrix. □

Corollary. A matrix is normal if and only if it possesses an orthonormal basis of eigenvectors.

Proof. Let $\mathbf{D} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$ be a diagonal matrix. The columns of the unitary matrix $\mathbf{Q}$ form an orthonormal basis and, by virtue of the equality $\mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{D}$, are eigenvectors of the matrix $\mathbf{A}$. □

As we see, any normal matrix is similar to a diagonal one, and the similarity transformation is realized by means of a unitary matrix. In such cases one speaks of *unitary similarity*.

If $\mathbf{A}^* = f(\mathbf{A})$ for some polynomial $f(\lambda)$, then the matrix $\mathbf{A}$ is obviously normal. The converse is also true. Suppose that $\mathbf{A}$ has m pairwise distinct eigenvalues $\lambda_1, ..., \lambda_m$, and take as $f(\lambda)$ a polynomial of degree at most $m-1$ that takes the value $\overline{\lambda}_i$ at λ_i. Then

$$\mathbf{D}^* = f(\mathbf{D}) \Rightarrow \mathbf{A}^* = \mathbf{Q}\mathbf{D}^*\mathbf{Q}^* = \mathbf{Q}f(\mathbf{D})\mathbf{Q}^* = f(\mathbf{A}).$$

Problem 3. Prove that the spectral radius (the maximum modulus of the eigenvalues) of a normal matrix $\mathbf{A}$ admits the representation $\rho(\mathbf{A}) = \max_{\mathbf{x}\neq\mathbf{0}} \frac{|\mathbf{x}^*\mathbf{A}\mathbf{x}|}{|\mathbf{x}^*\mathbf{x}|}$.

Hermitian Matrices

Recall that a matrix $\mathbf{A}$ is called Hermitian if $\mathbf{A}^* = \mathbf{A}$ (see Tyrtyshnikov, §25.2). Obviously, any Hermitian matrix is normal.

Proposition. A normal matrix is Hermitian if and only if all of its eigenvalues are real.

Proof. $\mathbf{A}^* = \mathbf{A} \Leftrightarrow \mathbf{D}^* = \mathbf{D} \Leftrightarrow \overline{\lambda}_i = \lambda_i$, $1 \le i \le n$. □

Remark. For a real matrix $\mathbf{A}$ we have $\mathbf{A}^* = \mathbf{A}^T$, so the real Hermitian matrices are exactly the real symmetric matrices, $\mathbf{A}^T = \mathbf{A}$; by the proposition, their eigenvalues are real. The real symmetric case is the *spectral theorem* of MA1522: a real square matrix $\mathbf{A}$ is orthogonally diagonalizable, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ with $\mathbf{P}$ orthogonal and $\mathbf{D}$ diagonal, if and only if $\mathbf{A}$ is symmetric. One direction is the theorem on the diagonalization of real symmetric matrices proved above; conversely, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ implies $\mathbf{A}^T = \mathbf{P}\mathbf{D}^T\mathbf{P}^T = \mathbf{P}\mathbf{D}\mathbf{P}^T = \mathbf{A}$.

Problem 4. It is known that $\mathbf{A}^2 = \mathbf{A}$ and $\mathrm{Null}(\mathbf{A}) \perp \mathrm{Col}(\mathbf{A})$ (orthogonality with respect to the standard inner product). Prove that $\mathbf{A} = \mathbf{A}^*$.

Problem 5. Given a subspace $V \subset \mathbb{C}^n$. Prove that among all matrices $\mathbf{A}$ such that $\mathrm{Col}(\mathbf{A}) = V$ and $\mathbf{A}^2 = \mathbf{A}$, the smallest value of the 2-norm is attained for some Hermitian matrix $\mathbf{A}$, and moreover for only one.

Problem 6. Prove that for any Hermitian matrix $\mathbf{H}$ the matrix $\mathbf{Q} = (\mathbf{I} - i\mathbf{H})^{-1}(\mathbf{I} + i\mathbf{H})$ is unitary. Can every unitary matrix be represented in this way?

Problem 7. Hermitian matrices $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$ are such that $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{B}^2 = \mathbf{B}$. Prove that if $\|\mathbf{A} - \mathbf{B}\|_2 < 1$, then $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{B})$.

6.4. Application of Diagonalization: Markov Chains

Spectral Radius BEYOND MA1522

The set of eigenvalues of a matrix is often also called its *spectrum*. The largest modulus of the eigenvalues of a matrix $\mathbf{A}$ is called its *spectral radius*. Notation: $\rho(\mathbf{A})$.

Proposition. For the spectral radius, the estimate $\rho(\mathbf{A}) \le \|\mathbf{A}\|$ holds, where $\|\cdot\|$ is an arbitrary matrix norm.

Proof. Let $\mathbf{Ax} = \lambda\mathbf{x}$, $\mathbf{x} \neq \mathbf{0}$. Then $\|\lambda\mathbf{x}\| = |\lambda|\|\mathbf{x}\| = \|\mathbf{Ax}\| \le \|\mathbf{A}\|\|\mathbf{x}\| \Rightarrow |\lambda| \le \|\mathbf{A}\|$. □

Problem 1. Prove that the spectral radius is obtained as the limit $\rho(\mathbf{A}) = \lim_{k\to\infty} \left\|\mathbf{A}^k\right\|^{1/k}$, where $\|\cdot\|$ is an arbitrary fixed matrix norm. (By the theorem on upper triangular form (see Tyrtyshnikov, §30.6), it suffices to consider the case of an upper triangular matrix $\mathbf{A}$.)

Problem 2. Prove that $\rho(\mathbf{A}) = \inf \left\|\mathbf{X}^{-1}\mathbf{AX}\right\|_p$, where the infimum is taken over all invertible matrices $\mathbf{X}$, and $\|\cdot\|_p$ is the operator norm induced by the vector p-norm.

Problem 3. For the entries of square matrices $\mathbf{A}$ and $\mathbf{B}$, the inequalities $0 \le a_{ij} \le b_{ij}$, $1 \le i, j \le n$, hold. Prove that $\rho(\mathbf{A}) \le \rho(\mathbf{B})$.

Problem 4. All entries of a square matrix $\mathbf{A}$ are nonnegative, and the sums of the entries in each row are equal to the same number λ. Prove that λ is the eigenvalue of $\mathbf{A}$ of largest modulus.

Remark. In the general case, a matrix may have no nonnegative eigenvalues, so $\rho(\mathbf{A})$ need not be an eigenvalue of the matrix $\mathbf{A}$. However, for any *nonnegative matrix*—a matrix all of whose entries are nonnegative—it is known that the spectral radius is necessarily also an eigenvalue of it, and as an associated eigenvector one can choose a vector with nonnegative entries (this is the fundamental result of the theory of nonnegative matrices, known as the *Perron–Frobenius theorem*). A possible outline of the proof in the case of a *positive matrix*—a matrix all of whose entries are positive—is,

for instance, the following: (1) the minimum value of the norm $\left\|\mathbf{D}^{-1}\mathbf{A}\mathbf{D}\right\|_\infty$ on the set of all diagonal matrices $\mathbf{D}$ with positive diagonal is *attained* at some diagonal matrix $\hat{\mathbf{D}}$ (why?); (2) in the matrix $\hat{\mathbf{A}} \equiv \hat{\mathbf{D}}^{-1}\mathbf{A}\hat{\mathbf{D}}$, the sums of the moduli of the entries in each row are identical and consequently equal to $\rho(\mathbf{A})$ (if there are rows with different sums, then one can find a diagonal matrix $\mathbf{D}$ for which $\left\|\mathbf{D}^{-1}\hat{\mathbf{A}}\mathbf{D}\right\|_\infty < \left\|\hat{\mathbf{A}}\right\|_\infty$); (3) the vector formed from the diagonal entries of $\hat{\mathbf{D}}$ is an eigenvector of $\mathbf{A}$ associated to the eigenvalue $\rho(\mathbf{A})$.

A stochastic matrix $\mathbf{P}$ of a Markov chain (a square matrix whose columns are probability vectors) is a nonnegative matrix whose columns, rather than rows, sum to 1; Problem 4 therefore applies to its transpose $\mathbf{P}^T$, with $\lambda = 1$.

6.5. Singular Value Decomposition

The Operator Norm BEYOND MA1522

Proposition 1. The set $\mathcal{L}(V, W)$ of all bounded linear transformations from V to W is a vector space (over the field common to V and W).

Proof. Let $\|T(\mathbf{x})\|_W \le c_1\|\mathbf{x}\|_V$ and $\|S(\mathbf{x})\|_W \le c_2\|\mathbf{x}\|_V$. Then for any scalars α and β,

$$\|(\alpha T + \beta S)(\mathbf{x})\|_W \le c\|\mathbf{x}\|_V, \quad c = |\alpha|c_1 + |\beta|c_2. \qquad \square$$

Proposition 2. The quantity

$$\|T\| \equiv \sup_{\|\mathbf{x}\|_V = 1} \|T(\mathbf{x})\|_W, \quad T \in \mathcal{L}(V, W), \tag{1}$$

is a norm on the vector space $\mathcal{L}(V, W)$.

Proof. Obviously, the quantity $\|T\|$ has a finite value and is, of course, nonnegative. If $\|T\| = 0$, then $\|T(\mathbf{x})\|_W = 0$ on the unit sphere $\|\mathbf{x}\|_V = 1$. Hence $\|T(\mathbf{x})\|_W = 0 \ \ \forall \mathbf{x} \in V \Rightarrow T(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in V \Rightarrow T = 0$. Positive homogeneity follows from the equality

$$\|\alpha T(\mathbf{x})\|_W = |\alpha|\|T(\mathbf{x})\|_W,$$

and the triangle inequality follows from the inequality

$$\|(\alpha T + \beta S)(\mathbf{x})\|_W \le |\alpha|\|T(\mathbf{x})\|_W + |\beta|\|S(\mathbf{x})\|_W. \qquad \square$$

Definition. The norm (1) for linear transformations $T \in \mathcal{L}(V, W)$ is called the *operator norm* or the norm *subordinate* to the vector norms $\|\cdot\|_V, \|\cdot\|_W$.

Proposition 3. If V is a finite-dimensional space, then every linear transformation $T : V \to W$ is bounded and $\|T\| = \|T(\mathbf{x}_0)\|_W$ for some vector $\mathbf{x}_0 \in V$ (depending on T) with norm $\|\mathbf{x}_0\|_V = 1$.

Proof. Let $\{\mathbf{e}_1, ..., \mathbf{e}_n\}$ be a basis for V and $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{e}_i$. Then $\|\mathbf{x}\|_{(e)} \equiv \sum_{i=1}^n |\alpha_i|$ is a norm on V, equivalent to any other norm, including the norm $\|\mathbf{x}\|_V$. Therefore, for some $c > 0$,

$$\|\mathbf{x}\|_{(e)} \le c\|\mathbf{x}\|_V \quad \forall \mathbf{x} \in V.$$

Consequently,

$$\|T(\mathbf{x})\|_W \le \sum_{i=1}^n |\alpha_i| \max_{1\le i\le n} \|T(\mathbf{e}_i)\|_W = \|\mathbf{x}\|_{(e)} \max_{1\le i\le n} \|T(\mathbf{e}_i)\|_W \le$$

$$\le \left(c \max_{1\le i\le n} \|T(\mathbf{e}_i)\|_W\right) \|\mathbf{x}\|_V \quad \forall \mathbf{x} \in V.$$

To prove the existence of $\mathbf{x}_0$, it suffices to take into account the compactness of the unit sphere in a finite-dimensional space, the continuity of the function $\|T(\mathbf{x})\|_W$ on it, and Weierstrass's theorem. □

If $1 \le p \le \infty$ is fixed and the Hölder p-norm is chosen as the norm in the space $\mathbb{C}^n$, then the corresponding operator norm of a matrix $\mathbf{A}$ (regarded as the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$) is customarily denoted by $\|\mathbf{A}\|_p$.

Problem 1. Let $\mathbf{A} = \left(a_{ij}\right)_{m\times n}$. Prove that

$$\|\mathbf{A}\|_\infty = \max_{1\le i\le m} \sum_{j=1}^n |a_{ij}|, \quad \|\mathbf{A}\|_1 = \max_{1\le j\le n} \sum_{i=1}^m |a_{ij}|.$$

Problem 2. Let $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$. Prove that $\left\|\mathbf{u}\mathbf{v}^T\right\|_2 = \|\mathbf{u}\|\|\mathbf{v}\|$.

Singular Value Decomposition of a Matrix

In the 1870s, independently and almost simultaneously, Beltrami (1873) and Jordan (1874) discovered that any square matrix can be reduced to diagonal form by multiplying on the left and on the right by unitary matrices. Various questions connected with this discovery, including its generalizations, subsequently became the subject of a whole series of investigations. It would not be a great exaggeration to say that this fact turned out to be tremendously useful and one of the most sought-after results in matrix theory and applications of linear algebra.

In reality, the same holds for a rectangular matrix as well. By multiplication by unitary matrices, it is reduced to a rectangular matrix of the same dimensions having zeros everywhere except at the entries with indices $i = j$. We shall call such matrices *diagonal rectangular* matrices.

Thus, we are dealing with a decomposition of the form

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \tag{2}$$

where $\mathbf{A}$ is a given $(m \times n)$-matrix, $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices of orders m and n respectively,[17] and $\mathbf{\Sigma}$ is a diagonal rectangular $(m \times n)$-matrix having in the positions $(1,1), (2,2), \ldots$ the nonnegative numbers

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{\min(m,n)}.$$

The decomposition (2) is called the *singular value decomposition* of the matrix $\mathbf{A}$. The numbers σ_i are called the *singular values* of the matrix $\mathbf{A}$.

Theorem. A singular value decomposition $\mathbf{A} = \mathbf{U\Sigma V}^*$ exists for any complex rectangular matrix $\mathbf{A}$. If $\mathbf{A}$ is real, then the matrices $\mathbf{U}$ and $\mathbf{V}$ can be chosen to be real (that is, orthogonal), and then $\mathbf{A} = \mathbf{U\Sigma V}^T$.

Proof. The case $\mathbf{A} = \mathbf{0}$ is trivial, so we shall assume that $\mathbf{A} \neq \mathbf{0}$. Set $\sigma_1 = \|\mathbf{A}\|_2 = \sup_{\mathbf{x}\neq\mathbf{0}} \|\mathbf{Ax}\|/\|\mathbf{x}\|$. By the compactness of the unit sphere in $\mathbb{C}^n$, the continuity of the norm, and Weierstrass's theorem, there exists a vector $\mathbf{x}_1$ such that $\|\mathbf{Ax}_1\| = \|\mathbf{A}\|_2$ and $\|\mathbf{x}_1\| = 1$. Let $\mathbf{y}_1 = \mathbf{Ax}_1/\|\mathbf{Ax}_1\|$. Thus,

$$\mathbf{Ax}_1 = \sigma_1 \mathbf{y}_1, \quad \|\mathbf{x}_1\| = \|\mathbf{y}_1\|. \tag{3}$$

Let us complete $\mathbf{x}_1$ and $\mathbf{y}_1$ to orthonormal bases and form the unitary matrices

$$\mathbf{U}_1 = (\mathbf{y}_1 \;\; \mathbf{y}_2 \;\; \cdots \;\; \mathbf{y}_m), \quad \mathbf{V}_1 = (\mathbf{x}_1 \;\; \mathbf{x}_2 \;\; \cdots \;\; \mathbf{x}_n).$$

According to (3), the matrix $\mathbf{A}_1 \equiv \mathbf{U}_1^* \mathbf{A} \mathbf{V}_1$ has only one nonzero entry in the first column, equal to σ_1:

$$\mathbf{A}_1 = \mathbf{U}_1^* \mathbf{A} \mathbf{V}_1 = \begin{pmatrix} \sigma_1 & \mathbf{z}^* \\ \mathbf{0} & \mathbf{A}_2 \end{pmatrix}.$$

By the unitary invariance of the spectral norm (see Tyrtyshnikov, §27.7), $\|\mathbf{A}_1\|_2 = \sigma_1$. Therefore,

$$\sigma_1 \left\| \begin{pmatrix} \sigma_1 \\ \mathbf{z} \end{pmatrix} \right\| \geq \left\| \mathbf{A}_1 \begin{pmatrix} \sigma_1 \\ \mathbf{z} \end{pmatrix} \right\| \geq \sigma_1^2 + \|\mathbf{z}\|^2 \quad \Rightarrow \quad \sigma_1^2 \geq \sigma_1^2 + \|\mathbf{z}\|^2 \quad \Rightarrow$$

$$\Rightarrow \quad \mathbf{z} = \mathbf{0} \quad \Rightarrow \quad \mathbf{A}_1 = \begin{pmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 \end{pmatrix}.$$

Next we argue by induction. If a singular value decomposition $\mathbf{A}_2 = \mathbf{U}_2 \mathbf{\Sigma}_2 \mathbf{V}_2^*$ is already available for $\mathbf{A}_2$, then a singular value decomposition for $\mathbf{A}$ is easily found. For this it suffices to take

$$\mathbf{U} = \mathbf{U}_1 \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_2 \end{pmatrix}, \quad \mathbf{V} = \mathbf{V}_1 \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_2 \end{pmatrix},$$

[17] The full textbook writes this decomposition as $\mathbf{A} = \mathbf{V\Sigma U}^*$, with the letters $\mathbf{U}$ and $\mathbf{V}$, and the singular vectors $\mathbf{u}_i$ and $\mathbf{v}_i$, interchanged; this edition follows MA1522, where a real matrix has $\mathbf{A} = \mathbf{U\Sigma V}^T$. (With the textbook's lettering, the Russian original inadvertently gives the orders of $\mathbf{U}$ and $\mathbf{V}$ as m and n.)

note that the matrices $\mathbf{U}$ and $\mathbf{V}$ are unitary (as products of unitary matrices), and verify that the equality

$$\mathbf{U}^*\mathbf{A}\mathbf{V} = \begin{pmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_2 \end{pmatrix}$$

holds.

It remains to note that the induction begins with constructing the singular value decomposition for matrices consisting of a single column or a single row.

Let $\mathbf{A} = (\mathbf{a}) \in \mathbb{C}^{m\times 1}$ be a column matrix. In this case we find in $\mathbb{C}^m$ an orthonormal basis $\mathbf{u}_1, ..., \mathbf{u}_m$ starting with $\mathbf{u}_1 = \mathbf{a}/\|\mathbf{a}\|$. Then

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \quad \mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m), \quad \mathbf{\Sigma} = (\|\mathbf{a}\|, 0, ..., 0)^T, \quad \mathbf{V} = (1) \in \mathbb{C}^{1\times 1}.$$

For a row matrix, the singular value decomposition is obtained by transposition. □

Corollary 1. The spectral norm of a matrix is equal to its largest singular value.

Corollary 2. Let the matrix $\mathbf{A}$ be invertible and let σ_n be its smallest singular value. Then $\left\|\mathbf{A}^{-1}\right\|_2 = 1/\sigma_n$.

Now it is clear that the largest singular value of a matrix and the smallest singular value of an invertible matrix are uniquely determined. The same holds for the entire set of singular values, but we shall prove this later. The singular value decomposition and its important consequences deserve a more thorough discussion, to which we shall return later in this section.

Problem 3. Prove the inequality $\|\mathbf{A}\|_F \le \sqrt{\operatorname{rank}(\mathbf{A})}\|\mathbf{A}\|_2$.

Problem 4. Prove that

$$\max_{\mathbf{A}\in\mathbb{C}^{n\times n}, \mathbf{A}\neq\mathbf{0}} \frac{\|\mathbf{A}\|_2}{\|\mathbf{A}\|_1} = \sqrt{n}.$$

Problem 5. Let $\mathbf{A} = \left(a_{ij}\right)$ and $\mathbf{D} = \left(d_{ij}\right)$ be complex matrices of order n, where $\mathbf{D}$ is a diagonal matrix with entries $d_{ii} = a_{ii}$ for $1 \le i \le n$. Prove that if $\|\mathbf{A}\|_2 = \|\mathbf{D}\|_2$, then the number of zero entries in the matrix $\mathbf{A}$ is at least $2n - 2$.

Problem 6. Let $\mathbf{L}$ be the lower triangular matrix with lower triangular part taken from a matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$. Prove that

$$\|\mathbf{L}\|_2 \le \log_2(2n)\|\mathbf{A}\|_2.$$

Singular Values and Singular Vectors

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$. Then $\mathbf{A}^*\mathbf{A} \in \mathbb{C}^{n\times n}$ is a Hermitian nonnegative definite matrix:

$$(\mathbf{A}^*\mathbf{A})^* = \mathbf{A}^*(\mathbf{A}^*)^* = \mathbf{A}^*\mathbf{A};$$

$$\mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{x} = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x}\rangle = \|\mathbf{A}\mathbf{x}\|^2 \ge 0 \quad \forall \mathbf{x} \in \mathbb{C}^n.$$

Therefore, all of its eigenvalues are nonnegative.

The nonnegative square roots of the eigenvalues of the matrix $\mathbf{A}^*\mathbf{A}$ are called the *singular values* of the matrix $\mathbf{A}$. It is customary to number the singular values $\sigma_i = \sigma_i(\mathbf{A})$ in nonincreasing order:

$$\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_r > \sigma_{r+1} = ... = \sigma_n = 0.$$

We shall assume that $\mathbf{A}$ has r nonzero singular values.

Let $\mathbf{v}_1, ..., \mathbf{v}_n$ be an orthonormal basis of eigenvectors of the matrix $\mathbf{A}^*\mathbf{A}$ such that

$$\mathbf{A}^*\mathbf{A}\mathbf{v}_i = \begin{cases} \sigma_i^2 \mathbf{v}_i, & 1 \leq i \leq r, \\ \mathbf{0}, & r+1 \leq i \leq n. \end{cases}$$

Set $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i$, $1 \leq i \leq r$. Then $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$. Let us extend the set $\mathbf{u}_1, ..., \mathbf{u}_r$ by vectors $\mathbf{u}_{r+1}, ..., \mathbf{u}_m$ to an orthonormal basis in $\mathbb{C}^m$. Note also that for $j \geq r+1$,

$$\mathbf{A}^*\mathbf{A}\mathbf{v}_j = \mathbf{0} \Rightarrow \mathbf{v}_j^*\mathbf{A}^*\mathbf{A}\mathbf{v}_j = 0 \Rightarrow \big(\mathbf{A}\mathbf{v}_j\big)^*\big(\mathbf{A}\mathbf{v}_j\big) = 0 \Rightarrow \big\|\mathbf{A}\mathbf{v}_j\big\| = 0 \Rightarrow \mathbf{A}\mathbf{v}_j = \mathbf{0}.$$

As a result, we obtain

$$\mathbf{A}(\mathbf{v}_1 \ \cdots \ \mathbf{v}_n) = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m) \begin{pmatrix} \sigma_1 & & 0 \\ & \ddots & \\ & & \sigma_r \\ 0 & & 0 \end{pmatrix} \Rightarrow \mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma},$$

where $\mathbf{V} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$ and $\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m)$ are unitary matrices, and $\mathbf{\Sigma}$ is a diagonal rectangular matrix of the same dimensions as the matrix $\mathbf{A}$.

The columns of the matrices $\mathbf{U}$ and $\mathbf{V}$ form *singular bases* of the matrix $\mathbf{A}$. The columns of $\mathbf{V}$ are called *right singular vectors*, and the columns of $\mathbf{U}$ are called *left singular vectors* of the matrix $\mathbf{A}$. The relationship between the singular vectors and the nonzero singular values is established by the relations

$$\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i, \quad \mathbf{A}^*\mathbf{u}_i = \sigma_i \mathbf{v}_i, \quad 1 \leq i \leq r.$$

Moreover,

$$\mathbf{A}\mathbf{v}_i = \mathbf{0}, \quad r+1 \leq i \leq n, \quad \mathbf{A}^*\mathbf{u}_i = \mathbf{0}, \quad r+1 \leq i \leq m.$$

Thus, we have proved that *for any matrix* $\mathbf{A} \in \mathbb{C}^{m\times n}$ *there holds the equality*

$$\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma} \tag{4}$$

for some unitary matrices $\mathbf{V} \in \mathbb{C}^{n\times n}$, $\mathbf{U} \in \mathbb{C}^{m\times m}$ *and a diagonal rectangular matrix of dimensions* $m \times n$ *with numbers* $\sigma_i \geq 0$ *for* $i = j$. Writing (4) in the form

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \tag{5}$$

we obtain a representation of the matrix called its *singular value decompo-*

sition.[18]

If a decomposition (5) with unitary matrices $\mathbf{U}$ and $\mathbf{V}$ has been obtained in some way, then $\mathbf{A}^*\mathbf{A} = \mathbf{V}(\mathbf{\Sigma}^*\mathbf{\Sigma})\mathbf{V}^*$. Therefore, if $\mathbf{\Sigma}$ is a diagonal rectangular matrix with nonnegative entries, then its nonzero entries are uniquely determined.

For a real matrix $\mathbf{A}$ we have $\mathbf{A}^*\mathbf{A} = \mathbf{A}^T\mathbf{A}$, all the vectors above can be chosen real, and (5) becomes the decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ with orthogonal $\mathbf{U}$ and $\mathbf{V}$; this is the construction of the singular value decomposition used in MA1522.

Problem 7. Find the singular value decomposition of the $(2 \times n)$-matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 & ... & 1 \\ 1 & 1 & ... & 1 \end{pmatrix}$.

Consequences of the Singular Value Decomposition

1) *The number of nonzero singular values r equals the rank of the matrix $\mathbf{A}$.*

2) *The singular value decomposition of the conjugate transpose matrix has the form*

$$\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^*.$$

3) $\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{u}_1, ..., \mathbf{u}_r\}$, $\mathrm{Null}(\mathbf{A}) = \mathrm{span}\{\mathbf{v}_{r+1}, ..., \mathbf{v}_n\}$.

4) $\mathrm{Col}(\mathbf{A}^*) = \mathrm{span}\{\mathbf{v}_1, ..., \mathbf{v}_r\}$, $\mathrm{Null}(\mathbf{A}^*) = \mathrm{span}\{\mathbf{u}_{r+1}, ..., \mathbf{u}_m\}$.

As a corollary, one can obtain representations of the spaces as orthogonal sums:

$$\mathbb{C}^n = \mathrm{Null}(\mathbf{A}) \oplus \mathrm{Col}(\mathbf{A}^*), \quad \mathbb{C}^m = \mathrm{Null}(\mathbf{A}^*) \oplus \mathrm{Col}(\mathbf{A}).$$

5) $\mathbf{A} = \sum_{k=1}^{r} \sigma_k \mathbf{u}_k \mathbf{v}_k^*$, $\mathbf{A}^* = \sum_{k=1}^{r} \sigma_k \mathbf{v}_k \mathbf{u}_k^*$.

6) *If $m = n = r$ (the matrix $\mathbf{A}$ is invertible), then*

$$\mathbf{A} = \sum_{k=1}^{n} \sigma_k \mathbf{u}_k \mathbf{v}_k^*, \quad \mathbf{A}^{-1} = \sum_{k=1}^{n} \frac{1}{\sigma_k} \mathbf{v}_k \mathbf{u}_k^*.$$

7) *Let $\sigma_1 \geq ... \geq \sigma_n$ be the singular values of an invertible matrix $\mathbf{A}$. Then $\sigma_n^{-1} \geq ... \geq \sigma_1^{-1}$ are the singular values of the matrix $\mathbf{A}^{-1}$.*

8) $\|\mathbf{A}\|_2 = \sigma_1$, $\|\mathbf{A}\|_F = \sqrt{\sigma_1^2 + ... + \sigma_r^2}$.

Proof. The spectral and Frobenius norms are unitarily invariant. Therefore $\|\mathbf{A}\|_2 = \|\mathbf{\Sigma}\|_2$ and $\|\mathbf{A}\|_F = \|\mathbf{\Sigma}\|_F$. Obviously, $\|\mathbf{\Sigma}\mathbf{x}\| \leq \sigma_1 \|\mathbf{x}\|$; equality is attained if $\mathbf{x}$ has 1 in the first position and 0 in the remaining ones.

It is also clear that $\|\mathbf{\Sigma}\|_F = \sqrt{\sigma_1^2 + ... + \sigma_r^2}$. □

Problem 8. Given a square matrix with norm $\|\mathbf{A}\|_2 \leq 1$. Prove that there exist square matrices $\mathbf{B}, \mathbf{C}, \mathbf{D}$ such that the matrix $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ is unitary.

[18] It was obtained by an altogether different method earlier in this section, in the subsection "Singular Value Decomposition of a Matrix".

Problem 9. Let $\mathbf{A} = \mathbf{A}^T \in \mathbb{C}^{n\times n}$. Prove that the matrix $\mathbf{A}$ possesses a singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ with the additional condition $\mathbf{V}^* = \mathbf{U}^T$.

Best Low-Rank Approximations BEYOND MA1522

In each matrix $\sigma_k \mathbf{u}_k \mathbf{v}_k^*$, the entry in position (i, j) may be regarded as a function of i and j with separated discrete variables i and j: $f(i, j) = f_1(i) f_2(j)$. Thus, writing $\mathbf{A}$ in the form $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ describes a certain special way of separating variables in each term of the sum or, in matrix terminology, a skeleton decomposition of the matrix $\mathbf{A}$—moreover, with the important additional property of orthonormality of the sets $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ and $\{\mathbf{v}_1, ..., \mathbf{v}_r\}$.

The special value and breadth of applications of the singular value decomposition are due above all to the fact that it provides a simple and reliable mechanism for eliminating the "least significant information" from a matrix—by approximating it with a sum of a *smaller number of summands* with separated variables i and j. What is at issue is finding an element of best approximation for a given matrix $\mathbf{A}$ on a rather complicated set—the set of matrices whose rank is bounded by a given number.

Theorem (Best Low-Rank Approximations)**.** Let a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ be given by a singular value decomposition of the form

$$\mathbf{A} = \sum_{l=1}^{r} \sigma_l \mathbf{u}_l \mathbf{v}_l^*,$$

and let us agree to set $\sigma_{r+1} = 0$. Let an integer $1 \le k \le r$ be given. Then

$$\min_{\substack{\operatorname{rank}(\mathbf{B})\le k \\ \mathbf{B}\in\mathbb{C}^{m\times n}}} \|\mathbf{A} - \mathbf{B}\|_2 = \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2,$$

where $\mathbf{A}_k = \sum_{l=1}^{k} \sigma_l \mathbf{u}_l \mathbf{v}_l^*$.

Proof. Let $\operatorname{rank}(\mathbf{B}) \le k$. Then $\dim(\operatorname{Null}(\mathbf{B})) \ge n - k$. Consider the subspace $W = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_{k+1}\}$ spanned by the leading singular vectors. By Grassmann's theorem,

$$\dim(\operatorname{Null}(\mathbf{B}) \cap W) = \dim(\operatorname{Null}(\mathbf{B})) + \dim(W) - \dim(\operatorname{Null}(\mathbf{B}) + W) \ge$$
$$\ge (n - k) + (k + 1) - n = 1.$$

Therefore, there exists a nonzero vector $\mathbf{z} \in \operatorname{Null}(\mathbf{B}) \cap W$. We may assume that $\|\mathbf{z}\| = 1$. Taking into account that

$$\mathbf{z} = \sum_{l=1}^{k+1} \alpha_l \mathbf{v}_l, \quad \sum_{l=1}^{k+1} |\alpha_l|^2 = 1,$$

we find

$$\|\mathbf{A} - \mathbf{B}\|_2 \ge \|(\mathbf{A} - \mathbf{B})\mathbf{z}\| = \|\mathbf{A}\mathbf{z}\| = \sqrt{\sum_{l=1}^{k+1} |\alpha_l|^2 \sigma_l^2} \ge \sigma_{k+1}.$$

At the same time,

$$\mathbf{A} - \mathbf{A}_k = \sum_{l=k+1}^{r} \sigma_l \mathbf{u}_l \mathbf{v}_l^* \Rightarrow \|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1}. \qquad \square$$

Chapter 7. Linear Transformations

7.1. Introduction to Linear Transformations

Mappings, Functions, Operators

All three words in the title of this section mean the same thing—we are speaking of a rule according to which to each element x of a set X there is assigned a uniquely determined element $y = f(x)$ of a set Y. Specifying such a rule is equivalent to choosing a subset

$$\Gamma = \{(x, f(x)) : x \in X\} \subset X \times Y,$$

called the *graph* of the mapping (function, operator) f.

The element $y = f(x)$ is called the *image* of the element x, and x is called a *preimage* of the element y under the mapping f. To emphasize that f acts from X into Y, one writes $f : X \to Y$; the set X is called the *domain* and Y the *codomain* of f.

The set $f(X) \equiv \{y : y = f(x) \text{ for some } x \in X\}$ is called the *range* (or *image*) of the mapping f.

If $M \subset Y$, then the set $f^{-1}(M) \equiv \{x : f(x) \in M\}$ is called the *complete preimage* of the set M. If $M = \{y\}$, one writes $f^{-1}(y) = f^{-1}(M)$.

A mapping $f : X \to Y$ is called *invertible* if there exists a mapping $g : Y \to X$ such that $f(g(y)) = y \ \forall y \in Y$ and $g(f(x)) = x \ \forall x \in X$. In this case, g is called the *inverse mapping* for f, and one writes $g = f^{-1}$.

A mapping f is called *bijective* if for every $y \in Y$ the complete preimage $f^{-1}(y)$ consists of exactly one element; that is, f is both *injective*, or *one-to-one* (each $f^{-1}(y)$ has at most one element: $f(x_1) = f(x_2)$ implies $x_1 = x_2$), and *surjective*, or *onto* (each $f^{-1}(y)$ is nonempty: $f(X) = Y$). (The full textbook calls a bijective mapping "one-to-one"; in this edition, as in the course, one-to-one means injective.) It is easy to show that invertibility is equivalent to being bijective.

Linear Transformations

Any matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ can be naturally regarded as a mapping that sends a vector $\mathbf{x} \in \mathbb{C}^n$ to the vector $\mathbf{Ax} \in \mathbb{C}^m$. Obviously, this mapping possesses the *linearity property*:[19]

$$\mathbf{A}(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha\mathbf{Ax} + \beta\mathbf{Ay} \quad \forall\alpha, \beta \in \mathbb{C}, \quad \forall\mathbf{x}, \mathbf{y} \in \mathbb{C}^n.$$

[19]This property can also be viewed as the property of *preserving operations* under a mapping of one vector space into another space over the same field. Such mappings are called *homomorphisms*.

(For a real $m \times n$ matrix $\mathbf{A}$ this is the linear transformation $T_{\mathbf{A}} : \mathbb{R}^n \to \mathbb{R}^m$, $T_{\mathbf{A}}(\mathbf{u}) = \mathbf{A}\mathbf{u}$, defined by $\mathbf{A}$.)

The linearity property holds for many very important mappings in vector spaces whose elements are functions united by some common feature (continuity, differentiability, etc.). Above all, one should mention mappings associated with differentiation and integration of functions. Thus, we have sufficient grounds to study the linearity property from a more general standpoint.

Definition. Let V and W be arbitrary vector spaces over the same field $\mathbb{P}$. A mapping $T : V \to W$ with the property

$$T(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y}) \quad \forall \alpha, \beta \in \mathbb{P}, \quad \forall \mathbf{x}, \mathbf{y} \in V$$

is called a *linear transformation* (or *linear operator*) from V to W. For linear transformations the argument is often written without parentheses, $T\mathbf{x} = T(\mathbf{x})$; in this edition we keep the parentheses, as the course does.

Matrix of a Linear Transformation

Consider a linear transformation $T : V_n \to V_m$, where V_n and V_m are vector spaces of dimensions n and m (over a common field $\mathbb{P}$).

Fix some basis $S = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ of V_n and some basis $C = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$ of V_m. By the linearity of the transformation T,

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = x_1 T(\mathbf{u}_1) + ... + x_n T(\mathbf{u}_n). \tag{1}$$

Therefore T is completely determined by its action on the basis vectors $\mathbf{u}_1, ..., \mathbf{u}_n$. Let us expand the images of the basis vectors with respect to the basis of the codomain:

$$T(\mathbf{u}_j) = a_{1j}\mathbf{v}_1 + ... + a_{mj}\mathbf{v}_m, \quad j = 1, ..., n. \tag{2}$$

From (1) and (2) we obtain

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = (a_{11}x_1 + ... + a_{1n}x_n)\mathbf{v}_1 + ... + \\ +(a_{m1}x_1 + ... + a_{mn}x_n)\mathbf{v}_m.$$

Consequently,

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = y_1\mathbf{v}_1 + ... + y_m\mathbf{v}_m \quad \Leftrightarrow$$

$$\Leftrightarrow \quad \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} a_{11} & ... & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{m1} & ... & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

The matrix appearing on the right is called the *matrix of the linear transformation* T in the pair of bases S and C (its representation with respect to S and C); we denote it by $[T]_{S,C}$. Its j-th column is $\left[T(\mathbf{u}_j)\right]_C$, and the equivalence above reads $[T(\mathbf{x})]_C = [T]_{S,C}[\mathbf{x}]_S$ for every $\mathbf{x} \in V_n$.

For $T : \mathbb{R}^n \to \mathbb{R}^m$ and the standard basis $C = E$ of $\mathbb{R}^m$ (where $[\mathbf{w}]_E = \mathbf{w}$), this matrix is the *representation of T with respect to the basis S*, denoted in

the course by $[T]_S$; if, in addition, $S = E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$, it is the *standard matrix* $\mathbf{A} = [T]_E$ of T:

$$[T]_S = [T]_{S,E} = \begin{pmatrix} T(\mathbf{u}_1) & T(\mathbf{u}_2) & \cdots & T(\mathbf{u}_n) \end{pmatrix},$$
$$\mathbf{A} = [T]_E = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix}.$$

Thus $T(\mathbf{x}) = [T]_S[\mathbf{x}]_S$ and $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. (Note that the course's $[T]_S$ is $[T]_{S,E}$, not $[T]_{S,S}$.)

Thus, any fixed pair of bases generates three isomorphisms (in the sense of Section 7.2):

$$V_n \leftrightarrow \mathbb{P}^n, \quad V_m \leftrightarrow \mathbb{P}^m, \quad \mathcal{L}(V_n, V_m) \leftrightarrow \mathbb{P}^{m\times n},$$

where $\mathcal{L}(V_n, V_m)$ is the vector space of all linear transformations from the space V_n into the space V_m, and $\mathbb{P}^{m\times n}$ is the vector space of all $m \times n$ matrices with entries from the field $\mathbb{P}$. From this, in particular, it is seen that the dimension of the space of linear transformations $\mathcal{L}(V_n, V_m)$ is equal to mn.

Example. Let $D : V \to W$ be the differentiation transformation on the space of polynomials $V = W$ of degree 2 and lower. Consider the basis $S = \{1 + t, 1 - t, t^2\}$ of V and the basis $C = \{1, t, t^2\}$ of W. Obviously,

$$D(1 + t) = 1, \quad D(1 - t) = -1, \quad D(t^2) = 2t.$$

Therefore, in the pair of bases S and C, the matrix of the linear transformation D has the form

$$[D]_{S,C} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

What will be the matrix of the same transformation if the same basis S is chosen in $W = V$? For this we need to find the expansions of the images of the vectors $1 + t$, $1 - t$, t^2 of S with respect to the same vectors $1 + t$, $1 - t$, t^2:

$$\left.\begin{aligned} D(1 + t) &= \tfrac{1}{2}(1 + t) + \tfrac{1}{2}(1 - t), \\ D(1 - t) &= -\tfrac{1}{2}(1 + t) - \tfrac{1}{2}(1 - t), \\ D(t^2) &= (1 + t) - (1 - t) \end{aligned}\right\} \quad \Rightarrow \quad [D]_{S,S} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Composition of Linear Transformations

The product of linear transformations $T_1 : V_n \to V_m$ and $T_2 : V_m \to V_k$ is defined as the composition of mappings: T_2T_1 is the transformation from V_n to V_k defined by the rule $(T_2T_1)(\mathbf{x}) = T_2(T_1(\mathbf{x}))$. It is elementary to verify that the product of linear transformations is a linear transformation.

Let $\mathbf{A} = [T_1]_{S,C}$ be the matrix of a linear transformation T_1 in a pair of bases S and C, and let $\mathbf{B} = [T_2]_{C,D}$ be the matrix of a linear transformation T_2 in a pair of bases C and D. Then *the product of the matrices* $\mathbf{BA}$ *is the*

matrix of the product of the transformations T_2T_1 in the pair of bases S and D: $[T_2T_1]_{S,D} = [T_2]_{C,D}[T_1]_{S,C}$.

The proof reduces to a direct verification. Note that the full textbook actually began with the definition of matrix multiplication (reproduced here in Sections 2.1 and 2.2) and, in effect, with a discussion of the composition of linear mappings!

Change of Bases

Let $[T]_{S,C}$ be the matrix of a linear transformation T in a pair of bases $S = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ and $C = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$. How can one find the matrix $[T]_{S',C'}$ of the same transformation in another pair of bases: $S' = \{\mathbf{u}'_1, ..., \mathbf{u}'_n\}$ and $C' = \{\mathbf{v}'_1, ..., \mathbf{v}'_m\}$?

Consider the equalities

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = y_1\mathbf{v}_1 + ... + y_m\mathbf{v}_m,$$
$$T(z_1\mathbf{u}'_1 + ... + z_n\mathbf{u}'_n) = w_1\mathbf{v}'_1 + ... + w_m\mathbf{v}'_m.$$

According to the definition of the matrices $[T]_{S,C}$ and $[T]_{S',C'}$, we find

$$[T]_{S,C}\mathbf{x} = \mathbf{y}, \quad [T]_{S',C'}\mathbf{z} = \mathbf{w},$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_m \end{pmatrix}.$$

Next we write

$$\mathbf{u}'_j = p_{1j}\mathbf{u}_1 + ... + p_{nj}\mathbf{u}_n, \quad 1 \le j \le n,$$
$$\mathbf{v}'_i = q_{1i}\mathbf{v}_1 + ... + q_{mi}\mathbf{v}_m, \quad 1 \le i \le m,$$

and introduce the transition matrices

$$\mathbf{P} = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \vdots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} q_{11} & \cdots & q_{1m} \\ \vdots & \vdots & \vdots \\ q_{m1} & \cdots & q_{mm} \end{pmatrix}.$$

The columns of $\mathbf{P}$ are $\left[\mathbf{u}'_j\right]_S$, so $\mathbf{P}$ is the transition matrix from S' to S, and $\mathbf{Q}$ is the transition matrix from C' to C (see Section 3.8; some books, including the full textbook, name these matrices the other way round). Then $\mathbf{x} = \mathbf{Pz}$ and $\mathbf{y} = \mathbf{Qw}$. Consequently, $[T]_{S,C}(\mathbf{Pz}) = \mathbf{Qw} \Rightarrow \left(\mathbf{Q}^{-1}[T]_{S,C}\mathbf{P}\right)\mathbf{z} = \mathbf{w} \Rightarrow$

$$[T]_{S',C'} = \mathbf{Q}^{-1}[T]_{S,C}\mathbf{P}. \qquad (*)$$

For $T : \mathbb{R}^n \to \mathbb{R}^m$ with $C = C' = E$ (so that $\mathbf{Q} = \mathbf{I}_m$) and $S' = E$, formula $(*)$ becomes $\mathbf{A} = [T]_S\mathbf{P}$, where $\mathbf{A}$ is the standard matrix of T and $\mathbf{P}$ is the transition matrix from E to S.

Recall the definition of equivalent matrices (Section 2.5): $\mathbf{A}$ and $\mathbf{B}$ are called *equivalent* if $\mathbf{B} = \mathbf{MAN}$ for some invertible $\mathbf{M}$ and $\mathbf{N}$.

Proposition 1. Matrices are equivalent if and only if they are matrices of one and the same linear transformation in some pairs of bases.

Proof. Obviously, $(*)$ means the equivalence of the matrices $[T]_{S',C'}$ and $[T]_{S,C}$. If $\mathbf{B} = \mathbf{MAN}$, then $\mathbf{M}$ and $\mathbf{N}$ can be regarded as transition matrices for different pairs of bases. □

Corollary. For matrices of the same size to be matrices of one and the same linear transformation in some pairs of bases, it is necessary and sufficient that they have the same rank.

Let $\mathbf{A}$ be the matrix of a linear transformation T in some pair of bases. If $r = \operatorname{rank}(\mathbf{A})$, then $\mathbf{A}$ is equivalent to the matrix $\mathbf{B} = \left(b_{ij}\right)$ in which $b_{11} = ... = b_{rr} = 1$ and all other entries are equal to 0. Consequently, there exists a pair of "canonical" bases in which T is determined by the matrix $\mathbf{B}$.

Thus, by choosing a pair of bases, the matrix of a linear transformation can acquire a form so simple that it may turn out to be almost useless for studying the individual properties of the given transformation. Therefore, studying a transformation cannot, in general, be reduced to studying its matrix.

If $V_n = V_m$, then it becomes possible to take $C = S$. Owing to the fact that images and preimages are considered in one and the same basis, the matrix of the transformation now contains everything needed for any detailed study of it. The same holds for any other pair of bases S' and C', provided $C' = S'$. In this case, of course, $\mathbf{Q} = \mathbf{P} \Rightarrow$

$$[T]_{S',S'} = \mathbf{P}^{-1}[T]_{S,S}\mathbf{P}. \qquad (**)$$

Matrices $\mathbf{A}$ and $\mathbf{B}$ are called *similar* if $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$ for some invertible matrix $\mathbf{P}$. Obviously, the following proposition holds.

Proposition 2. Matrices are similar if and only if they are matrices of one and the same linear transformation in some bases, provided identical bases are chosen in the common domain and codomain.

Problem 1. Given arbitrary numbers $\beta_1, ..., \beta_{n-1}$, prove that any bidiagonal $n \times n$ matrix $\mathbf{A}$ of the form

$$\mathbf{A} = \begin{pmatrix} \alpha_1 & 1 & & & 0 \\ & \alpha_2 & 1 & & \\ & & \ddots & \ddots & \\ & & & \alpha_{n-1} & 1 \\ 0 & & & & \alpha_n \end{pmatrix}$$

is similar to some $n \times n$ matrix $\mathbf{B} = \left(b_{ij}\right)$ in which $b_{11} = \beta_1, ..., b_{n-1,n-1} = \beta_{n-1}$.

7.2. Range and Kernel of a Linear Transformation

Kernel and Range of a Linear Transformation
The set $\ker(T) \equiv \{\mathbf{x} \in V_n : T(\mathbf{x}) = \mathbf{0}\}$ is called the *kernel* of the linear transformation T, and the set $\mathrm{R}(T) \equiv \{\mathbf{y} \in V_m : \mathbf{y} = T(\mathbf{x}) \text{ for some } \mathbf{x} \in V_n\}$ is called its *range* (or *image*). The dimension of the range is called the *rank* of the transformation T, and the dimension of the kernel is called its *nullity* (or *defect*). Notation: $\mathrm{rank}(T) = \dim(\mathrm{R}(T))$, $\mathrm{nullity}(T) = \dim(\ker(T))$.

Proposition 1. The kernel of a linear transformation $T : V \to W$ is a subspace of V, and its range is a subspace of W.

Proof. Let $\mathbf{x}, \mathbf{y} \in \ker(T)$. Then $T(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y}) = \alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0} \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in \ker(T)$. Let $\mathbf{x}, \mathbf{y} \in \mathrm{R}(T)$. Then $\mathbf{x} = T(\mathbf{u})$ and $\mathbf{y} = T(\mathbf{v})$ for some $\mathbf{u}, \mathbf{v} \in V \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} = T(\alpha\mathbf{u} + \beta\mathbf{v}) \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in \mathrm{R}(T)$. □

Theorem (Rank–Nullity Theorem)**.** Let V be finite-dimensional. Then

$$\dim(\ker(T)) + \dim(\mathrm{R}(T)) = \dim(V).$$

Proof. Let $\dim(\ker(T)) = k$ and let $\mathbf{v}_1, ..., \mathbf{v}_k$ be a basis of the subspace $\ker(T)$. Complete it with some vectors $\mathbf{v}_{k+1}, ..., \mathbf{v}_n$ to a basis of V. Obviously,

$$\mathrm{R}(T) = \mathrm{span}\{T(\mathbf{v}_{k+1}), ..., T(\mathbf{v}_n)\}.$$

It remains to prove that the vectors $T(\mathbf{v}_{k+1}), ..., T(\mathbf{v}_n)$ are linearly independent.

Suppose $\alpha_{k+1}T(\mathbf{v}_{k+1}) + ... + \alpha_n T(\mathbf{v}_n) = \mathbf{0} \Rightarrow T(\alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n) = \mathbf{0} \Rightarrow \alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n \in \ker(T) \Rightarrow \alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n = \beta_1\mathbf{v}_1 + ... + \beta_k\mathbf{v}_k$ for some scalars $\beta_1, ..., \beta_k \Rightarrow \alpha_{k+1} = ... = \alpha_n = 0$. □

Remark. This theorem could have been left unproved, since it is a consequence of a fact we already know: for any matrix $\mathbf{A}$, the sum of its rank and its nullity (the dimension of its nullspace $\mathrm{Null}(\mathbf{A})$, the solution space of the homogeneous system $\mathbf{Ax} = \mathbf{0}$) equals the number of its columns. We know that the rank coincides with the dimension of the column space $\mathrm{Col}(\mathbf{A})$, and the latter is obviously the range of $\mathbf{A}$ (regarded as the transformation of multiplication by the given matrix).

Proposition 2. Suppose a linear transformation T acts from V to V. Then its kernel and range are invariant under T.

Proof. The invariance of the kernel is obvious, since every vector in it is mapped to $\mathbf{0}$.

Let $\mathbf{x} \in \mathrm{R}(T)$. Then $\mathbf{x} = T(\mathbf{u}) \Rightarrow T(\mathbf{x}) = T(T(\mathbf{u})) \Rightarrow T(\mathbf{x}) \in \mathrm{R}(T)$. □

Problem 1. For two linear transformations acting from V to W, the sum of their kernels coincides with V. Prove that the range of the sum of these

transformations equals the sum of their ranges. Does this hold in the case of three transformations?

The Inverse of a Linear Transformation

A transformation $T : V \to W$ is called *invertible* if there exists a transformation $S : W \to V$ such that $T(S(\mathbf{y})) = \mathbf{y} \quad \forall \mathbf{y} \in W$ and $S(T(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in V$. In this case S is called the *inverse transformation* for T.

Proposition. If a linear transformation is invertible, then the inverse transformation is also linear.

Proof. Any vectors $\mathbf{y}_1, \mathbf{y}_2 \in W$ can be represented in the form $\mathbf{y}_1 = T(\mathbf{x}_1)$, $\mathbf{y}_2 = T(\mathbf{x}_2)$. Therefore,

$$S(\alpha\mathbf{y}_1 + \beta\mathbf{y}_2) = S(\alpha T(\mathbf{x}_1) + \beta T(\mathbf{x}_2)) = S(T(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2)) = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2.$$

It remains to take into account that

$$\mathbf{x}_1 = S(\mathbf{y}_1), \quad \mathbf{x}_2 = S(\mathbf{y}_2). \qquad \square$$

Theorem. Let $T : V \to W$ be a linear transformation, and let V and W be finite-dimensional spaces of the same dimension. The transformation T is invertible if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof. Let $\dim(V) = \dim(W) = n$. By the rank–nullity theorem, if $\dim(\ker(T)) = 0$, then $\dim(\mathrm{R}(T)) = n$. This means that for each vector $\mathbf{y} \in W$ there exists $\mathbf{x} \in V$ such that $T(\mathbf{x}) = \mathbf{y}$ (T is surjective). Moreover, such a vector $\mathbf{x}$ is unique (otherwise the kernel would contain a nonzero vector), so T is also injective. Define the transformation $S : W \to V$ by the rule $S(\mathbf{y}) = \mathbf{x}$. Then $T(S(\mathbf{y})) = \mathbf{y}$ and $S(T(\mathbf{x})) = \mathbf{x} \Rightarrow S$ is the inverse transformation for T.

Conversely, if it is known that T is an invertible transformation, then its kernel can only be zero (if $\mathbf{x}_1 \neq \mathbf{x}_2$ and $T(\mathbf{x}_1) = T(\mathbf{x}_2)$, this contradicts the invertibility of the transformation T). $\square$

Remark. If a linear transformation $T : V \to W$ is invertible, then necessarily $W = \mathrm{R}(T)$, that is, T is surjective. At the same time, the condition $W = \mathrm{R}(T)$ is not sufficient for the invertibility of T.

Problem 2. Linear transformations S and T are such that the transformation $S + T$ is invertible. Prove that the transformations $P = (S + T)^{-1}S$ and $Q = (S + T)^{-1}T$ commute.

Problem 3. A transformation $T : \mathbb{C}^n \to \mathbb{C}^n$ preserves inner products. Prove that T is an invertible transformation. Must it be linear?

Problem 4. A linear transformation $T : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ preserves the determinant of a matrix. Prove that every such transformation is invertible.

Isomorphism of Vector Spaces BEYOND MA1522

Two real vector spaces V and W are called *isomorphic* if there exists a bijective mapping $\Phi : V \to W$ preserving the operations in the following sense:

$$\Phi(\mathbf{a}+\mathbf{b}) = \Phi(\mathbf{a}) + \Phi(\mathbf{b}), \quad \Phi(\alpha\mathbf{a}) = \alpha\Phi(\mathbf{a}) \quad \forall \mathbf{a}, \mathbf{b} \in V, \forall \alpha \in \mathbb{R}.$$

The mapping Φ itself—a bijective linear transformation from V to W—is then called an *isomorphism*.

Note that the operations of vector addition and scalar multiplication on the left-hand and right-hand sides of these equalities are, generally speaking, *different*! The operations on the left act in V, while the operations on the right act in W. Nevertheless, once it is established that the spaces are isomorphic, this means that they are indistinguishable from the standpoint of the *properties of the operations*.

Proposition. $\Phi(\mathbf{0}) = \mathbf{0}, \quad \Phi(-\mathbf{a}) = -\Phi(\mathbf{a}) \quad \forall \mathbf{a} \in V.$

Proof. $\Phi(\mathbf{0}) = \Phi(\mathbf{0}+\mathbf{0}) = \Phi(\mathbf{0}) + \Phi(\mathbf{0})$. Add to both sides the vector $\mathbf{b} = -\Phi(\mathbf{0})$ (the vector opposite to $\Phi(\mathbf{0})$):

$$\mathbf{0} = \mathbf{b} + \Phi(\mathbf{0}) = (\mathbf{b} + \Phi(\mathbf{0})) + \Phi(\mathbf{0}) = \mathbf{0} + \Phi(\mathbf{0}) = \Phi(\mathbf{0}) \Rightarrow \Phi(\mathbf{0}) = \mathbf{0}. \quad \square$$

Obviously, on the set of all real vector spaces, isomorphism generates an equivalence relation. Importantly, investigations carried out for one space V carry over immediately to any space isomorphic to it. For example, vectors $\mathbf{a}_1, ..., \mathbf{a}_n \in V$ are linearly dependent if and only if the vectors $\Phi(\mathbf{a}_1), ..., \Phi(\mathbf{a}_n)$ are linearly dependent.

Theorem. Any real vector space V of dimension $n = \dim(V)$ is isomorphic to $\mathbb{R}^n$.

Proof. Choose some basis $S = \{\mathbf{a}_1, ..., \mathbf{a}_n\}$ of the space V and define the mapping Φ as follows:

$$\Phi(\mathbf{v}) = [\mathbf{v}]_S = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

where $x_1, ..., x_n$ are the coefficients of the expansion of the vector $\mathbf{v}$ in this basis:

$$\mathbf{v} = x_1\mathbf{a}_1 + ... + x_n\mathbf{a}_n.$$

Preservation of the operations is verified in an obvious manner. $\square$

Corollary. Any finite-dimensional real vector spaces of the same dimension are isomorphic.